ICC-1 high yield, all modules

Year 3, 2026-2027. Each module: one page per lecture, then one summary per Specification Grid topic.

Module	Grid hours	Lectures
Cardiology	77	17
Chest	77	19
Vertical subjects	54	34
Vascular surgery	39	10
Hematology	19	11

ICC-1 Cardiology: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. ECG interpretation and reporting [070, 086]
2. ECG workshop and ECG revision [086]
3. Tachyarrhythmias and bradyarrhythmias [078, 079] (lecture and LGW)
4. Valvular heart disease and heart murmurs [073]
5. Infective endocarditis [019] (lecture and LGW)
6. Heart failure [072]
7. Cardiomyopathies [063]
8. Pericardial diseases [063]
9. Coronary artery disease [071]
10. Acute coronary syndrome LGW [028, 064]
11. Hypertension [090]
12. Dyslipidaemia LGW [029]
13. Acute heart failure [028, 061] (lecture and LGW)
14. Aortic dissection [076]
15. Chest pain and cardiac presentations [048, 055]
16. Cardiology history taking and examination [099]
17. Clinical cardiac cases S, T, U [039, 040] and Diagnosis unclear [002]

Part 2: topic summaries

1. Arrhythmias and ECG interpretation
2. Valvular heart disease and infective endocarditis
3. Chronic heart failure, cardiomyopathies and pericardial disease
4. Ischaemic heart disease, dyslipidaemia and hypertension
5. Cardiac emergencies (acute heart failure, aortic dissection)
6. Cardiac symptoms, presentations and clinical reasoning

PART 1: LECTURE

ECG interpretation and reporting [070, 086]

Topic: Arrhythmias and ECG interpretation

Leads and paper

- 10 electrodes (4 limb, 6 chest) give 12 leads; right leg = ground. V4 = 5th ICS midclavicular; V1/V2 = 4th ICS right/left sternal edge.
- Small square 0.04 s; large square 0.2 s. Check settings first: 25 mm/s, 10 mm = 1 mV, aVR negative.

Normal values and axis

- PR 120-200 ms (3-5 small squares); QRS < 110 ms; QTc long > 440 ms (men) / > 460 ms (women); > 500 ms = torsades risk.
- Transition V3-V4. At/before V2 = counterclockwise (RV enlargement); after V4 = poor R progression (COPD, old anteroseptal MI).
- Normal axis -30 to +90. Lead I +, III - = LAD (conduction abnormalities); I -, III + = RAD (RVH); both - = extreme.

Ischaemia and infarction (not with LVH or BBB)

- STE in 2 contiguous leads ≥ 1 mm; in V2-V3 ≥ 2 mm men ≥ 40, ≥ 2.5 mm men < 40, ≥ 1.5 mm women. ST depression ≥ 0.5 mm.
- Pathological Q: > 1 mm wide and > 2 mm deep, or > 25% of the next R.
- Age: 0-6 h STE/hyperacute T; 6-24 h Q + STE; 24 h-3 wk Q, ST at baseline, inverted T; > 3 wk Q only. STE > 3 wk, no pain, normal enzymes = LV aneurysm.
- Anteroseptal V1-V4; anterolateral V3-V6, I, aVL; posterior = tall R, ST depression, tall T in V1-V3; RV = STE V3R-V4R. Inferior MI: reciprocal change in lateral leads.

Chambers and others

- LAE: broad notched P in II, deep negative P in V1. RAE: tall peaked P > 2.5 mm in II.
- LVH: S V1 + R V5 > 35 mm (or R aVL ≥ 11 mm); strain V5-V6. RVH: R in V1 > 7 mm, deep S V6; strain V1-V2.
- Low voltage + electrical alternans = massive pericardial effusion.
- U waves = low K; long QT = low K, Ca, Mg; short QT = high Ca; sagging ST = digitalis.
- Pacing spike before P = atrial, before QRS = ventricular, both = dual chamber.

Source: ECG interpretation [014.1]; ECG reporting [016.1]

PART 1: LECTURE

ECG workshop and ECG revision [086]

Topic: Arrhythmias and ECG interpretation

Chest pain: life-threatening four

- Always exclude acute MI, aortic dissection, pulmonary embolism and tension pneumothorax.
- Check BP and pulse volume in both arms before antithrombotics: antiplatelets, anticoagulants and thrombolytics are absolutely contraindicated in dissection.
- STEMI: aspirin 300 mg + clopidogrel 300 mg, analgesia, nitrates, urgent primary PCI (preferred) or thrombolysis if pain < 12 h. Do not wait for enzymes: minutes mean muscle.
- Non-ST elevation ACS: about 50% have a normal or non-diagnostic ECG; diagnose on typical pain plus risk factors. Diffuse ST depression with STE in aVR is managed as NSTE-ACS (urgent angiography).
- Treat NSTE-ACS with aspirin, clopidogrel, nitrates, beta-blocker, enoxaparin 1 mg/kg 12-hourly, CCU, enzymes, echo.

Classic cases

- PE (OCP, unilateral leg swelling): sinus tachycardia, RBBB, RAD, S1Q3T3. Confirm with CT pulmonary angiography; anticoagulate.
- Aortic dissection: tearing pain to the back, unequal pulses and BP, early diastolic murmur (aortic regurgitation); ECG may show LVH with strain. Echo and CT aortography, then urgent surgery.
- Pericarditis: pleuritic pain after viral illness, rub; widespread concave STE with PR depression, reciprocal changes in aVR and V1. NSAIDs + colchicine.
- Tamponade: Beck's triad (hypotension, raised JVP, muffled sounds); echo shows RV collapse. Urgent pericardiocentesis.
- Hyperkalaemia (ESRD): peaked T, long PR, broad bizarre QRS, sine wave. IV calcium over 10 min, glucose-insulin, bicarbonate, urgent dialysis.

Rhythm cases

- Wide-complex tachycardia with structural heart disease = VT until proved otherwise: unstable, synchronised DC shock; stable, IV amiodarone.
- Regular narrow tachycardia ~200, no P = SVT: IV verapamil or beta-blocker. Irregular narrow, no P = AF: rate control + anticoagulation.
- Complete heart block: regular, variable PR, slow (~35): atropine, then pacemaker.
- Heart block: uniform long PR, all conducted = 1st degree; uniform PR with dropped beats = Mobitz II; variable PR, irregular = Mobitz I.

Conduction and pre-excitation

- LAFB: LAD, qR in I/aVL, rS in inferior leads. LPFB: RAD, rS in I/aVL, qR inferior. Bifascicular = RBBB + hemiblock; trifascicular = bifascicular + 1st degree block.
- WPW: short PR, delta wave, wide QRS. Dominant R in V1 = type A (left pathway); dominant S in V1 = type B (right pathway).

Source: ECG workshop [015.2]; ECG revision [017.5]

PART 1: LECTURE

Tachyarrhythmias and bradyarrhythmias [078, 079] (lecture and LGW)

Topic: Arrhythmias and ECG interpretation

Identify the pacemaker

- Narrow QRS < 120 ms = supraventricular; wide ≥ 120 ms = ventricular. Rate: regular = 300 / large squares in R-R; irregular = QRS in 30 large squares x 10.
- Sinus: P upright in II, negative in aVR. Junctional: inverted P in II, upright in aVR.
- Sinus arrhythmia: rate varies with breathing (R-R difference ≥ 120 ms). MAT: ≥ 3 P shapes, rate > 100 (< 100 = wandering pacemaker).
- VT: wide, regular, AV dissociation, T opposite QRS. Torsade: twisting polymorphic VT with long QT. VF: chaotic, no P/QRS/T.

Atrial fibrillation

- Pulmonary vein foci (300-600/min); no atrial contraction, so LA appendage thrombus and stroke.
- Paroxysmal < 7 days; persistent > 7 days; long-standing > 1 year; permanent = rhythm attempts stopped.
- Unstable: 5000 IU UFH then synchronised DC shock. Stable: rate control first-line (or rhythm) + OAC.
- LVEF < 40%: no verapamil/diltiazem (beta-blocker or digitalis). Structural disease: no flecainide/ propafenone (amiodarone).
- Before cardioversion: TEE to exclude thrombus or 3 weeks OAC; OAC for 4 weeks after.
- Warfarin for metallic valve or mitral stenosis regardless of score. Otherwise CHA2DS2-VA 2 = OAC, 1 = consider; DOACs preferred. No antiplatelets; LAA occlusion if OAC contraindicated.

Other tachycardias

- SVT: unstable = synchronised DC; stable = IV adenosine, beta-blocker or verapamil/diltiazem.
- VT: unstable = synchronised DC; stable = IV amiodarone; long term amiodarone or ICD.

Heart block and bundle branch block

- 1st degree: PR > 200 ms, all P conducted. Mobitz I: PR lengthens until a drop (AV node; monitor). Mobitz II: fixed PR, sudden drop, wide QRS (infranodal; pace).
- Complete block: narrow escape 50-60 (His, reliable); broad escape 15-40 (His-Purkinje, Stokes-Adams; permanent pacemaker).
- Symptomatic block: correct cause, IV atropine, temporary then permanent pacing.
- LBBB: QS/rS in V1, broad notched R in I, V5-V6. RBBB: RSR' in V1, slurred S in V6.
- WPW: short PR, delta wave; dominant R in V1 = type A (left), dominant S = type B (right).

Source: Tachy-Bradyarrhythmias lecture [018.1/019.1]; LGW Parts 1 and 2 [020.2]

PART 1: LECTURE

Valvular heart disease and heart murmurs [073]

Topic: Valvular heart disease and infective endocarditis

Aortic stenosis (AS)

- Causes: senile calcification, bicuspid valve, rheumatic. Asymptomatic for years, then exertional syncope, angina, dyspnoea; untreated death in 2-3 years once symptoms start.
- Slow-rising small pulse; sustained, non-displaced apex; harsh diamond-shaped ejection systolic murmur at aortic area radiating to carotids.
- Severity: late-peaking murmur, soft/absent A2, reversed S2 splitting. Loudness is NOT a guide.
- CXR: small heart, post-stenotic dilated ascending aorta. ECG: LVH with strain.
- Valve replacement if severe + symptoms; if asymptomatic, operate for EF <50%, symptoms/ hypotension on exercise test, peak velocity >5.5 m/s. TAVI if unfit for surgery.

Aortic regurgitation (AR)

- Collapsing pulse, De Musset, Quincke, Duroziez (severe AR), pistol-shot femorals; apex displaced down and out.
- Soft early diastolic murmur at left sternal edge, leaning forward in expiration; Austin Flint = apical mid-diastolic rumble from the jet hitting the anterior mitral cusp.
- Surgery: acute severe AR (e.g. endocarditis), symptoms, EF ≤50%, or LV dilated (EDD >70 mm / ESD >50 mm).
- Tissue valves for elderly or when anticoagulation must be avoided; not for young (rapid degeneration).

Mitral stenosis (MS)

- Almost always rheumatic. Dyspnoea, haemoptysis, malar flush, AF with emboli.
- Tapping apex, loud S1, opening snap, low-pitched mid-diastolic rumble: bell at apex, left lateral, expiration. Graham Steell = PR from pulmonary hypertension.
- CXR: LA enlargement, double shadow. ECG: P mitrale, AF.
- AF: rate control or cardioversion PLUS anticoagulation. Intervene if area ≤1.5 cm^2 with symptoms or pulmonary hypertension.
- Balloon valvuloplasty: TOE first to exclude LA thrombus; contraindicated with heavy calcification, more than mild MR, LA thrombus.

Mitral regurgitation and right-sided lesions

- MR: soft S1, pansystolic murmur apex to axilla, S3. MVP: mid-systolic click then late systolic murmur.
- Surgery: symptomatic severe MR, or asymptomatic with ESD >40 mm or EF <60%. Papillary/ chordal rupture = emergency valve surgery.
- TR: large cv waves, pulsatile liver, pansystolic murmur at lower left sternal edge louder on inspiration; usually functional (RV dilatation).

Auscultation tips

- Time murmurs with the carotid pulse: starts with it = systolic; starts after it = diastolic.

Source: [073] Valvular heart disease; [073] Heart murmurs auscultation LGW

PART 1: LECTURE

Infective endocarditis [019] (lecture and LGW)

Topic: Valvular heart disease and infective endocarditis

Pathogenesis and risk

- Needs two things: damaged endocardium (turbulent flow → platelet-fibrin clot) plus bacteraemia; bacteria adhere and multiply to form vegetations that embolise and trigger immune complexes.
- Risks: prosthetic valves, prior IE, congenital and rheumatic heart disease, degenerative aortic valve, severe MVP, IV drug use, intravascular devices, invasive procedures. Can affect normal valves.
- People who inject drugs: tricuspid valve in about 50%; right-sided lesions give septic pulmonary emboli.

Organisms

- Native valve: viridans streptococci most common (54%), then S. aureus (19%), enterococci.
- Early prosthetic (<6 months): S. aureus, coagulase-negative staph (S. epidermidis). Late prosthetic: same as native.
- Culture negative (up to 20%): prior antibiotics, fastidious strep, HACEK, Coxiella, Candida; use prolonged/fungal cultures and paired serology.

Clinical features

- Fever in >95%, fatigue, breathlessness; acute presentation usually heart failure.
- Signs: splinter haemorrhages, Osler nodes, Janeway lesions, conjunctival haemorrhages, Roth spots, splenomegaly.

Diagnosis

- Blood cultures are the key test: at least 2, ideally 3 sets BEFORE antibiotics, 8-10 mL per bottle.
- Echo: TTE then TOE. Also FBC, U&E, CRP, urine microscopy, CXR.
- 2023 ESC criteria. Major: typical organism in 2 separate cultures; positive imaging. Minor: predisposition, fever >38 °C, embolic vascular (Janeway, conjunctival haemorrhage), immunological (Osler, Roth, GN, RF), microbiology.
- Definite: 2 major, 1 major + 3 minor, or 5 minor. Possible: 1 major + 1-2 minor, or 3-4 minor.

Treatment

- Empirical IV high dose after cultures: native = amoxicillin + gentamicin; prosthetic = vancomycin + gentamicin; then tailor to organism and MIC.
- Surgery: heart failure from valve damage, uncontrolled infection (abscess, heart block, fungal/resistant organism), embolic risk (large vegetation, recurrent emboli).
- Remove infected lines/devices; dental review.
- Prophylaxis for high-risk dental procedures: single dose 30-60 min before, e.g. amoxicillin 2 g orally (adults).

Source: [019] Infective Endocarditis simplified; [019] IE LGW Answers

PART 1: LECTURE

Heart failure [072]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Definition and pathophysiology

- Clinical syndrome of cardinal symptoms (breathlessness, ankle swelling, fatigue) ± signs, from structural/functional heart abnormality raising intracardiac pressures and/or lowering cardiac output.
- CO = SV × HR. Myocardial loss activates SNS and RAAS (harmful remodelling) and natriuretic peptides (protective, from stretch).
- Treatment aim: block SNS and RAAS, enhance natriuretic peptides.
- Causes: myocardial (CAD, hypertension, cardiomyopathy), valvular, pericardial, arrhythmia, high-output states (anaemia, thyrotoxicosis, Paget's, AV fistula, sepsis), volume overload.

Clinical features

- Left-sided: dyspnoea, orthopnoea, PND (most specific), bibasal fine crackles.
- Right-sided: leg oedema, RUQ pain, raised JVP, hepatojugular reflux, ascites. Cor pulmonale = RV failure from lung disease.
- Apex: LV enlargement = downward and outward, localised; RV = outward, diffuse, systolic retraction.
- S3 (just after S2): volume overload; normal in children, pregnancy, athletes. S4 (just before S1): stiff ventricle (hypertension, CAD); not normal even in elderly.

Classification

- LVEF: HFrEF ≤40%, HFmrEF 41-49%, HFpEF ≥50%.
- NYHA I none; II ordinary activity; III less-than-ordinary activity; IV symptoms at rest.
- ACC/AHA: A at risk; B structural disease, no symptoms; C symptomatic; D advanced.

Investigations

- Natriuretic peptides: normal level in an untreated patient virtually excludes HF. Rule-out: BNP <100, NT-proBNP <300 pg/mL.
- Echo is the key test of cardiac function. Normal ECG makes HF unlikely. CXR finds other causes of breathlessness.
- Iron studies (ferritin, transferrin saturation): about a third are iron deficient.
- Cardiac MRI: poor echo windows; infiltrative disease (amyloid, sarcoid), myocarditis.

Treatment

- Prevention first; lifestyle: salt restriction, exercise, stop smoking/alcohol, weight.
- Diuretics improve symptoms only.
- Four prognostic pillars: ACEI/ARB/ARNI, MRA, beta-blocker, SGLT2 inhibitor.
- Sacubitril inhibits neprilysin, raising natriuretic peptides. SGLT2i increase distal sodium delivery, lowering preload and afterload.
- Devices/surgery: revascularisation, valve replacement, biventricular pacemaker, LVAD, transplant.

Source: [072] Heart Failure, Dr Mina Samy

PART 1: LECTURE

Cardiomyopathies [063]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Classification

- Myocardial diseases affecting mechanical or electrical function. ESC groups: HCM, arrhythmogenic (ACM), DCM, restrictive (RCM), unclassified (LVNC, Tako-tsubo); each familial or non-familial.
- DCM causes mnemonic APCDEF MN: Alcohol, Pregnancy, Calcium/phosphate low, Drugs (anthracycline), Endocrine (thyroid), Familial, Fast HR, Myocarditis, Nutritional (thiamine, carnitine, selenium).

Hypertrophic cardiomyopathy (HCM)

- Inherited (mostly autosomal dominant, sarcomere genes MYH7, MYBPC3) hypertrophy without AS or hypertension; 1 in 500; commonest cause of sudden death in the young.
- Asymmetric septal hypertrophy; ~25% have dynamic LVOT obstruction (SAM of anterior mitral leaflet). AF causes rapid deterioration/pulmonary oedema.
- Signs: double apical impulse (S4), jerky carotid pulse, ejection murmur louder with standing/Valsalva, softer with squatting; MR pansystolic murmur.
- ECG: LVH, ST-T changes, inferolateral Q waves. Echo usually diagnostic; CMR shows fibrosis and separates HCM from infiltration.
- SCD risk factors: wall >30 mm, family SCD <50 y, NSVT on Holter, unexplained syncope, flat/hypotensive BP response on exercise. Two or more: ICD; lower risk: amiodarone.
- Symptoms: beta-blocker and/or verapamil; disopyramide if LVOT obstruction; dual-chamber pacing, alcohol septal ablation (risk: heart block, MI) or surgical myectomy. Avoid vasodilators.

ARVC

- Fibro-fatty replacement of RV: early VT/sudden death, later RV or biventricular failure.
- ECG: T inversion V1-V3, epsilon waves, RBBB; Holter RV ectopics/NSVT; CMR shows fibro-fat. Task Force Criteria.
- Beta-blocker first line; amiodarone/sotalol if symptomatic; ICD if life-threatening; transplant occasionally.

DCM and acquired forms

- Dilated ventricles, systolic dysfunction, normal wall thickness; HF, arrhythmia, emboli, sudden death. Exclude CAD (angiography/CTCA) if >40 y or risk factors; biopsy not routine. Treat as HF; CRT/ICD in NYHA III-IV; transplant.
- Peripartum: last trimester to 5 months postpartum; obese, multiparous, >30 y, pre-eclampsia; ~half recover in 6 months. Tachycardiomyopathy: cardioversion/ablation allows recovery.
- Tako-tsubo: older women, MI-like, catecholamine excess, normal coronaries, apical ballooning; recovers in 4-6 weeks.

Restrictive and LVNC

- RCM: small/normal ventricles, bi-atrial enlargement, near-normal EF, impaired filling; amyloid commonest. Raised JVP, Kussmaul's sign, S3/S4.
- Amyloid clues: macroglossia, periorbital bruising, neuropathy, carpal tunnel, proteinuria, AF, effusions; voltage-mass mismatch; speckled myocardium on echo.
- Biopsy often useful (unlike other CMs); cath separates from constriction. Treat HF/AF (anticoagulate); tafamidis for familial amyloid; liver transplant in familial type.
- LVNC: spongy trabeculated LV apex; HF, emboli, arrhythmia, sudden death.

Source: ICC-1 [063] Cardiomyopathies 2026-2027

PART 1: LECTURE

Pericardial diseases [063]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Pericardium and causes

- Double-layered fibro-serous sac: visceral (epicardium) and parietal (fibrous + serous) layers; normally 15-25 ml fluid; blood supply from internal mammary arteries.
- Causes (RHIM + malignancy): radiotherapy; haemopericardium (trauma, aortic dissection); infection (viral, bacterial, TB, fungal), inflammatory (connective tissue disease, rheumatic fever, Dressler's, drugs), idiopathic; metabolic (uraemia, hypothyroidism); malignancy.

Acute pericarditis

- Pleuritic-type chest pain relieved by leaning forward.
- Friction rub: rough scratching, louder leaning forward, not tied to cardiac cycle, often absent.
- ECG: widespread saddle-shaped ST elevation (not aVR), PR depression, T inversion; low voltage and electrical alternans if large effusion.
- CXR normal unless large effusion (globular heart); echo to exclude significant effusion; bloods guided by likely cause.
- Treatment: NSAID; add colchicine to reduce symptoms and recurrence; steroids if autoimmune or resistant; antimicrobials for bacterial/TB/fungal; dialysis for uraemic; pericardiocentesis if tamponade.

Effusion and tamponade

- Tamponade: raised intrapericardial pressure limits ventricular filling and cardiac output; rapid accumulation raises pressure fast.
- Symptoms: breathlessness, dizziness, syncope, chest pain.
- Signs: tachycardia, hypotension, weak pulses, pulsus paradoxus, absent apex, faint heart sounds, raised JVP, cold peripheries (slide also lists Kussmaul's sign).
- Beck's triad: raised JVP + hypotension + quiet heart sounds.
- Pulsus paradoxus: inspiratory systolic BP fall >10 mmHg (normal <10) from marked septal shift to the left.
- ECG low voltage + electrical alternans; CXR globular heart; echo is the best non-invasive test.
- Bloods: thyroid, ANA, Quantiferon; clotting and platelets before pericardiocentesis. Fluid: cytology, GeneXpert PCR for TB.
- Treatment: emergency percutaneous pericardiocentesis; surgical drainage if not accessible, traumatic or bloody; pleuro-pericardial window for recurrent (e.g. malignant) effusion; treat cause.

Constrictive and adhesive pericarditis

- Constrictive: TB is the commonest cause (per lecture); pericardial knock; CT/MRI most sensitive; main differential is restrictive cardiomyopathy.
- Adhesive: rare, few manifestations; may follow rheumatic fever (adhesion to mediastinum/chest wall).

Source: ICC-1 [063] Pericardial Diseases 2026-2027

PART 1: LECTURE

Coronary artery disease [071]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Pathophysiology and risk

- CAD = CHD = IHD: atherosclerotic narrowing of coronaries; leading cause of death worldwide. Stable CAD: fixed 80-90% stenosis; enough flow at rest, ischaemia when demand rises.
- ACS: plaque often only 30-50% but unstable; rupture/erosion exposes vWF, platelet aggregation, thrombus.
- Ischaemia = supply/demand imbalance. Low supply: atherosclerosis, spasm, thrombus/embolus, vasculitis; hypoxia, anaemia, hypotension. High demand: exercise, tachycardia, hypertension, LVH.
- Modifiable: smoking, diabetes, hypertension, dyslipidaemia, inactivity, obesity. Non-modifiable: age, sex, family history, ethnicity.

Coronary anatomy

- LAD ("widow maker", ~60% of LV): anterior septum, anterior and lateral walls, apex.
- LCX: left AV groove; lateral wall (obtuse marginals), posterior wall in 10-20%.
- RCA: right AV groove; RA, RV, posterior LV wall via PDA in 80%. SA node: RCA 65%, LCX 25%. AV node: RCA 90%, LCX 10%.

Presentations

- Typical pain: retrosternal, to jaw/arm/back/epigastrium, triggered by exertion/emotion, relieved by rest/SL nitrate. Angina equivalents: dyspnoea, fatigue, faintness.
- ACS spectrum: UA (troponin negative), NSTEMI (troponin positive; ST depression/T changes/ normal), STEMI (ST elevation or new LBBB).
- Variant (Prinzmetal): rest angina, midnight-early morning, transient ST elevation.
- High-risk stable angina: post-MI angina, poor effort tolerance, ischaemia at low workload, left main/3-vessel disease, poor LV function.

Investigations

- Resting ECG/echo usually normal in stable CAD. Labs: CBC, U&E, lipids, glucose, hs-troponin.
- Pathway: ECG, then CTCA (1st line), then MPS/stress echo/stress MRI (2nd), then invasive angiography (3rd).
- Calcium score 0: obstructive CAD very unlikely (~0.1%/yr); >100: high risk (>2%/yr).
- Thallium perfusion scan: exercise best; dobutamine if unable to exercise.

Treatment

- Nitrates: dilate veins/arteries, cut preload/afterload; headache, flushing, hypotension; nitrate-free period; never with sildenafil.
- Beta-blockers: lower HR, BP, contractility; bradycardia, bronchospasm, cold peripheries, impotence; never stop abruptly (withdrawal syndrome).
- CCBs: oedema, headache, flushing; short-acting nifedipine worsens ischaemia in MI.
- ACEi/ARB: cough, hyperkalaemia, renal impairment. Statins: cramps, raised liver enzymes. Antiplatelets: bleeding, gastritis.
- Nicorandil (K channel activator): no tolerance. Ivabradine (If): slows sinus node, no negative inotropy, safe in HF. Interventional: PCI or CABG.

Source: ICC-1 [071] CAD NGU; CAD Dr Mina Samy 2026-2027

PART 1: LECTURE

Acute coronary syndrome LGW [028, 064]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Suspecting ACS

- Check history, risk factors (e.g. smoking), previous IHD, prior tests.
- Suggestive: pain in chest/arms/back/jaw >15 min; pain with nausea, vomiting, sweating, breathlessness or haemodynamic instability; new pain or abrupt worsening of stable angina at little/no exertion.
- If ECG is non-diagnostic: serial ECGs, compare old ECGs, extra leads; consider PE, aortic dissection, pneumonia.

Immediate management

- Pain relief: GTN and/or IV opioid.
- Aspirin 300 mg single loading dose (unless allergic), then continue indefinitely; other antiplatelets only in hospital.
- Resting 12-lead ECG; send to hospital before arrival.
- Oxygen only if SpO2 <94%: target 94-98%; COPD at risk of hypercapnia 88-92% until blood gas.
- Monitor: pain, pulse, BP, rhythm, SpO2, repeated ECGs, pain-relief effect.

Diagnosis

- Troponin repeat 10-12 h after symptom onset (session answer).
- ACS = rise/fall of troponin with at least one value above 99th percentile URL plus ischaemia: symptoms, new ST-T change or LBBB, new Q waves, or new wall motion loss on imaging.

STEMI cases

- Acute chest pain in ED: 12-lead ECG first.
- Inferior STEMI: right coronary artery. Anterior STEMI: LAD.
- Send troponin I, CBC, glucose, urea/creatinine, clotting.
- Antiplatelet loading dose, then urgent angiography and angioplasty (PCI).

Other chest pain cases

- Prinzmetal (vasospastic): rest/night pain, transient ST elevation. Confirm ischaemia and risk stratify (echo, stress ECG, CTCA, labs). Treat: CCB, nitroglycerine, statin, stop smoking.
- Effort angina (chronic coronary syndrome): labs, stress ECG, echo, CTCA. Treat: aspirin, beta-blocker, ACEi if echo abnormal, nitrate or K-channel opener, statin, lifestyle. Angiography/stenting only if severe ischaemia or medical failure.
- Chest wall inflammatory pain: hours long, worse with movement/cough, tender, normal ECG. CBC, CXR, echo; reassure, anti-inflammatory + antacid, review in 1 week.

Source: ICC-1 [028,064] ACS LGW tutor and student notes 2026-2027

PART 1: LECTURE

Hypertension [090]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Definitions (ESC 2024)

- Office BP: non-elevated <120/70; elevated 120-139/70-89; hypertension ≥140/90 mmHg (persistent).
- Out-of-office thresholds: home ≥135/85; ABPM daytime ≥135/85, night ≥120/70, 24-h ≥130/80.
- Primary 90% (gradual, >40 y); secondary 10% (often <40 y, abrupt/severe, potentially curable).
- Resistant HTN: above target on ≥3 classes incl. a diuretic, at maximal tolerated doses.
- Isolated systolic HTN (age >50, arterial stiffness): treat as usual but keep DBP ≥60 mmHg (coronary flow). Consider aortic regurgitation.
- Orthostatic hypotension: fall >20 systolic or >10 diastolic within 1-3 min of standing; use standing BP to guide treatment.

Measurement and diagnosis

- Confirm with out-of-office BP (HBPM/ABPM preferred); both detect white-coat and masked HTN.
- Both arms at first visit, use the higher arm; palpate pulse for AF; manual auscultation in AF and pregnancy.
- Screening >160/100: re-evaluate within days to weeks (≤1 month). >180/110: exclude hypertensive emergency.

Secondary causes and work-up

- Suspect if: <40 y with grade 2+, acute worsening, grade 3/emergency, resistant, extensive organ damage, endocrine or OSA features.
- Primary aldosteronism → aldosterone-to-renin ratio (hypokalaemia). Renovascular → renal Doppler/CT/MR angiography (late onset, flash pulmonary oedema, abdominal bruit). Phaeo → metanephrines. OSA → polysomnography. Cushing's → urinary cortisol/dexamethasone test. Coarctation → echo/aortic CTA.
- Organ damage: fundoscopy, ECG/echo (LVH), creatinine, urine albumin-creatinine ratio.
- Already high risk (no SCORE2 needed): ASCVD/HF, organ damage, CKD, DM, familial hypercholesterolaemia. Otherwise SCORE2 (40-69) / SCORE2-OP (≥70); ≥10% = increased risk.

Treatment

- Hypertension ≥140/90: drugs immediately + lifestyle; start low-dose single-pill dual: ACEi/ARB + CCB or thiazide.
- Uncontrolled at 1-3 months → triple (ACEi/ARB + CCB + diuretic) → add spironolactone.
- Elevated BP: lifestyle; drugs only if high risk and BP ≥130/80 after 3 months; start with monotherapy.
- Monotherapy also for age ≥85, frailty, symptomatic orthostatic hypotension.
- Beta-blockers only with compelling indication (arrhythmia, CAD, HF, pregnancy).
- Targets: SBP 120-129, DBP 70-79 (pregnancy: not <80); lenient <140 in elderly/frail.
- Lifestyle: sodium ~2 g/day, ≥150 min/week exercise, BMI 20-25, DASH/Mediterranean diet, alcohol <100 g/week, stop smoking.

Source: ICC-1 [090b] Hypertension 2026-2027

PART 1: LECTURE

Dyslipidaemia LGW [029]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Causes and risk

- Familial hypercholesterolaemia (FH): common autosomal dominant disorder; very high LDL, tendon xanthomas, premature CHD; test siblings and children.
- Drug causes: antiretrovirals, beta-blockers, oral contraceptives, corticosteroids, thiazides.
- LDL-C and non-HDL-C (TC - HDL) track CVD risk; HDL protects; chylomicrons raise TG but not CVD risk.
- TG >500 mg/dL → risk of acute pancreatitis (an emergency).

Measurement and screening

- Fasting affects only TG; fast if TG >400 mg/dL or FH.
- Friedewald (TG <400): LDL-C = TC - (HDL + TG/5).
- After MI lipids fall: measure within 24 h or 1 month later.
- Opportunistic screening: family history of CHD or lipid disorder, xanthoma/xanthelasma, arcus <40 y, DM, HTN, obesity, pancreatitis, dialysis.

Risk categories and targets

- SCORE2 (40-69 y) / SCORE2-OP (70-89 y): age, sex, smoking, TC, HDL, SBP.
- Not used in ASCVD (very high), CKD, DM, FH, LDL >190.
- Extreme risk: recurrent event on maximal statin, or polyvascular disease → LDL <40.
- LDL goals: low <116; moderate <100; high <70 + ≥50% fall; very high <55 + ≥50% fall.
- Non-HDL goals: moderate <130; high <100; very high <85.

Management

- Step I maximal statin → II + ezetimibe or bempedoic acid → III triple (add PCSK9 mAb) → IV quadruple.
- Post-MI on moderate statin: switch to high intensity (atorvastatin 40-80 mg) first.
- Statin intolerance: bempedoic acid (low muscle risk), other statin, lower dose + ezetimibe, alternate-day/weekly.
- PCSK9 mAbs (alirocumab, evolocumab) only for very high risk; inclisiran = siRNA.
- Statin monitoring: baseline lipids, liver enzymes, CK; lipids 3-monthly to target then yearly; stop if ALT >3× or CK >10× normal.
- TG <500 on statin: icosapent ethyl preferred (TG 135-499); fenofibrate if >200.
- TG ≥500: diabetes control, no alcohol, caloric restriction, fibrates/omega-3, volanesorsen (FCS >750), plasmapheresis.

Source: ICC-1 [029] Dyslipidemia 2026-2027

PART 1: LECTURE

Acute heart failure [028, 061] (lecture and LGW)

Topic: Cardiac emergencies (acute heart failure, aortic dissection)

Definition and causes

- AHF: worsening signs and symptoms as the heart cannot pump enough blood for the body's needs at normal filling pressure.
- Causes: ischaemic heart disease, valve disease, AF, hypertension, acute kidney disease (cardiomyopathy in chronic HF).

Pulmonary oedema

- Fluid leaks into interstitium/alveoli; Starling balance of capillary hydrostatic vs plasma oncotic pressure.
- Cardiogenic: sudden rise in pulmonary venous (hydrostatic) pressure. Non-cardiogenic: increased capillary permeability.
- Other types: high-altitude, neurogenic, rapid aspiration of massive effusion/pneumothorax, after cardiopulmonary bypass.

Clinical picture and diagnosis

- Exertional dyspnoea, orthopnoea, PND, pink frothy sputum, tachypnoea, tachycardia.
- Bilateral crackles, wheeze ("cardiac asthma"), raised JVP, pitting oedema.
- Tests: CBC, renal and liver function, ABG, cardiac enzymes, pro-BNP; ECG for ischaemia; echo (EF, wall motion); CXR (oedema, effusions, cardiomegaly).
- Differentials: COPD, chronic bronchitis, ILD, pneumothorax, PE, anxiety, anaemia (e.g. iron deficiency from menorrhagia).

Treatment

- Aims: stabilise, find cause and precipitant, rapid symptom relief.
- Oxygen if SpO2 <90%; NIPPV cuts intubation, ventilation and mortality.
- IV loop diuretic (furosemide 40 mg) = cornerstone; check response at 2-4 h, then raise dose/ frequency.
- IV nitrates: venodilators, lower preload and LV filling pressure. Nitroprusside: potent arterial dilator, afterload reduction.
- Hypoperfusion or shock (e.g. BP 60/40, no urine): inotropes (dobutamine, dopamine) and/or mechanical support.
- Ultrafiltration: removes isotonic plasma water across a membrane without changing electrolytes.
- Morphine IV 5 mg only for pain if present.
- Long-term survival: ACEi/ARB, beta-blocker, aldosterone antagonist.

Source: ICC-1 [028,061] Acute heart failure; [061] AHF LGW (2026-2027)

PART 1: LECTURE

Aortic dissection [076]

Topic: Cardiac emergencies (acute heart failure, aortic dissection)

Pathophysiology and risk

- Bleeding within the tunica media; an initial intimal tear is found in 90%. The false lumen usually extends anterograde and may re-enter the true lumen or rupture through the adventitia.
- Rare but catastrophic: untreated early mortality up to 1% per hour; peak age about 63; male:female 2:1.
- Raised wall stress: **hypertension (72%)**, cocaine/stimulants, trauma (including coronary intervention), deceleration injury, weight lifting, coarctation.
- Medial abnormality: Marfan, Ehlers-Danlos, Turner, bicuspid aortic valve; vasculitis (giant cell, Takayasu, RA, syphilis); pregnancy, polycystic kidney disease.

Classification and presentation

- Stanford: **type A involves the ascending aorta**; type B does not.
- Sudden severe **tearing** pain, maximal at onset: retrosternal (type A) or interscapular (type B); may migrate or disappear.
- Signs: **pulse deficit / radio-radial delay**, unequal arm BP, early diastolic murmur (acute AR), pulmonary oedema or shock.
- **Chest pain + stroke** (carotid) or **chest pain + acute limb ischaemia** (iliac) at the same time = suspect dissection.
- Complications: acute AR, MI (usually **RCA**), haemopericardium/tamponade, heart failure, syncope, stroke, paraplegia, mesenteric/renal/limb ischaemia.

Investigations

- **CT aortic angiogram = definitive**; TOE and MRI are alternatives. TTE has inferior accuracy; chest X-ray is not diagnostic.
- ECG: 20% of type A show ischaemia. D-dimer <500 ng/mL may rule out dissection only in low-risk patients.

Management and prognosis

- All patients: ICU, monitoring, arterial line, urinary catheter, IV morphine.
- BP: IV beta-blocker, IV calcium antagonist or IV sodium nitroprusside; **target SBP 100-120 mmHg**.
- **Type A = surgical emergency** (mortality rises 1-2% per hour without surgery); graft replacement.
- **Type B = medical** (tight BP control, serial imaging). Stent or operate if: end-organ ischaemia, ongoing extension (ongoing pain), aneurysmal expansion, Marfan.
- Type A: 40% die in the first 24 h without surgery; 96% alive at 1 year after surgery. Type B: 85% alive at 1 year on medical therapy.

Source: [076] Aortic dissection

PART 1: LECTURE

Chest pain and cardiac presentations [048, 055]

Topic: Cardiac symptoms, presentations and clinical reasoning

History structure

- Introduce yourself, confirm name, get consent. Let the patient tell the story in their own words; avoid leading questions.
- Analyse each complaint: onset, course, duration, aggravating and relieving factors, associated symptoms.
- Cardiac symptoms: chest pain, palpitations, dyspnoea, cough, haemoptysis, leg oedema, syncope.

Chest pain

- First question: is it an emergency? Life-threatening: **coronary ischaemia, aortic dissection, pneumothorax, pulmonary embolism**.
- **Ischaemic:** central/retrosternal/epigastric, crushing, radiates to neck, jaw, left arm; brought on by effort or stress; relieved by rest or nitrates; minutes (longer = infarction); sweating, vomiting.
- **Pericarditis:** localised, stitching, radiates to back; worse on inspiration and lying; **relieved by leaning forward**; malaise.
- **Musculoskeletal/pleuritic:** localised, stitching, worse on breathing, coughing and movement; no radiation.
- Palpitations: time, mode of onset and offset, relation to exertion, duration.

Breathlessness and heart failure

- **Orthopnoea:** needs to sit up; count the pillows. **PND:** wakes about 2 h after sleep and opens the window for air.
- Leg oedema in heart failure builds up during the day; eased by raising the legs.
- **Dyspnoea + cough + leg oedema = heart failure.**
- NYHA: I none; II slight limitation; **III marked limitation, comfortable only at rest**; IV symptoms at rest.

Acute coronary syndrome (055)

- Crushing chest pain + sweating: **ECG first**.
- ACS = unstable angina (troponin -ve), NSTEMI (troponin +ve), STEMI or new LBBB (ST elevation, troponin +ve); all share plaque rupture and similar first treatment.
- Initial drugs: **aspirin 300 mg + clopidogrel 300 mg + enoxaparin 1 mg/kg**.
- Complete heart block after MI (pulse 28): temporary pacing + revascularisation in the cath lab.
- Temporary pacing: asystole, symptomatic bradycardia not responding to atropine, Mobitz II, complete heart block, drug overdose (digoxin, beta-blocker, verapamil). Permanent pacing: persisting symptomatic bradycardia, complete AV block.

Source: [048] Chest pain and cardiac presentations; [055] Cardiopresentations

PART 1: LECTURE

Cardiology history taking and examination [099]

Topic: Cardiac symptoms, presentations and clinical reasoning

History

- Symptom groups: pulmonary congestion (cough, dyspnoea, orthopnoea, PND, haemoptysis), systemic congestion (RUQ pain, oedema, swelling), low output (fatigue, syncope, oliguria), chest pain, palpitations, embolic, constitutional, pressure (dysphagia, hoarseness).
- Risk factors: hypertension, smoking, hyperlipidaemia, diabetes, obesity, family history. **Premature CAD:** patient = man <45, woman <55; family = male <55, female <65.
- Drugs: contraceptive pill raises clot risk; cocaine causes coronary spasm. Finish with ICE and a summary.

JVP

- Patient at **45 degrees**, head turned to the left; use the right IJV (more reliable than EJV, no valves).
- JVP = vertical height above the **sternal angle** (normal up to 3 cm). **CVP = JVP + 5 cm** (normal 8 cm H2O).
- Venous pulsation: 2 waves, inward, not palpable, occludable, falls on inspiration and sitting up, rises with abdominal pressure.
- 'a' wave absent in **AF**; **cannon a** in AV dissociation (complete heart block, VT); large 'a' in pulmonary hypertension or tricuspid stenosis; **large v (systolic expansion) = TR**.
- Raised JVP: right heart failure, tamponade, constriction, RV infarct, fluid overload; **SVC obstruction = non-pulsatile**. Hepatojugular reflux positive if a sustained rise of 4 cm or more.

Precordium and apex

- Normal apex: left 5th space, midclavicular line. Out and down = LV dilatation; diffuse with systolic retraction = RV enlargement.
- **Hyperdynamic** = volume overload (AR, MR, VSD, PDA); **heaving** = pressure overload (AS, coarctation, hypertension); **tapping** = MS.
- Milking test: dilated vein refills from above = SVC obstruction; from below = IVC obstruction.

Auscultation

- Bell (cone) for low-pitched sounds (MS rumble, S3, S4); diaphragm for high-pitched sounds.
- Loud S1 in MS; soft S1 in MR. **Wide fixed split S2 = ASD**; reversed split = LBBB, AS, HCM; narrow split = pulmonary hypertension.
- Metallic click with S1 = mitral valve replacement; with S2 = aortic valve replacement.
- Murmurs: pansystolic (MR, TR, VSD), ejection (AS, PS), early diastolic (AR, PR), mid-diastolic (MS, TS), continuous (PDA). Right-sided murmurs get louder on inspiration; MS and MR in the left lateral position; AR leaning forward in expiration.

Source: [099] History taking; CVS examination; Student notes; Neck veins

PART 1: LECTURE

Clinical cardiac cases S, T, U [039, 040] and Diagnosis unclear [002]

Topic: Cardiac symptoms, presentations and clinical reasoning

Case S: heart failure and dilated cardiomyopathy

- Congestive cardiac failure: dyspnoea, raised JVP, **S3 gallop**, displaced apex, crackles, hepatomegaly, ascites, oedema.
- Causes: coronary disease (two-thirds), hypertension, valve disease, cardiomyopathies, infiltration, high-output states.
- Chest X-ray: cardiomegaly, bat-wing shadowing, upper lobe diversion, septal lines, effusions. **TTE is the first-line imaging test.**
- Acute treatment: sit up, oxygen (SaO2 at least 95%), IV morphine + anti-emetic, **IV furosemide** with or without nitrates, fluid restriction, daily weights.
- Long-term (once LV impairment is confirmed): **ACEi/ARB + beta-blocker + aldosterone antagonist**; second-line hydralazine/ISDN, digoxin; CRT, ICD, LVAD, transplant.
- DCM causes: idiopathic, familial, **alcohol, anthracyclines**, post-myocarditis, peripartum, tachycardia, infections (HIV, Chagas), thiamine/selenium deficiency. Post mortem: nutmeg liver.

Case T: NSTEMI

- Unstable angina = troponin negative; **NSTEMI = troponin positive without ST elevation**; STEMI = ST elevation.
- Immediate: oxygen, morphine, nitrates, **aspirin 300 mg then 75 mg, clopidogrel 300 mg then 75 mg**, anticoagulant. Later: beta-blocker, ACEi, statin, angiography.
- Risk scores: GRACE (in-hospital and 6-month mortality), TIMI, HEART.
- **Ongoing pain or worsening ST depression → urgent inpatient angiography and PCI.**

Case U: infective endocarditis

- Fever + valve lesion + a recent procedure (dental) + splenomegaly + new murmur.
- **Osler's nodes** (tender, finger pulps); **Janeway lesions** (non-tender, palms and soles); splinter haemorrhages; Roth's spots.
- **At least 3 blood culture sets from different sites before antibiotics**; echo. Bloods: anaemia, raised CRP/ESR, low C3.
- New **AV block in aortic valve IE = abscess** reaching the AV node.
- **IV antibiotics for at least 4-6 weeks**; consult surgeons early.

When the diagnosis is unclear

- Symptoms without an organic cause are common and real for the patient.
- Investigate only if the result will change management: tests bring anxiety, harm, radiation (CT chest = 330 chest X-rays), incidental findings and cost.
- Explaining: accept the symptoms are real, give a positive explanation, don't list diagnoses you then exclude, agree a **safety-net**.

Source: [039,040] Cases S, T, U; [002] Diagnosis unclear

PART 2: TOPIC SUMMARY

Arrhythmias and ECG interpretation

Read every ECG in the same order

- Check settings (25 mm/s, 10 mm = 1 mV), then rate, rhythm (find the pacemaker), axis, intervals (PR 120-200 ms, QRS, QTc), chambers, ST-T, Q waves.
- Rate: regular = 300 / large squares between R waves; irregular = QRS in 30 large squares x 10.

Narrow versus wide

Finding	Think of	Unstable	Stable
Irregular, no P	AF	Heparin then synchronised DC	Rate control + anticoagulation
Regular, no P, ~150-200	SVT	Synchronised DC	Adenosine, beta-blocker or verapamil
Wide, regular, AV dissociation	VT (assume VT if structural disease)	Synchronised DC	IV amiodarone
Chaotic	VF	Shock	

AF essentials

- Stroke risk: CHA2DS2-VA 2 or more = anticoagulate; DOACs preferred; warfarin for metallic valve or mitral stenosis.
- LVEF under 40%: no verapamil/diltiazem. Before cardioversion: TOE or 3 weeks of anticoagulation, and 4 weeks after.

Heart block ladder

- 1st degree: long PR, all conducted. Mobitz I: PR lengthens until a drop (AV node, watch). Mobitz II: fixed PR, sudden drop (below the node, pace). Complete: P and QRS independent; broad slow escape = pacemaker.
- Bundle branch: LBBB broad notched R in I, V5-V6; RBBB RSR' in V1. WPW: short PR + delta wave.

Ischaemia and other patterns

- STEMI territories: anterior (LAD) V1-V4; inferior (RCA) II, III, aVF; posterior = tall R and ST depression V1-V3.
- Pericarditis: widespread saddle ST elevation + PR depression. Large effusion: low voltage + electrical alternans.
- Electrolytes: peaked T and broad QRS = high K; U waves = low K; long QT = low K, Ca, Mg.
- LVH: S V1 + R V5 over 35 mm. PE: tachycardia, RBBB, right axis, S1Q3T3.

PART 2: TOPIC SUMMARY

Valvular heart disease and infective endocarditis

Murmur map

Lesion	Murmur	Key signs
Aortic stenosis	Ejection systolic, aortic area to carotids	Slow-rising pulse, soft A2; late peak = severe
Aortic regurgitation	Early diastolic, left sternal edge, sitting forward	Collapsing pulse, displaced apex, Austin Flint
Mitral stenosis	Mid-diastolic rumble, apex, bell, left lateral	Tapping apex, loud S1, opening snap, malar flush, AF
Mitral regurgitation	Pansystolic, apex to axilla	Soft S1, S3; MVP = click + late systolic
Tricuspid regurgitation	Pansystolic, lower left sternal edge, louder on inspiration	Large v waves, pulsatile liver

When to intervene

- AS: severe with symptoms (TAVI if unfit for surgery). AR: symptoms, EF 50% or less, or dilated LV. MS: area 1.5 cm^2 or less with symptoms; balloon only after TOE excludes LA thrombus. MR: symptoms, or ESD over 40 mm or EF under 60%.
- MS with AF: anticoagulate as well as control rate.

Endocarditis: link to valve disease

- Damaged valve + bacteraemia = vegetation. Native valve: viridans strep commonest. Early prosthetic: S. aureus, S. epidermidis. Drug users: tricuspid valve.
- Fever + new murmur + embolic or immune signs (Osler, Janeway, splinters, Roth spots, splenomegaly).
- At least 2-3 blood culture sets before antibiotics, then echo (TTE, then TOE). Definite: 2 major, 1 major + 3 minor, or 5 minor.
- Empirical: native = amoxicillin + gentamicin; prosthetic = vancomycin + gentamicin; at least 4-6 weeks IV.
- Surgery: heart failure, uncontrolled infection (abscess, new heart block), large vegetation or recurrent emboli.

PART 2: TOPIC SUMMARY

Chronic heart failure, cardiomyopathies and pericardial disease

Heart failure core

- Syndrome of breathlessness, ankle swelling and fatigue from structural or functional heart disease. SNS and RAAS harmful; natriuretic peptides protective.
- Classify by EF: reduced 40% or less, mildly reduced 41-49%, preserved 50% or more. NYHA I-IV for symptoms.
- Tests: normal BNP/NT-proBNP in an untreated patient virtually excludes HF; echo is the key test; check iron.
- Prognostic pillars: ACEI/ARB/ARNI + beta-blocker + MRA + SGLT2 inhibitor. Diuretics relieve symptoms only.

Cardiomyopathies at a glance

Type	Key feature	Watch for
HCM	Asymmetric septal hypertrophy, autosomal dominant	Sudden death in the young; murmur louder on standing/ Valsalva; avoid vasodilators
DCM	Dilated, thin-walled, poor systole	Alcohol, anthracyclines, peripartum, myocarditis; treat as HF
ARVC	Fibro-fatty RV	VT, T inversion V1-V3, epsilon wave
Restrictive	Stiff, normal size, big atria	Amyloid: low voltage with thick walls
Tako-tsubo	Apical ballooning after stress	Normal coronaries, recovers in weeks

Pericardium

- Acute pericarditis: sharp pain relieved by leaning forward, rub, saddle ST elevation + PR depression; NSAID + colchicine.
- Tamponade: Beck's triad (high JVP, low BP, quiet heart sounds), pulsus paradoxus over 10 mmHg; echo; urgent pericardiocentesis.
- Constrictive: pericardial knock; TB commonest per the lecture; CT/MRI; differential = restrictive cardiomyopathy.

PART 2: TOPIC SUMMARY

Ischaemic heart disease, dyslipidaemia and hypertension

One disease, three risk factors

- Atherosclerosis drives stable angina (fixed 80-90% stenosis) and ACS (unstable plaque that ruptures, often only 30-50%).
- Modifiable risk: smoking, diabetes, hypertension, dyslipidaemia, inactivity, obesity.

Coronary syndromes

	Troponin	ECG	Plan
Unstable angina	Negative	ST depression, T change or normal	Antiplatelets, anticoagulant, risk score
NSTEMI	Positive	No ST elevation	Same, angiography (urgent if ongoing pain)
STEMI	Positive	ST elevation or new LBBB	Primary PCI (thrombolysis if PCI not possible, within 12 h)

- First: ECG within minutes, aspirin 300 mg, pain relief, oxygen only if SpO2 under 94%.
- Stable angina pathway: ECG, then CT coronary angiography, then stress imaging, then invasive angiography. Treat with aspirin, statin, beta-blocker, nitrate; PCI/CABG for severe ischaemia.
- Arteries: LAD = anterior; RCA = inferior and AV node (90%).

Hypertension (ESC 2024)

- Hypertension 140/90 or more; confirm with home or ambulatory BP. Suspect secondary causes if under 40, resistant or severe.
- Start a single pill of two drugs (ACEi/ARB + CCB or thiazide), then triple, then add spironolactone. Target systolic 120-129.

Lipids

- LDL goals: very high risk under 55 (and 50% fall), high under 70. Steps: maximal statin, add ezetimibe, add PCSK9 antibody.
- Stop statin if ALT over 3x or CK over 10x normal. Familial hypercholesterolaemia: very high LDL, tendon xanthomas, screen family. TG over 500 = pancreatitis risk.

PART 2: TOPIC SUMMARY

Cardiac emergencies (acute heart failure, aortic dissection)

First question for any chest pain: which of the four killers?

- Acute MI, aortic dissection, pulmonary embolism, tension pneumothorax.
- Check pulses and BP in both arms before giving antiplatelets, anticoagulants or thrombolysis: all are contraindicated in dissection.

Acute heart failure

- Breathless, orthopnoea, pink frothy sputum, bilateral crackles, raised JVP.
- Tests: ECG, echo, chest X-ray, ABG, troponin, pro-BNP, renal function.
- Treat: sit up; oxygen if SpO2 under 90%; non-invasive ventilation; IV furosemide (review response at 2-4 h); IV nitrates; inotropes only for hypoperfusion or shock; morphine only for pain.

Aortic dissection

- Tearing pain maximal at onset, to the back; pulse deficit or unequal arm BP; new early diastolic murmur (AR).
- Chest pain together with stroke or a cold limb = think dissection. Can cause inferior MI (RCA) and tamponade.
- CT aortic angiogram is definitive. Lower systolic BP to 100-120 with IV beta-blocker.
- Stanford A (ascending) = emergency surgery. Stanford B = medical, unless organ ischaemia, extension or expansion.

	Acute heart failure	Aortic dissection	STEMI
Key clue	Crackles, high JVP	Pulse deficit	ST elevation
Key test	Echo, BNP	CT aortogram	ECG
First drug	IV furosemide	IV beta-blocker	Aspirin

PART 2: TOPIC SUMMARY

Cardiac symptoms, presentations and clinical reasoning

History

- Let the patient tell the story; analyse each symptom (onset, course, duration, triggers, relief, associated features).
- Ischaemic pain: central, crushing, to neck/jaw/arm, on effort, eased by rest or nitrates. Pericardial: sharp, relieved by leaning forward. Musculoskeletal: localised, worse on movement and pressure.
- Heart failure clues: orthopnoea (count the pillows), PND, evening leg swelling. Premature CAD in family: male under 55, female under 65.

Examination

- JVP at 45 degrees, right internal jugular, height above the sternal angle (normal up to 3 cm). Absent a wave = AF; cannon a = complete heart block; large v = TR; non-pulsatile raised = SVC obstruction.
- Apex: displaced = LV dilatation; heaving = pressure load; hyperdynamic = volume load; tapping = MS.
- Heart sounds: loud S1 in MS; wide fixed split S2 = ASD; reversed split = LBBB or AS; S3 = volume overload; S4 = stiff ventricle.

Case patterns to recognise

- Breathless + raised JVP + S3 + crackles = heart failure; TTE first; ACEi/ARB + beta-blocker + MRA.
- Crushing pain + troponin rise without ST elevation = NSTEMI; aspirin + clopidogrel + anticoagulant; urgent angiography if pain continues.
- Fever + new murmur after dental work = endocarditis; 3 blood culture sets first; new AV block = aortic root abscess.

When the diagnosis is unclear

- Symptoms are real even without an organic cause. Test only if the result changes management.
- Give a positive explanation, avoid listing excluded diseases, and agree a safety net.

ICC-1 Chest: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. Community-acquired pneumonia [018] (lecture and LGW)
2. Bacterial infections of the lower respiratory tract [020] (Moodle topic)
3. Diagnosis of TB [058]
4. Treatment of TB [094] (lecture and WHO Moodle topic)
5. Extrapulmonary TB [114]
6. Non-tuberculous mycobacteria [115]
7. Lung abscess and bronchiectasis [118]
8. Bronchogenic carcinoma [067a, 067b] (lecture and LGW)
9. Asthma [005] (lecture, LGW and CBD)
10. COPD [009] (lecture and case)
11. Wheezy chest [047] LGW
12. Interstitial lung disease [095] (lecture and LGW)
13. Occupational lung disease [109] LGW
14. Respiratory emergencies [105]
15. Respiratory cardinal symptoms [108]
16. Chest history taking and examination [100]
17. Pulmonary function and spirometry [062]
18. Pleural diseases [117] LGW
19. Chest case reviews [098] and cases W, X, Y, Z [021, 022]

Part 2: topic summaries

1. Infections (pneumonia, TB, abscess, bronchiectasis)
2. Lung malignancies
3. COPD, asthma, wheezy chest
4. Interstitial lung disease
5. Respiratory emergencies
6. Respiratory symptoms, presentations and clinical reasoning

PART 1: LECTURE

Community-acquired pneumonia [018] (lecture and LGW)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Definitions and pathogenesis

- Pneumonia = acute infection with clinical and/or radiological consolidation of part of one or both lungs. No consolidation on CXR = acute bronchitis (winter, viral).
- CAP = acquired in community; HAP = 48 h or more after admission; VAP = 48 h or more on ventilator. Different organisms and antibiotics.
- Routes: inhalation (viruses, Legionella, Mycoplasma); aspiration (stroke, alcohol, hypnotics; oral flora, anaerobes); colonisation (bronchiectasis, CF; Pseudomonas); blood spread (IVDU, cannula, dialysis; S. aureus, Gram-negatives).
- Commonest organisms: S. pneumoniae, atypicals, H. influenzae. Resistant-organism risk: hospital + IV antibiotics in last 90 days, previous MRSA/Pseudomonas, comorbidities.

Diagnosis and severity

- Elderly: vague symptoms (confusion, GI upset, no fever). Signs: dullness, bronchial breathing, crackles. Consolidation = bacterial; ground-glass = atypical.
- Routine: CXR (or LUS/CT), CBC, ABG, BUN (diagnosis + severity).
- IDSA/ATS severe CAP = 1 major (septic shock on vasopressors; ventilation) or 3 minor (RR ≥30, fluid-responsive hypotension, temp <36°C, confusion, WBC <4,000, platelets <100,000, BUN ≥20 mg/dL, PaO2/FiO2 ≤250, multilobar). Room air FiO2 = 0.21.
- Sputum Gram stain/culture + blood cultures only if in hospital AND severe or resistance risk. Urinary antigen: Legionella travel/outbreak or severe CAP.
- Procalcitonin: start antibiotics in confirmed CAP regardless of level.

Treatment

- CURB-65 0-1 home; 2 ward; 3-5 or severe IDSA/ATS = ICU.
- Outpatient, no comorbidity: amoxicillin, doxycycline or macrolide. With comorbidity: amox-clav/ceftriaxone + macrolide or doxycycline, or fluoroquinolone alone.
- Inpatient: beta-lactam + macrolide (or + fluoroquinolone). Add anti-MRSA (vancomycin, linezolid) or anti-Pseudomonas (pip-tazo, meropenem) only with risk factors; adjust by culture.
- Clindamycin only if abscess/empyema suspected; oseltamivir if influenza PCR positive.
- Steroid only: severe CAP + shock, P:F <300, or rapid progression (≤3 days).
- Duration: 3-4 days non-severe and stable; 5+ days severe. Stability at 72 h: T ≤37.8, HR ≤100, RR ≤24, SBP ≥90, SpO2 ≥90%, normal mentation and intake. No routine follow-up CXR if better in 5-7 days.

Special types and prevention

- Legionella: water aerosols, haemoptysis (1/3), confusion, diarrhoea, bradycardia. Klebsiella: diabetics/alcoholics, RUL, bulging fissure.
- Influenza vaccine yearly from 6 months; LAIV not in >65, pregnancy, immunosuppression (give IIV/RIV). Pneumococcal vaccine ≥65 or high-risk 2-64.

Source: Pneumonia lecture [IM-PULMO.13.1]; CAP LGW [IM-PULMO.14.2]

PART 1: LECTURE

Bacterial infections of the lower respiratory tract [020] (Moodle topic)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Upper airway and streptococci

- Group A Streptococcus (S. pyogenes) throat infection; toxin-associated complication = scarlet fever.
- Child with fever, drooling, stridor + swollen epiglottis on lateral X-ray = acute bacterial epiglottitis; commonest cause Haemophilus influenzae.
- S. pneumoniae: alpha haemolysis on blood agar; causes CAP, meningitis, acute otitis media, septicaemia.

Tuberculosis basics

- BCG vaccine scar = protection against M. tuberculosis.
- Mycobacteria seen with Ziehl-Neelsen stain or auramine fluorescent stain.
- Rifampicin darkens (orange-red) urine.
- Isoniazid: peripheral neuropathy, hepatitis. Ethambutol: visual disturbance. Pyrazinamide: hepatotoxicity.

Hospital-acquired pneumonia

- Risks: time on ventilator, rise in gastric pH, nasogastric tube, time in hospital, extremes of age, chronic respiratory disease.
- Organisms: Staphylococcus aureus, E. coli, Pseudomonas aeruginosa, Acinetobacter.

Community-acquired pneumonia tests

- Lobar CAP: sputum culture, blood culture (aerobic + anaerobic bottles), pneumococcal urinary antigen.
- Blood culture helps for S. pneumoniae, H. influenzae, S. aureus.

Atypical pneumonias

- Mycoplasma: young adult, cough, headache, myalgia, tiredness; aerosol spread; about 15% of adult CAP.
- No cell wall: no Gram staining; resistant to beta-lactams and glycopeptides.
- X-ray variable; culture positive in only 30-60%; gold standard = fourfold rise in paired titres (e.g. 1:40 to 1:1024).
- Treat with protein/DNA synthesis drugs (tetracyclines, macrolides, quinolones); macrolides most active.
- Legionnaires' disease: travel (e.g. Spain), fever, myalgia, confusion; bilateral interstitial shadowing; sputum Gram-negative rods with many white cells. Treat macrolide ± rifampicin.

Source: Bacterial Infections of the LRT tutorial [IM-PULMO.16.7]

PART 1: LECTURE

Diagnosis of TB [058]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Principles and radiology

- No TB without the bacillus; no healthy carriers; any material with tubercle bacilli is pathologic.
- Infection evidence: hilar nodes, calcified primary focus, positive TST. Disease needs direct or indirect evidence of active disease.
- CXR: TB mimics any chest disease; commonest = apical infiltrates/cavitation, mediastinal nodes, pleural effusion, miliary shadows.

Samples and smear

- Samples: sputum, laryngeal swab, gastric lavage, body fluids (pleural, ascitic, CSF), bronchoscopy specimens, tissue biopsy.
- Sputum: 3 consecutive early-morning samples.
- Ziehl-Neelsen: simple, rapid, cheap; misses low bacillary loads. Fluorescent: low-power large field, quicker, more positives; but costly, needs maintenance and steady power.
- False positive: food particles, precipitated stain, scratches, fibres, contamination. False negative: poor collection/storage/staining, loss of acid-fastness (sunlight, UV, heat, >1 week hot storage).

Culture and molecular tests

- Media: egg-based (Lowenstein-Jensen), agar-based (Middlebrook 7H10/7H11), liquid (7H9).
- Culture: more sensitive, species identification, drug resistance; slow (4-8 weeks), so MGIT/Bactec used (Bactec: positive if growth index ≥10).
- PCR: rifampicin resistance = rpoB; high-level isoniazid = katG; low-level isoniazid = inhA promoter.

Immune-based tests

- ADA: pleural >70 IU/L highly suggestive, <40 excludes; ratio ≥2 or lymphocyte/neutrophil >0.75 supports TB. CSF cut-off 8 U/L; ascitic >36 U/L.
- IGRA (QuantiFERON, ELISPOT): ESAT-6 and CFP-10, not shared with BCG; one visit, result in 24 h, no reader bias, no boosting; blood processed within 12 h.
- TST: 5 TU PPD (0.1 mL) intradermal, read at 48-72 h, induration in mm (0 mm if none); PPD refrigerated, dark, away from light. Shared with BCG, boosting, two visits.
- Positive TST: ≥5 mm HIV, recent contact, old healed TB on CXR, transplant/immunosuppressed; ≥10 mm recent arrivals, IV drug users, high-risk settings, lab staff, diabetes, silicosis, CRF, steroids, malignancy; ≥15 mm no risk factor.
- Latent TB: immune response without active disease; diagnose by TST or IGRA (neither clearly preferred).

Source: Diagnosis of TB lecture [IM-PULMO.17.1]

PART 1: LECTURE

Treatment of TB [094] (lecture and WHO Moodle topic)

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Aims and regimen

- Aims: cure, prevent death, relapse, transmission and drug resistance.
- First-line: isoniazid (H), rifampicin (R), pyrazinamide (Z), ethambutol (E), streptomycin (S).
- WHO new pulmonary TB: 2HRZE/4HR (6 months of rifampicin); 2HRZE/6HE phased out; daily dosing optimal. Retreatment: 2HRZES/1HRZE/5HRE.
- Intensive phase: kill dividing bacilli, non-infectious, prevent resistance. Continuation phase: kill dormant bacilli, prevent relapse.
- MDR-TB = resistant to rifampicin and isoniazid.

Monitoring and side effects

- Clinical review; sputum at 2 months and at completion; AST, ALT, bilirubin, CBC twice weekly for 2 weeks, at 1 month, then monthly.
- Major (stop and refer): skin rash (stop all); jaundice/hepatitis from H, R, Z (stop all, reintroduce one at a time); streptomycin deafness, vertigo, nystagmus, low urine output (stop S); ethambutol visual loss, red-green first (stop E).
- Minor (continue): GI upset (small meals/bedtime); pyrazinamide gout-like joint pain (NSAIDs, aspirin, paracetamol); isoniazid neuropathy (pyridoxine 50-75 mg daily) and drowsiness; rifampicin orange urine (warn); intermittent rifampicin flu-like illness (go daily).

TB with HIV

- HIV raises latent-to-active risk 12-20 times; atypical presentation, lower-zone infiltrates, false-negative smears.
- Problems: poor tolerance and adherence, MDR common; 90% of reactions in first 2 months.
- ART: start 2-8 weeks after TB treatment if CD4 <200 (WHO file: as soon as possible within 8 weeks); no need to wait for TB treatment to end. Barriers: toxicity, interactions, pill burden, IRIS.

Special groups

- Pregnancy: oral first-line drugs safe; streptomycin avoided (fetal ototoxicity). Congenital TB presents in weeks 2-3 with hepatosplenomegaly and respiratory distress.
- Renal: rifampicin safest (bile excretion); isoniazid, pyrazinamide standard dose; aminoglycosides reduced if CrCl <50; avoid cycloserine, PAS; ethambutol needs levels or 48 h interval.
- Liver: H, R, Z hepatotoxic (H/R and R/Z synergistic); monitor LFTs closely.

Source: Treatment of TB [IM-PULMO.18.1]; WHO TB treatment guidelines 4th ed. [IM-PULMO.21.7]

PART 1: LECTURE

Extrapulmonary TB [114]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Spread and key forms

- Mostly haematogenous spread; direct extension from an adjacent organ is less common. General symptoms: fever, malaise, weight loss.
- Miliary TB: lesion erodes into a vessel. Commonest in children < 4, elderly, immunocompromised. Lungs and bone marrow most affected (anaemia, low platelets, leukemoid reaction); may present as FUO. CXR: thousands of 2-3 mm nodules evenly spread in both lungs.
- TB meningitis: most serious form; children 0-5, elderly, immunocompromised; the one form BCG is believed to prevent. Low-grade fever, unremitting headache, then drowsiness with focal signs, then coma. Focal signs = tuberculoma; stroke from vessel thrombosis.

Other sites

- Genitourinary: kidney (fever, back pain, pyuria) to bladder; men: prostate, seminal vesicles, epididymis (scrotal mass); women: salpingo-oophoritis (pelvic pain, sterility, ectopic). Psoas spread can give an anterior thigh abscess.
- Peritonitis: common in alcoholics with cirrhosis; mild or mimics acute abdomen.
- Pericarditis: from mediastinal nodes or pleura; can cause tamponade and heart failure.
- Lymphadenitis: posterior cervical and supraclavicular nodes; may form a draining fistula.
- Pott's disease: starts in a vertebral body, spreads to the next with disc-space narrowing; collapse, cord compression (paraplegia), paravertebral abscess.
- Skin: lupus vulgaris (face nodules); prosector's wart (direct inoculation). GI TB from M. bovis milk; may mimic appendicitis.

Diagnosis

- Smear, culture, biopsy/aspirate, PCR on CSF, urine, pleural, pericardial or joint fluid. Few organisms, so cultures often negative: use NAAT.
- TST may be negative at first; repeat in a few weeks. Blood culture positive in ~50% of disseminated TB (often HIV).
- TB meningitis CSF: lymphocytes, glucose < 50% of serum, high protein, ADA > 8 U/L.
- MRI preferred for spine; CT/US for abdominal and GU disease. Bone marrow or liver biopsy if miliary TB suspected and all else negative.
- Granuloma on biopsy, or positive TST/IGRA + unexplained lymphocytic fluid: treat without proof (treatment trial).

Treatment

- Same regimens as lung TB, 6-9 months; meningitis 9-12 months; bone/joint 9 months.
- In meningitis, slides replace ethambutol with streptomycin.
- Steroids in serositis (pericarditis, peritonitis, pleuritis, meningitis) to prevent fibrosis, unless resistance suspected.
- Surgery: drain empyema, tamponade or CNS abscess; close bronchopleural fistula; resect bowel; Pott's with deficit or deformity. Not needed for lymphadenitis except diagnosis.

Source: Extrapulmonary TB [114] (IM-PULMO.19.2)

PART 1: LECTURE

Non-tuberculous mycobacteria [115]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Basics

- Mycobacteria other than M. tuberculosis and M. leprae (atypical, environmental). Spread by inhalation, ingestion or skin penetration; tap water is a major reservoir.
- Runyon groups (slow growers 1-3): 1 photochromogens (M. kansasii, M. marinum); 2 scotochromogens (M. scrofulaceum); 3 non-photochromogens (M. avium complex, MAC). Group 4 fast growers, culture within 7 days (M. fortuitum, M. chelonae, M. abscessus).
- Risk: lung disease (COPD, bronchiectasis, CF, ABPA), trauma/surgery/cosmetic procedures not responding to standard antibiotics, breast prostheses, HIV, transplant, TNF-alpha inhibitors and other immunosuppressants.

Clinical syndromes

- 85% pulmonary; rest lymph nodes, skin/soft tissue, disseminated. Indolent course; fever the commonest symptom.
- Lung disease: MAC, then M. kansasii and M. abscessus. Fibrocavitary form = upper-lobe cavities like TB; nodular bronchiectatic form = bronchiectasis + nodule clusters, often right middle lobe and lingula.
- Lymphadenitis: healthy children 1-5 years, one-sided cervical node, no systemic symptoms; MAC 80%, then M. scrofulaceum. Differentiate from TB lymphadenitis.
- Disseminated: immunocompromised; MAC 90%, the main NTM in HIV.
- Skin: nodules in a line along lymphatics; fish tank granuloma = M. marinum; Buruli ulcer = M. ulcerans.

Diagnosis

- Lung: symptoms + cavity or nodules on CXR/HRCT + other diagnoses excluded (all), plus one of: two positive sputum cultures, one positive bronchial wash/lavage, or positive lung biopsy with typical histology.
- AFB stain (fluorochrome or Ziehl-Neelsen) cannot tell MTB from NTM. Culture of tissue is optimal; PCR and biopsy help.

Treatment

- MAC lung: macrolide (azithromycin/clarithromycin) + ethambutol + rifampin, for 12 months after culture conversion.
- M. kansasii: rifampin + ethambutol + isoniazid + pyridoxine, 12 months after culture conversion.
- Surgery: drainage/debridement + > 3 months of combined drugs in some cases; resect solitary nodules; cervical lymphadenectomy cures child NTM lymphadenitis.

Source: NTM [115] (IM-PULMO.20.2)

PART 1: LECTURE

Lung abscess and bronchiectasis [118]

Topic: Infections (pneumonia, TB, abscess, bronchiectasis)

Lung abscess

- Localised pus or necrotic debris in lung tissue forming a cavity with an air-fluid level.
- Routes: aspiration (commonest: dental infection, stroke, alcohol, frequent vomiting; anaerobes); inhalation/necrotizing pneumonia (S. aureus/MRSA, Strep. milleri, Klebsiella, Pseudomonas); colonisation of diseased lung (bronchiectasis, CF, bullae, infarcts); blood spread (IV drugs, cannulas, haemodialysis, abdominal or urinary sepsis).
- 90% polymicrobial, anaerobes usual (e.g. Bacteroides fragilis); TB, fungi, parasites less common.
- Acute < 6 weeks (poorly defined wall); chronic > 6 weeks (well-defined fibrous wall).
- Starts like pneumonia, then large sputum volumes; chronic: weight loss, clubbing.
- Differential of a cavity: cavitating cancer, localised empyema, cavitary pneumoconiosis, hydatid cyst.
- Tests: sputum and blood culture, CBC (anaemia, leukocytosis); bronchoscopy for foreign body or tumour, BAL, biopsy.
- Treatment: antibiotics (clindamycin, meropenem) 4-6 weeks; percutaneous US/CT-guided drainage if antibiotics fail; lobectomy if > 6 cm, symptoms > 12 weeks, failed medical treatment or suspected cancer.

Bronchiectasis: causes and features

- Irreversible abnormal dilatation of bronchi with chronic inflammation, chronic sputum, recurrent infection and airflow limitation.
- Causes: infection (pneumonia from measles or adenovirus, TB); immune (HIV, ABPA, ?lupus/RA); bronchomalacia; immotile cilia; cystic fibrosis; obstruction (cancer, nodes).
- Patterns: cylindrical, varicose, cystic. Mechanisms: atelectasis, pressure of secretions, traction fibrosis.
- Colonisers: H. influenzae (commonest), S. pneumoniae, Klebsiella, Pseudomonas (rapidly progressive, severe).
- Persistent cough, voluminous purulent sputum, haemoptysis (can be massive), clubbing, coarse crackles that change with cough.

Bronchiectasis: tests and treatment

- CXR not sensitive (ring, tramline shadows). HRCT is the gold standard (signet ring). Spirometry obstructive; sputum culture; test for cause (e.g. TB).
- Antibiotics, mucolytics, physiotherapy; ICS and bronchodilators only with asthma/COPD; lobectomy for localised disease.
- First Pseudomonas: ciprofloxacin 750 mg every 12 h for 2 weeks, then inhaled colistin for at least 3 months.
- Non-Pseudomonas with 3 or more exacerbations a year: culture-based antibiotic + long-term macrolide (azithromycin) for at least 3 months.

Source: Lung abscess and bronchiectasis [118] (IM-PULMO.15.2)

PART 1: LECTURE

Bronchogenic carcinoma [067a, 067b] (lecture and LGW)

Topic: Lung malignancies

Basics

- 95% of primary lung tumours; peak age 50-70; old smoker with chest symptoms = cancer until proven otherwise.
- Risks: smoking (pack-years; 80% of cancers in smokers), pollution, genetics, asbestos, old lung scarring.
- Central tumours: commoner, squamous cell, early mediastinal spread, cough, dyspnoea, haemoptysis, seen at bronchoscopy. Peripheral: adenocarcinoma, early pleural spread, pleurisy and chest pain.

Local spread

- Bronchial: changed cough, haemoptysis, localized wheeze, collapse, unresolved pneumonia, abscess.
- Pancoast tumour: rib erosion (shoulder pain to axilla), subclavian artery (unequal pulses), brachial plexus (hand wasting), sympathetic chain (Horner's: ptosis, miosis, anhidrosis, enophthalmos).
- Mediastinal: trachea (stridor, brassy cough), oesophagus (dysphagia), left RLN (hoarseness), phrenic (hiccups, then diaphragm palsy), SVC (facial plethora, neck veins, arm swelling, collaterals; classically SCLC).

Paraneoplastic (mostly SCLC)

- SCLC: SIADH, ectopic ACTH. Squamous: hypercalcaemia (PTH-like), hypoglycaemia. Large cell/adeno: gynaecomastia.
- Migrating thrombophlebitis, anaemia, eosinophilia; Eaton-Lambert, neuropathy, cerebellar ataxia, dermatomyositis; clubbing and hypertrophic osteoarthropathy (painful periostitis, subperiosteal new bone); acanthosis nigricans; nephrotic syndrome.

Investigations

- Bloods, calcium, glucose, sodium, ACTH; CXR, CT chest; metastasis search: abdominal US/CT, CT brain, bone scan, PET-CT.
- Tissue is the issue: bronchoscopy for central; CT/US-guided biopsy for peripheral; node biopsy; thoracoscopy if effusion.

Staging (NSCLC TNM 8th)

- T1 up to 3 cm; T2 3-5 cm (T2a 3-4, T2b 4-5) or main bronchus, visceral pleura, atelectasis; T3 5-7 cm or chest wall, phrenic, parietal pericardium; T4 over 7 cm or carina, mediastinum, heart, great vessels, oesophagus, vertebra.
- N1 ipsilateral hilar/peribronchial; N2 ipsilateral mediastinal/subcarinal; N3 contralateral, or any scalene/supraclavicular.
- M1a contralateral lung, pleural nodules, malignant effusion; M1b single extrathoracic; M1c multiple.
- Metastasis sites: adrenals, brain, liver, bone.

Treatment

- NSCLC: surgery if resectable with no metastases, plus radiotherapy/chemotherapy; overall 5-year survival ~20%.
- SCLC: limited vs extensive (not TNM); chemotherapy (very sensitive at first); surgery usually contraindicated; 10-20 months treated vs 2-4 untreated.

Source: Bronchogenic carcinoma lecture [067a] and LGW [067b]

PART 1: LECTURE

Asthma [005] (lecture, LGW and CBD)

Topic: COPD, asthma, wheezy chest

Definition and diagnosis

- Heterogeneous disease with chronic airway inflammation; defined by **variable** symptoms (dyspnea, cough, wheeze, chest tightness) plus **variable** expiratory airflow. COPD = persistent symptoms and obstruction.
- Airway hyper-responsiveness is the key feature. Typical: more than one symptom, worse at night, with viral infections, exercise, laughing, cold air, allergens.
- Asthma less likely with: isolated cough, chronic sputum, chest pain, dyspnea with dizziness/paresthesia, stridor.
- Atopy history and nasal polyps support but are not essential.

Tests

- Spirometry (may be normal outside attacks): FEV1/FVC < 80%; reversibility = FEV1 rise of **12% and 200 mL** 10-15 min after salbutamol 200-400 mcg.
- PEF diurnal variability > 10% = asthma: (highest - lowest) / mean x 100; twice daily, best of 3.
- Bronchial challenge (histamine/methacholine): FEV1 fall of **20% or more**. Others: raised FeNO, blood/sputum eosinophils.

Drugs

- **ICS = mainstay**; every controller must contain ICS; never LABA alone. Side effects: oral candidiasis, hoarseness, pneumonia at high dose.
- **ICS-formoterol** = preferred reliever and pre-exercise prophylaxis; Steps 1-2 = as-needed low-dose ICS-formoterol.
- OCS (prednisolone 40 mg 5-7 days) for exacerbations only. LTRA (montelukast) less effective than ICS; good for rhinitis; caution in neuropsychiatric illness.
- Beta-2 agonist side effects: tachycardia, tremor, anxiety, hypokalemia. Theophylline/aminophylline should **not** be used. IV MgSO4 only in severe exacerbations.

Control and severe asthma

- Cycle: confirm diagnosis, assess control, check triggers/comorbidities (GERD, rhinitis, obesity, smoking) and inhaler technique, adjust, review in 2-3 months; step down after 3 months of good control.
- Severe asthma with Th2 inflammation (sputum eos 2% or more, FeNO 20 ppb or more, blood eos 150/microL or more): biologics (omalizumab, benralizumab, dupilumab). Otherwise LAMA, azithromycin, bronchial thermoplasty.

Phenotypes and exacerbations

- Exercise-induced: symptoms 15-30 min after exercise. Aspirin-induced: 1-2 h after aspirin/NSAID, with rhinitis and sinusitis; avoid NSAIDs. Occupational: low PEF on workdays; avoid exposure.
- Severe attack: confusion, low BP, RR > 30, SpO2 < 90%, silent chest = ICU. Treat early with OCS, SABA + SAMA, oxygen to SpO2 > 93%.
- Discharge when SpO2 > 94% on air and PEF > 60-80%; follow up in 1-2 weeks.

Source: Asthma Lecture [005], Asthma LGW [005], Asthma CBD [005]

PART 1: LECTURE

COPD [009] (lecture and case)

Topic: COPD, asthma, wheezy chest

Definition and pathophysiology

- Heterogeneous condition: chronic respiratory symptoms from airway and/or alveolar abnormalities with **persistent, often progressive** airflow obstruction.
- Risk factors: smoking (about 50% of smokers), biomass fuel, alpha-1 antitrypsin deficiency, impaired lung development, childhood infection, long-standing asthma.
- Air trapping and V/Q mismatch cause low O2 and high CO2; chronic hypoxia leads to pulmonary hypertension and cor pulmonale.

Diagnosis = clinical + spirometry

- Starts in the 40s; persistent, progressive dyspnea is characteristic.
- **Post-bronchodilator FEV1/FVC < 70% is mandatory.** Signs have low sensitivity.
- Emphysema: thin, pursed lips, barrel chest, distant breath sounds, "pink". Chronic bronchitis: cough/sputum, obese, edema, cyanosis.
- CXR: hyperinflation, flat diaphragm, ribbon-shaped heart. CBC: secondary polycythemia.
- Screen all patients once for alpha-1 antitrypsin. Blood eos > 300 = ICS benefit.

Assessment and stable treatment

- Assess FEV1, mMRC dyspnea, exacerbations in the last year (2 or more, or 1 admission = high risk).
- All: smoking cessation, vaccines, rehabilitation.
- High exacerbation risk: LABA + LAMA; LAMA reduces exacerbations more than LABA.
- Exacerbations on LABA + LAMA with eos 300/microL or more: add ICS (triple). ICS + LABA alone not encouraged; avoid ICS if eos < 100, recurrent pneumonia or TB history.
- Still exacerbating on triple: azithromycin (250 mg/day for 1 year; arrhythmia, resistance) or roflumilast (needs **FEV1 < 50%**). Dupilumab blocks IL-4/IL-13.
- Theophylline not recommended unless inhalers unavailable: arrhythmia, convulsions, GI upset.

Oxygen

- LTOT 18-24 h/day if PaO2 < 55 mmHg (SpO2 < 88%), or < 60 mmHg with right heart failure or polycythemia; reassess every 2-3 months.
- High PCO2 (type II failure): target **SpO2 88-92%**; excess O2 can cause respiratory depression.

Acute exacerbation

- Worsening over < 14 days, usually infection. Exclude PE, pneumothorax, ACS.
- Antibiotics cover H. influenzae, S. pneumoniae, M. catarrhalis (ampicillin, macrolide, doxycycline); SABA + SAMA nebulizer; prednisolone 40 mg for 5 days; controlled O2; NIV/invasive ventilation if needed.
- Discharge when stable; check inhaler technique and O2 need; follow up in 1-4 weeks.

Source: COPD Lecture [009], COPD Case History [009]

PART 1: LECTURE

Wheezy chest [047] LGW

Topic: COPD, asthma, wheezy chest

Wheeze types

- Wheeze = continuous musical sound from vibration of narrowed airway walls; high or low pitched; inspiratory or expiratory.
- **All wheezes are obstruction, but not all wheezes are asthma.** Extra-thoracic obstruction limits inspiration; intra-thoracic limits expiration.
- **Stridor**: high-pitched; supraglottis to trachea (e.g. vocal cord dysfunction); often heard without stethoscope.
- **Rhonchi**: low-pitched, snoring-like; obstruction/secretions in large airways (COPD, bronchiectasis).

Causes

- Upper airway: post-nasal drip, vocal cord dysfunction, anaphylaxis (angioedema), foreign body, tracheal tumor.
- Lower airway: asthma, COPD, bronchiectasis, aspiration, heart failure ("cardiac asthma"), bronchogenic carcinoma (**localized wheeze**).
- Work-up: history, examination, spirometry with reversibility, challenge test, CXR/CT chest/CT neck, bronchoscopy.

Acute asthma

- Spirometry: usually too dyspneic. ABG only if SpO2 < 92%, life-threatening asthma or another diagnosis (e.g. DKA).
- CXR only if pneumothorax, pneumonia, life-threatening asthma or poor response.
- Treat: nebulized beta-2 agonist + ipratropium, prednisolone 20-40 mg for 5-7 days. **No routine antibiotics.**
- Discharge: symptoms improved, PEF > 60-80%, SpO2 > 94% on air, home support. Give reliever, start/step up controller, check technique, follow up in 2-7 days.

Acute exacerbation of COPD

- Older smoker, worsening cough and wheeze, prolonged expiration, hypoxemia.
- ABG example: pH 7.38, PaCO2 55, HCO3 32 = **compensated respiratory acidosis** (type II respiratory failure).
- CXR: hyper-expanded chest. Target SpO2 **88-92%** using a **Venturi mask**.
- Antibiotics: amoxicillin/clavulanate, macrolide or tetracycline. Sputum culture if frequent exacerbations or ventilated (resistant organisms, e.g. Pseudomonas).

Stridor plus wheeze

- Inspiratory stridor with expiratory wheeze in a smoker, with a paratracheal mass displacing the trachea = central airway compression; image with CT neck/chest and bronchoscopy.

Source: Wheezy Chest LGW [047]

PART 1: LECTURE

Interstitial lung disease [095] (lecture and LGW)

Topic: Interstitial lung disease

Basics

- Over 150 disorders with inflammation and fibrosis of the interstitium (between alveolar epithelium and capillary endothelium).
- Exposure-related: pneumoconioses (asbestosis, silicosis, coal), drugs (amiodarone, methotrexate, nitrofurantoin), connective tissue disease, radiation.
- IPF after 50; IPF and PLCH in smokers; sarcoidosis and HP less common in smokers. HP in bird breeders/farmers.

Clinical and tests

- Dyspnea is the main symptom; dry cough; clubbing; **Velcro inspiratory crackles**, basal first. Mid-inspiratory squeaks = HP.
- CXR can be misleadingly normal; **HRCT is the gold standard**.
- PFTs: **restrictive**, low TLC, RV, VC; normal or high FEV1/FVC; **low DLCO**. Resting hypoxemia from V/Q inequality; 6-minute walk unmasks desaturation.
- BAL clues: eos > 50% = eosinophilic pneumonia; rising RBCs + hemosiderin macrophages = DAH; milky fluid + foamy macrophages = PAP; Langerhans cells > 3%, CD1a/S100 = PLCH; CD4/CD8 < 1 = HP; 2.5 or more = sarcoidosis.
- Surgical biopsy preferred in most ILD; avoid honeycomb areas.

Patterns to recognize

- IPF: basal honeycombing, traction bronchiectasis; **pirfenidone**.
- Cysts: LAM (young women, thin-walled; **sirolimus, avoid pregnancy**); PLCH (smokers, bizarre shapes; **smoking cessation**, cladribine).
- PAP: **crazy paving**; whole-lung lavage, GM-CSF.
- DAH (e.g. GPA): alveolar filling; **plasmapheresis**.
- Radiation pneumonitis: ground glass on the irradiated side.

Sarcoidosis

- Non-caseating granulomas; age 20-40, female, black patients; lung in > 90%.
- **Löfgren**: fever, arthralgia, erythema nodosum, bilateral hilar nodes; no investigation, NSAIDs/rest only. **Heerfordt**: anterior uveitis, fever, parotid enlargement, facial palsy.
- CXR stages: 1 hilar nodes; 2 nodes + lung; 3 lung only; 4 fibrosis (not progressive).
- Hypercalcemia (10-20%) from macrophage vitamin D; ACE raised in > 75%, used for monitoring.
- Definitive diagnosis: biopsy with non-caseating granulomas. Prednisolone 0.5 mg/kg first line; add methotrexate, azathioprine or hydroxychloroquine.

General treatment

- Smoking cessation, oxygen, rehabilitation, vaccines, remove the offending agent; lung transplant for end-stage disease.

Source: ILD Lecture [095], ILD LGW [095]

PART 1: LECTURE

Occupational lung disease [109] LGW

Topic: Interstitial lung disease

Basics and general care

- Pneumoconiosis: inhaled dust/fibres taken up by alveolar macrophages → inflammation → lung fibrosis. Diagnosis needs a history of exposure.
- Primary prevention = reduce exposure. General measures: influenza + pneumococcal vaccines, stop smoking, bronchodilators, antibiotics for infections, oxygen for hypoxaemia.

Asbestos

- Exposure: mining/milling, boiler and pipe insulation, brake linings, roofing/floor products, shipyards (plumbers, pipe fitters).
- Pleural disease is most common: pleural plaques (localized calcified thickening at diaphragm level, usually incidental), diffuse pleural thickening, effusion, rounded atelectasis, mesothelioma.
- Rounded atelectasis: folded scarred pleura traps lung; comet tail sign; DD lung cancer (pseudotumour).
- Asbestosis: fibrosis of LOWER zones, 20-30 years after exposure (< 10 years if very heavy). Dyspnoea, cough, clubbing, bilateral inspiratory crackles, cor pulmonale.
- CXR reticulonodular opacities + honeycombing; restrictive PFTs (low volumes, low FVC, normal ratio), low DLCO, raised A-a gradient.
- Biopsy/sputum: ferruginous bodies (golden-brown barbell rods, Prussian blue).
- No specific treatment. Asbestos + smoking are synergistic for bronchogenic carcinoma (~75x risk): smoking cessation is key.

Silica

- Mining, quarrying, drilling, tunnelling, sandblasting, cement, glass and pottery. Pathology: silicotic nodule (concentric hyalinized collagen).
- Chronic (> 15 years): 1-10 mm UPPER lobe nodules → progressive massive fibrosis (PMF, > 10 mm); eggshell calcification of hilar nodes is characteristic.
- Acute (within months of massive exposure): lower-zone alveolar filling, rapid hypoxaemic failure, death.
- No effective therapy. TB complicates all forms → yearly tuberculin test.

Coal

- Coal macule (dust-laden macrophages round respiratory bronchioles) with focal emphysema; coal nodules are larger and palpable.
- Simple CWP: small rounded opacities < 1 cm, upper zones; PMF > 1 cm. Increased COPD risk. No cure, symptomatic only.
- Clubbing and crackles are NOT features: if present, look for another disease.
- Caplan syndrome: RA + coal/silica dust → nodular opacities, RF positive; treat as PMF + rheumatology protocol.

Source: Occupational Lung Disease LGW [109]

PART 1: LECTURE

Respiratory emergencies [105]

Topic: Respiratory emergencies

Red flags of life-threatening distress

- Altered mental state, severe cyanosis, pallor/sweating, 1-2 word dyspnoea, HR > 130, accessory muscles/retractions, absent breath sounds, stridor.

Pulmonary embolism: diagnosis

- Strong risk factors: hip/knee replacement, lower limb fracture, major trauma, spinal cord injury, previous VTE, recent admission (HF, AF, MI); also thrombophilia, OCP.
- Classic: sudden dyspnoea, pleuritic pain, tachycardia, hypoxia, often normal chest exam. ABG: low PaO2 (V/Q mismatch), low PaCO2.
- CXR usually normal; Hampton's hump (wedge opacity), Westermark, Fleischner, Palla's sign.
- D-dimer: good negative test (normal = excludes PE); raised value non-specific (cancer, hospital, infection, inflammation, pregnancy).
- CTPA = method of choice. V/Q scan (low radiation, no contrast): pregnancy, young women, contrast anaphylaxis, severe renal failure, low-probability outpatient with normal CXR.
- Leg compression ultrasound: incompressible vein = clot. Simplified PESI 0 = 1% vs ≥ 1 = 10.9% 30-day mortality.

Pulmonary embolism: treatment

- O2 to SaO2 93-98%; RV failure: fluid ≤ 500 ml, norepinephrine, dobutamine, ECMO.
- High-risk (shock): thrombolysis (rtPA, streptokinase, urokinase); alternatives catheter-directed treatment, surgical embolectomy. After thrombolysis in arrest, continue CPR 60-90 min.
- Absolute CI: stroke, CNS tumour, trauma/surgery/head injury < 3 weeks, bleeding diathesis, active bleeding. Relative: peptic ulcer, oral anticoagulant, pregnancy.
- Start LMWH/fondaparinux/UFH while awaiting tests if probability high/intermediate. Warfarin overlap ≥ 5 days and INR 2-3 for 2 days. NOACs: dabigatran (thrombin), apixaban/edoxaban/rivaroxaban (Xa).
- IVC filter: VTE + absolute CI to anticoagulation; recurrent PE despite anticoagulation; high-risk prophylaxis.

ARDS (Berlin)

- Within 1 week of insult (e.g. peritonitis, haemorrhage, burns); bilateral opacities not explained by effusion/collapse/nodules; not cardiac/fluid (echo, PAWP < 18); PaO2/FiO2 ≤ 300.
- Treat cause, neutral fluid balance, high FiO2, inhaled NO/epoprostenol, neuromuscular blockade, early high-dose steroids.

Atelectasis

- Types: obstructive (mucus plug, tumour, FB, tube), compressive (effusion, pneumothorax), cicatricial (scarring, post-TB), adhesive (no surfactant, ARDS).
- Collapse signs: retracted side, trachea TOWARDS lesion, dull, absent breath sounds. Postoperative: poor cough from pain, aspiration.
- Luftsichel sign (air crescent) = LUL collapse. DD of a dull silent side: effusion, fibrosis, collapse.

Source: Respiratory Emergencies [105]

PART 1: LECTURE

Respiratory cardinal symptoms [108]

Topic: Respiratory symptoms, presentations and clinical reasoning

Cough (six cardinal symptoms: cough, sputum, dyspnoea, chest pain, haemoptysis, wheeze)

- Acute < 3 weeks (bronchitis, pneumonia, PE); subacute 3-8 weeks (post-infection, asthma); chronic > 8 weeks (ILD, COPD).
- Larynx: painful, hoarse, stridor. Trachea: metallic, retrosternal soreness. Pneumonia: dry then productive. TB: weeks-months, blood-tinged. ILD: dry, persistent. Pleura: dry with sharp pain.
- GERD: VERY common; after meals, worse supine, heartburn/bitter taste. LVF: supine with orthopnoea.
- ACE inhibitors: dry cough in ~10%, 3-6 months after starting, more in women (bradykinin not broken down).
- Psychogenic: young female, barking, in front of audience; exclude organic causes. Complications: syncope, rib fracture, pneumothorax, hernia, incontinence.

Sputum

- Green = infection (myeloperoxidase); rusty = pneumococcus; red currant jelly = Klebsiella; black = coal miners, heavy smokers.
- 100 ml/day = bronchorrhoea: lung abscess, bronchiectasis, alveolar cell carcinoma. Offensive = anaerobes. Mucolytics: acetylcysteine, bromhexine.

Dyspnoea

- Sudden: PE, pneumothorax, foreign body, MI. Hours-days: pneumonia, asthma/COPD exacerbation. Weeks-months: ILD, heart failure. Grade with mMRC 0-4.
- Orthopnoea: LVF (commonest), ascites. PND ~2 h after sleep = cardiac; very early morning = asthma; worse on waking, eased by sputum = COPD.
- Better at weekends/holiday = occupational asthma; worse 15-30 min after exercise = exercise-induced asthma; tingling + light-headedness = psychogenic.

Chest pain

- Visceral pleura insensitive; pain from parietal pleura, chest wall, airways, diaphragm, mediastinum.
- Pleural: sharp, localized, worse on breathing/cough, refers to shoulder tip. Angina: crushing retrosternal, exertional, eased by rest/nitrate. Oesophageal: burning, meals, supine.
- Emergencies: MI, aortic dissection, PE, tension pneumothorax, mediastinitis, perforated viscus.

Haemoptysis and wheeze

- From below the vocal cords. False: GI blood (food particles, melaena) or nose/mouth/larynx.
- Common causes: cancer, TB, PE, bronchiectasis; also mitral stenosis, LVF, warfarin. Intermittent over years = bronchiectasis/COPD; daily = cancer, TB, abscess; single + pain + dyspnoea = PE/infarction.
- Massive > 150 ml/attack (slide figure): suffocation risk; resuscitate, cough suppressant, bronchoscopy, bronchial artery embolization.
- Wheeze: high-pitched, expiratory, small airways. Rhonchi: low-pitched, both phases, change with cough.

Source: Respiratory Cardinal Symptoms [108]

PART 1: LECTURE

Chest history taking and examination [100]

Topic: Respiratory symptoms, presentations and clinical reasoning

History

- Complaint in patient's words (open question); present history: onset, course, duration, severity, aggravating/relieving, associated symptoms.
- Cough: diurnal/seasonal variation. Sputum: colour, volume (1 cup = 50 ml), odour. Dyspnoea: MRC, orthopnoea/PND. Wheeze: relation to work. Haemoptysis: volume, attacks. Chest pain: SOCRATES.
- Also constitutional (fever, sweats, weight/appetite loss), congestive (oedema, RUQ pain, ascites, jaundice) and pressure symptoms (dysphagia, hoarseness).
- Drugs: oestrogen (PE), beta-blockers (bronchospasm), ACE inhibitors (cough), methotrexate/amiodarone (fibrosis); home oxygen.
- Social: smoking, cannabis, birds/pets, full occupational history. Family: atopy, cancer, TB. Travel: TB areas, long-haul flights (PE). Finish with ICE.

General examination

- Position 45 degrees, exposed to waist, examiner on right. RR normal 14-20; temp 36.5-37.2 C.
- Distress: accessory muscles, intercostal recession, pursed lips (stop early airway closure in expiration), nasal flaring, tripod position.
- Central cyanosis: lips and tongue. Hands: 3 Cs (clubbing, cyanosis, CRT) and 3 Ts (tremor, tar staining, temperature).
- Clubbing: 1st nail angle lost, 2nd parrot beak, 3rd drumstick, 4th hypertrophic osteoarthropathy. Flapping tremor = CO_2 retention.
- Legs: oedema (cor pulmonale), calf tenderness (DVT).

Inspection and palpation

- Normal AP:transverse = 5:7; barrel chest in emphysema. Limited side = abnormal side.
- Trachea TOWARDS upper lobe collapse/fibrosis or lobectomy; AWAY from tension pneumothorax or massive effusion.

Percussion

- Hyperresonant = pneumothorax; impaired = fibrosis; dull = consolidation, mass, collapse; stony dull = effusion.
- Kronig's isthmus dull: apical TB fibrosis, collapse, Pancoast tumour, consolidation, abscess, apical effusion.
- Traube's area dull: splenomegaly, big left liver lobe, tense ascites, left effusion, full stomach. Large: splenectomy, pneumothorax, pneumoperitoneum.

Auscultation

- Normal: vesicular (inspiration > expiration, no gap), equal, no added sounds.
- Bronchial breathing (3 Cs): consolidation, collapse with patent bronchus, superficial cavity; with increased VR (bronchophony, whispering pectoriloquy).
- Decreased VR and air entry: effusion, pneumothorax, collapse, mass; diffuse: emphysema, severe asthma, obesity.
- Wheeze: musical, louder on expiration; inspiratory = severe narrowing. Coarse early-mid crackles: oedema, bronchiectasis, abscess. Fine late: ILD, LVF, pneumonia.

Source: History and examination lecture [100]; student notes; history and examination checklists

PART 1: LECTURE

Pulmonary function and spirometry [062]

Topic: Respiratory symptoms, presentations and clinical reasoning

Spirometry basics

- Volume-time curve shows FVC and FEV1; flow-volume loop shows FVC, FEV1, FER (FEV1/FVC x 100) and PEF. Normally about 80% of FVC is blown out in the first second.
- Uses: dyspnoea work-up, pre-operative assessment, monitoring disease, occupational exposure, disability, response to treatment.
- Spirometry cannot measure RV, so it cannot give TLC or FRC. Volumes are in litres at BTPS.
- Results are compared with predicted values from age, sex, height, weight (and ethnicity).

Doing the test

- Before: no bronchodilators, no smoking for 1 h, no heavy exercise for 30 min, no heavy meal for 2 h, loose clothing.
- Nose clip, tight seal; full inspiration, blow hard and fast to RV. Effort-dependent: coach the patient.
- At least 3 acceptable blows (no more than 8); record best FVC, FEV1, PEF.
- Acceptable: quick forceful start, no cough or leak, smooth curve, exhalation at least 6 s with no volume change in the last 1 s; efforts agree within 5% or 100 mL.

Patterns

- Obstructive: low FEV1/FVC, low FEV1 and FEF25-75, normal or low FVC, normal or high TLC; scooped expiratory curve (asthma). Early collapse of flow = advanced obstruction (COPD).
- Restrictive: normal or high FEV1/FVC, low FVC, low TLC (confirm TLC in PFT lab); tall narrow loop (pulmonary fibrosis).
- Variable extrathoracic upper airway obstruction (goitre): inspiratory loop most flattened.
- Positive bronchodilator response: FEV1 up by 12% and 200 mL (asthma).
- Raised RV = air trapping; hyperinflation = TLC above 120%.
- Normal ranges: FEV1, FVC, TLC 80-120%; RV 75-120%; DLCO above 60%.
- COPD (FEV1/FVC below 0.7) by FEV1: GOLD 1 80% or more, 2 50-79%, 3 30-49%, 4 below 30%.

Lung volumes and gas transfer

- Helium dilution: FRC of ventilated lung only. Body plethysmography: thoracic gas volume including non-ventilated areas.
- DLCO uses CO (single breath-hold); helium gives alveolar volume (VA); KCO = TLCO / VA.
- DLCO low: ILD, emphysema, pneumonectomy, anaemia, V/Q mismatch. DLCO high: pulmonary haemorrhage (Goodpasture), polycythaemia.
- Restriction on spirometry: order DLCO next. Bird exposure + restriction + low DLCO = hypersensitivity pneumonitis: remove antigen + oral steroids.

Source: Pulmonary Function [062] slides and handout

PART 1: LECTURE

Pleural diseases [117] LGW

Topic: Respiratory symptoms, presentations and clinical reasoning

Pleural effusion: clinical

- Normal pleural space has 1-15 mL fluid. Dyspnoea is the commonest symptom; pleuritic pain suggests PE or inflammation; constant dull ache suggests chest wall invasion (carcinoma, mesothelioma).
- Signs from about 300 mL: stony dullness, reduced breath sounds and vocal resonance. Above 1000 mL: mediastinal shift away.
- Shift TOWARDS the effusion = bronchial obstruction by an endobronchial lesion (usually malignancy).

Causes and fluid analysis

- Transudates: heart failure, constrictive pericarditis, cirrhosis, hypoalbuminaemia, nephrotic syndrome, SVC syndrome.
- Exudates: parapneumonic, TB, malignancy, RA/SLE, PE, asbestos, pancreatitis, trauma, oesophageal perforation, radiation, drugs.
- Light's criteria (any one = exudate): protein ratio above 0.5; LDH ratio above 0.6; pleural LDH above 2/3 upper normal serum LDH.
- Protein above 2.9 g/dL = exudate. LDH above 1000 = empyema or malignancy. pH below 7.2 and glucose below 30 = empyema. Glucose 30-50 = malignancy or TB.
- Milky + TG above 110 mg/dL = chylothorax. Pleural Hct above 50% of blood Hct = haemothorax.
- Lymphocytes above 85% = TB, lymphoma. Eosinophils above 10% = air/blood in pleura, PE with infarction, parasites, drugs. WBC above 1000/mm3 = exudate.
- Malignant effusion: haemorrhagic, massive, rapidly reaccumulating.

Imaging and thoracentesis

- PA CXR detects 200-300 mL (blunted angle); lateral decubitus 75-100 mL. D-shaped opacity suggests empyema.
- Tap 2 rib spaces below dullness, guided by US or CT; check position with small needle first; give oxygen.
- Remove at most 1-1.5 L at once (re-expansion oedema, pneumothorax). Post-tap expiratory CXR to exclude pneumothorax.
- Relative contraindications: very small effusion, bleeding disorder or anticoagulation, mechanical ventilation, skin disease at site.
- Empyema: large-bore (20 F) chest tube is a must.

Haemothorax and pneumothorax

- Haemothorax: trauma or procedures. Large-bore drain, cross-match, correct clotting; above 1000 mL or above 200 mL/h for 3 h = call thoracic surgeon.
- Primary spontaneous: thin young men, no lung disease (Marfan, Ehlers-Danlos). Secondary: COPD, asthma, CF.
- Signs: reduced expansion, hyper-resonance, reduced breath sounds. CXR: no lung markings beyond a sharp lung edge.
- Tension: hypotension, tachycardia, raised JVP, shift away. Immediate needle decompression, 2nd intercostal space, midclavicular line, then chest tube.

Source: Pleural Diseases [117] LGW

PART 1: LECTURE

Chest case reviews [098] and cases W, X, Y, Z [021, 022]

Topic: Respiratory symptoms, presentations and clinical reasoning

Airways disease

- Child with recurrent wheeze, cough, dyspnoea, pulsus paradoxus = acute asthma. NSAIDs can trigger attacks. Clubbing is not a feature of asthma.
- COPD = airflow limitation that is not fully reversible (e.g. FEV1/FVC 47%, no reversibility, smoker). Moderate = FEV1 50-79% predicted.
- COPD does not cause clubbing: new clubbing in a smoker suggests bronchogenic carcinoma. Respiratory causes: lung cancer, mesothelioma, empyema, CF.
- Raised JVP, hepatomegaly and oedema in COPD suggest cor pulmonale.

Blood gases and respiratory failure

- Type 1 failure: PO2 below 60 mmHg with normal or low PCO2 (e.g. PE). Type 2: raised PCO2.
- pH 7.61, PCO2 22, HCO3 24 = uncompensated respiratory alkalosis (anxiety hyperventilation).
- pH 7.32, PCO2 60, HCO3 24 = respiratory acidosis (COPD exacerbation).

PE and ARDS

- After a long flight: Westermark sign, Hampton's hump, type 1 failure = PE.
- D-dimer is sensitive, not specific. Start LMWH + warfarin (INR 2-3) while CTPA is awaited.
- ARDS: acute, bilateral infiltrates, PaO2:FiO2 below 300, no heart failure. Mild 200-300, moderate 100-200, severe 100 or less. Aspiration is a trigger.

Infection and granulomas

- CURB-65: confusion, urea above 7, RR 30 or more, low BP, age 65 or more (one point each).
- Fibrocavitary upper lobe lesion, night sweats, haemoptysis, weight loss = TB. Treat 2 months RHZE then 4 months RH; check LFTs. Rifampicin turns urine orange-red.
- Caseating granulomas = TB; non-caseating granulomas + bilateral hilar nodes = sarcoidosis (restrictive PFTs, low DLCO).
- Asbestos (stripping ship insulation) + basal fine crackles = asbestosis.

Pleura and lung cancer

- High protein and LDH = exudate (e.g. bronchogenic carcinoma); heart failure, cirrhosis, nephrotic and SVC syndrome cause transudates.
- Empyema fluid: pH below 7.2, high LDH, low glucose.
- Suspected pneumothorax: chest X-ray first.
- Pancoast tumour: Horner syndrome (ptosis, miosis, anhidrosis); no exophthalmos.

Source: Chest Cases Review [098]; Cases WX and YZ [021, 022]

PART 2: TOPIC SUMMARY

Infections (pneumonia, TB, abscess, bronchiectasis)

Pneumonia

- CAP (community), HAP (48 h or more after admission), VAP (48 h or more on a ventilator) have different organisms and antibiotics. Commonest CAP organism: S. pneumoniae.
- Severity: CURB-65 (confusion, urea over 7, RR 30 or more, low BP, age 65 or more). 0-1 home; 2 ward; 3-5 or severe criteria = ICU.
- Outpatient, healthy: amoxicillin, doxycycline or macrolide. Inpatient: beta-lactam + macrolide. Add anti-MRSA or anti-Pseudomonas only with risk factors.
- Atypicals: Mycoplasma (young, headache, no cell wall, macrolide); Legionella (travel, water aerosols, confusion, diarrhoea, macrolide).

Tuberculosis

Step	Key facts
Diagnose	3 early-morning sputa; Ziehl-Neelsen or fluorescent smear; culture (slow, 4-8 weeks, gives resistance); PCR rpoB = rifampicin resistance
Latent TB	TST (read at 48-72 h; cut-offs 5, 10, 15 mm by risk) or IGRA (not affected by BCG)
Treat	2 months HRZE then 4 months HR; meningitis 9-12 months; bone 9 months
Side effects	Isoniazid neuropathy (pyridoxine), hepatitis (H, R, Z); ethambutol vision (red-green first); rifampicin orange urine; streptomycin ear damage (avoid in pregnancy)
Monitor	Sputum at 2 months and at the end; LFTs

- Extrapulmonary TB: miliary (2-3 mm nodules), meningitis (most serious; CSF lymphocytes, low glucose, high protein), Pott's spine (paraplegia), lymphadenitis. Steroids for TB pericarditis and meningitis.
- HIV: atypical TB; start ART 2-8 weeks after TB treatment if CD4 under 200.
- NTM: MAC commonest; child with one-sided neck node = cure by excision; fish tank granuloma = M. marinum. MAC lung: macrolide + ethambutol + rifampicin.

Suppurative lung disease

- Lung abscess: cavity with air-fluid level, usually aspiration (anaerobes). Antibiotics 4-6 weeks; drainage if failing; lobectomy if over 6 cm.
- Bronchiectasis: permanent bronchial dilatation, large purulent sputum, clubbing, coarse crackles. HRCT = gold standard (signet ring). First Pseudomonas: ciprofloxacin then inhaled colistin; frequent exacerbations: long-term azithromycin.

PART 2: TOPIC SUMMARY

Lung malignancies

Bronchogenic carcinoma at a glance

- Old smoker with chest symptoms = cancer until proven otherwise. Central = squamous (cough, haemoptysis); peripheral = adenocarcinoma (pleural pain).
- Local spread: Pancoast tumour (shoulder pain, hand wasting, Horner's: ptosis, miosis, anhidrosis); SVC obstruction (facial plethora; classically small cell); hoarseness (left recurrent laryngeal nerve); localised wheeze.

Cell type	Paraneoplastic clue
Small cell	SIADH (low sodium), ectopic ACTH, Lambert-Eaton
Squamous	Hypercalcaemia (PTH-like peptide)
Large cell / adeno	Gynaecomastia; clubbing and hypertrophic osteoarthropathy

- Tissue is the issue: bronchoscopy for central, CT-guided biopsy for peripheral; stage with CT chest, abdomen, brain and PET-CT.
- TNM: T1 up to 3 cm, T2 3-5 cm, T3 5-7 cm, T4 over 7 cm or mediastinal invasion; N1 hilar, N2 same-side mediastinal, N3 opposite side or supraclavicular; M1a opposite lung or malignant effusion.
- Non-small cell: surgery if resectable. Small cell: limited versus extensive, chemotherapy (surgery usually not possible).
- Clubbing in a smoker with COPD is not from COPD: look for cancer.

PART 2: TOPIC SUMMARY

COPD, asthma, wheezy chest

Asthma versus COPD

	Asthma	COPD
Key word	Variable	Persistent
Spirometry	FEV1/FVC under 80% with reversibility (FEV1 up 12% and 200 mL)	Post-bronchodilator FEV1/FVC under 70%
Mainstay	Inhaled steroid (never LABA alone); ICS-formoterol reliever	LABA + LAMA; add ICS if eosinophils 300 or more with exacerbations
Acute attack	SABA + ipratropium, prednisolone, oxygen to SpO2 over 93%	SABA + SAMA, prednisolone 40 mg 5 days, antibiotics, oxygen 88-92% via Venturi
Danger signs	Silent chest, confusion, SpO2 under 90%	Type 2 failure (high CO2)

- COPD long-term oxygen: PaO2 under 55 mmHg (or under 60 with cor pulmonale or polycythaemia), 18-24 h a day. Screen once for alpha-1 antitrypsin.
- Asthma phenotypes: exercise-induced, aspirin-induced (avoid NSAIDs), occupational (better at weekends).
- Not every wheeze is asthma: stridor = upper airway; localised wheeze = tumour or foreign body; cardiac asthma = heart failure.

PART 2: TOPIC SUMMARY

Interstitial lung disease

Shared features

- Dyspnoea, dry cough, clubbing, fine Velcro crackles at the bases. Restrictive PFTs with low DLCO. HRCT is the gold standard.

Disease	Clue	Treatment
IPF	Over 50, basal honeycombing	Pirfenidone
Sarcoidosis	Young, bilateral hilar nodes, non-caseating granulomas, raised ACE and calcium	Prednisolone; Löfgren needs only NSAIDs
Hypersensitivity pneumonitis	Birds or farming; mid-inspiratory squeaks; BAL CD4/CD8 under 1	Remove antigen, steroids
LAM	Young woman, thin-walled cysts	Sirolimus, avoid pregnancy
PLCH	Smoker, odd-shaped cysts	Stop smoking

Occupational lung disease

- Asbestos: pleural plaques (commonest), asbestosis in lower zones, mesothelioma; with smoking, lung cancer risk multiplies.
- Silica: upper lobe nodules, eggshell calcification of hilar nodes, TB risk (yearly test).
- Coal: simple CWP under 1 cm, PMF over 1 cm; clubbing and crackles are not features. Caplan = RA + dust nodules.

PART 2: TOPIC SUMMARY

Respiratory emergencies

Pulmonary embolism

- Sudden dyspnoea, pleuritic pain, hypoxia, often clear chest. Low PaO2 with low PaCO2.
- D-dimer excludes when negative; CTPA confirms (V/Q in pregnancy, contrast allergy, renal failure).
- High risk (shock) = thrombolysis unless contraindicated. Otherwise anticoagulate (LMWH, DOAC, or warfarin overlap to INR 2-3).

Other emergencies

- Tension pneumothorax: hypotension, trachea away, hyper-resonant: immediate needle decompression (2nd intercostal space, midclavicular line), then chest tube.
- ARDS: within a week of an insult, bilateral opacities, not cardiac, PaO2/FiO2 300 or less (mild 200-300, moderate 100-200, severe 100 or less).
- Collapse: trachea pulled towards the lesion; Luftsichel sign = left upper lobe collapse.
- Massive haemoptysis: protect the good lung, bronchoscopy, bronchial artery embolisation.

PART 2: TOPIC SUMMARY

Respiratory symptoms, presentations and clinical reasoning

Symptoms

- Cough: acute under 3 weeks, chronic over 8 weeks; common hidden causes = reflux, ACE inhibitors (bradykinin), post-nasal drip.
- Sputum: rusty = pneumococcus; red currant jelly = Klebsiella; over 100 mL a day = bronchiectasis or abscess.
- Haemoptysis: cancer, TB, PE, bronchiectasis; first exclude blood from nose or stomach.

Examination findings

Finding	Trachea	Percussion	Breath sounds
Consolidation	Central	Dull	Bronchial, increased resonance
Effusion	Away (if large)	Stony dull	Reduced
Collapse	Towards	Dull	Reduced
Pneumothorax	Away (tension)	Hyper-resonant	Reduced
Fibrosis	Towards (upper lobe)	Impaired	Fine crackles

Pleural fluid

- Light's criteria: exudate if protein ratio over 0.5, LDH ratio over 0.6, or LDH over two-thirds of the upper normal. Transudates: heart, liver, kidney failure.
- Empyema: pH under 7.2, low glucose, high LDH: large-bore chest drain. Lymphocytes over 85% = TB or lymphoma. Remove at most 1-1.5 L at a time.

Spirometry

- Obstructive: low FEV1/FVC; restrictive: normal or high ratio with low FVC and TLC. Low DLCO = ILD or emphysema; high DLCO = pulmonary haemorrhage.
- ABG: type 1 failure = low O2, normal or low CO2; type 2 = high CO2.

ICC-1 Vertical subjects: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. Disorders of blood flow [VM 035] (parts 1 and 2)
2. Atherosclerosis, aneurysms and ischaemic heart disease pathology [037a, 037b]
3. Cardiomyopathy and myocarditis histopathology [VM 036]
4. Pathology of hypertension [VM 090a]
5. Pathology of COPD, asthma and ILD [004] and lung pathology [033]
6. Pathology of pulmonary tuberculosis [010]
7. Pathology of infectious diseases [VM 113]
8. Pathology of lung tumours [VM 011, 067]
9. Basic chest X-ray interpretation [VM 080]
10. Radiology of TB [VM 085]
11. Arterial imaging [VM 015]
12. Venous imaging [VM 024]
13. Imaging of thromboembolic disease and PE [VM 091]
14. ICC-1 spotting: anatomy and radiology
15. Management of disturbed behaviour [VM 031] (lecture and LGW)
16. Depression, anxiety and obsession [VM 042]
17. Drug misuse presenting to A&E [VM 075]
18. Egyptian Mental Health Act and involuntary management [VM 084]
19. Obsessive compulsive disorder [VM 088]
20. Reactions to stress including PTSD [VM 089]
21. Care of the critically ill patient [VM 003]
22. People with learning disabilities [VM 014]
23. Self-harm and suicide [VM 065]
24. Retinal vascular diseases [VM 043]
25. Fundus examination skill lab [VM 043]
26. Leprosy [VM 116]
27. Filariasis [VM 121]
28. Poisoning lecture [VM 030]
29. Poisoning LGW and cases [VM 030, VM 050]
30. Clinical therapeutics: cardiology [VM 110]
31. Antiplatelet and anticoagulant therapy [VM 034]
32. Clinical therapeutics: chest [VM 111]
33. Drugs for anaemia [VM 112]
34. Biochemical assay of cholesterol [VM 029]

Part 2: topic summaries

1. Pathology
2. Radiology
3. Mental health
4. Ophthalmology
5. Dermatology
6. Poisoning and toxicology
7. Use of medicine
8. Biochemistry

PART 1: LECTURE

Disorders of blood flow [VM 035] (parts 1 and 2)

Topic: Pathology

Hyperemia, congestion, edema

- Hyperemia = active arteriolar dilatation (exercise, inflammation). Congestion = passive venous engorgement from impaired outflow, with edema.
- CVC lung (mitral stenosis, left HF): heavy, wet, frothy lungs; late = heart failure cells (hemosiderin macrophages) + fibrotic septa.
- CVC liver (right HF): nutmeg liver = congested central veins + centrilobular necrosis (red) and peripheral steatosis (yellow); late fibrosis.
- CVC spleen (main cause: cirrhosis): big heavy spleen, Gamna-Gandy fibrosiderotic nodules.
- Edema mechanisms: raised hydrostatic pressure (HF, venous obstruction), low plasma oncotic pressure (nephrotic, cirrhosis, malnutrition), increased permeability (inflammation = exudate), Na retention, lymphatic obstruction.
- Exudate: protein > 3 g%, SG > 1015, clots, cellular. Transudate: the opposite.
- Lymphedema: Milroy, filariasis, peau d'orange (breast cancer), post-mastectomy; non-pitting. Renal edema starts periorbital.

Thrombosis

- Virchow's triad: endothelial injury, altered flow (stasis, turbulence), hypercoagulability.
- Stasis = most common cause of venous thrombosis (DVT after immobilization, right HF). Turbulence: aneurysms, AF.
- Factor V Leiden: Va resistant to protein C; young recurrent venous thrombosis. Acquired: OCPs, pregnancy, surgery, Trousseau (pancreatic cancer).
- Lines of Zahn = layers of platelets/fibrin and RBCs; formed in flowing blood during life (heart, aorta).
- Venous thrombi almost always occlude and easily embolize; arterial thrombi form on ruptured plaques.
- Fate: lysis, embolization, propagation, organization and recanalization, incorporation into wall.

Embolism and infarction

- PE: massive = sudden death; medium = no effect in healthy lung but hemorrhagic infarct in congested lung (MS, left HF); repeated small = cor pulmonale.
- Paradoxical embolus: venous embolus crosses septal defect to systemic arteries.
- Systemic emboli (mural thrombus, aneurysm, plaque, vegetations) lodge in end arteries.
- Infarct = coagulative necrosis, except brain (liquefactive). Pale wedge-shaped: heart, spleen, kidney. Red: venous occlusion (torsion, strangulation), lung, dual supply, congested tissue.

Gangrene

- Dry: arterial occlusion, veins patent; mummified, slow putrefaction, mild toxaemia, clear demarcation and separation.
- Moist: artery + vein (crush, tourniquet, bed sores, diabetic, intestine); rapid putrefaction, severe toxaemia, no separation line.
- Gas gangrene: clostridia in deep wounds; crepitus.

Source: VM 035 Disorders of blood flow part 1 and part 2 (2026-2027)

PART 1: LECTURE

Atherosclerosis, aneurysms and ischaemic heart disease pathology [037a, 037b]

Topic: Pathology

Atherosclerosis: lesions and pathogenesis

- Patchy intimal thickening by lipid surrounded by connective tissue (atheroma); mainly large and medium muscular arteries.
- Fatty streak: earliest lesion; LDL + lipid-laden macrophages; appears in infancy; flat, asymptomatic, reversible.
- Simple plaque: fibrous cap (SMCs, collagen) over a soft core (cholesterol clefts, necrotic debris, foam cells); neovessels at the periphery; common at bifurcations.
- Sites in descending order: lower abdominal aorta, coronaries, popliteal, internal carotid, circle of Willis; upper limb usually spared.
- Complicated plaque: rupture/ulceration (thin cap) with thrombus and emboli; haemorrhage into plaque from neovessels; media atrophy causing aneurysm.
- Response to injury: endothelial dysfunction, LDL entry, monocytes become macrophages, oxidized LDL engulfed = foam cells, growth factors drive SMC proliferation.
- Major risk factors: age, male, family history, genetics, hyperlipidaemia, smoking, hypertension, diabetes.

Consequences

- Coronary: angina, MI. Femoral: intermittent claudication (partial), dry gangrene (complete). Cerebral: TIA, stroke. Mesenteric: abdominal claudication, bowel infarction.

Aneurysms

- True (all 3 layers), false (walled-off extravascular haematoma), dissection (blood in media).
- Atherosclerotic AAA: elderly, fusiform, below renal arteries, with mural thrombus. Elective surgery if > 5 cm or enlarging; emergency surgery for rupture/dissection (50% mortality).
- Berry: circle of Willis; media weakness; ADPKD, Ehlers-Danlos IV, Marfan, smoking, HTN; causes SAH.
- Charcot-Bouchard: lenticulostriate vessels; hypertension. Syphilitic: ascending aorta, endarteritis of vasa vasorum. Mycotic: infection, often septic emboli.
- Dissection: intimal tear, blood tracks in media; HTN (men 40-60), Marfan; not atherosclerosis. Type A ascending, Type B distal to arch.

Ischaemic heart disease and MI

- 90% due to coronary atherosclerosis; significant stenosis > 75%.
- Transmural (occlusion, commonest) vs subendocardial (inner third; lysed thrombus or shock, circumferential).
- Necrosis begins in subendocardium; contractility lost in 60 s, necrosis by 30 min.
- Timeline: 1-3 d coagulation necrosis + neutrophils; 3-7 d yellow centre, hyperaemic rim; 10-14 d granulation tissue; > 2 mo dense scar.
- Complications: arrhythmia, acute failure, rupture (2-4 d, tamponade), papillary muscle rupture (MR), pericarditis (day 2-3), mural thrombus, Dressler's, ventricular aneurysm (late).

Source: 037a ATH and aneurysms; VM 037b Pathology of IHD (2026-2027)

PART 1: LECTURE

Cardiomyopathy and myocarditis histopathology [VM 036]

Topic: Pathology

Definition

- Cardiomyopathy (CM) = intrinsic myocardial defect or myocardial disease from a non-cardiovascular cause.
- Damage from hypertension, coronary disease, valvular or congenital disease, or myocarditis is not strictly CM ("ischaemic CM" is a clinical label).

Dilated CM (most common)

- All chambers dilated; big globular heart, thin balloon-like walls; dilatation masks hypertrophy.
- Systolic dysfunction (hypokinetic heart).
- Microscopy: hypertrophied fibres with dark nuclei, interstitial fibrosis, T-lymphocytes.
- Causes: genetic 30-40% (dystrophin: Duchenne/Becker), alcohol, anthracyclines (adriamycin), previous myocarditis, nutritional, haemochromatosis, peripartum, sarcoidosis.
- Effects: progressive heart failure, mural thrombus and embolism; 50% die within 2 years; transplant.

Hypertrophic CM

- Massive hypertrophy without dilatation; asymmetric septal hypertrophy; banana-shaped cavity; hyperkinetic LV; diastolic dysfunction.
- Mitral leaflet hits septum in systole: outflow obstruction, subaortic fibrosis.
- Microscopy: myocyte hypertrophy, pleomorphic nuclei, myocyte disarray, interstitial fibrosis.
- Genetic (50% familial AD): beta-myosin heavy chain most often, then myosin-binding protein C, troponin T.
- Commonest cause of sudden death in young people during vigorous exercise; syncope, AF, mitral endocarditis. Treatment: medical, pacing, septal myomectomy, alcohol ablation.

Restrictive and other CM

- Restrictive: infiltrated firm myocardium; diastolic dysfunction; near-normal systolic function; normal-sized ventricles; biatrial dilatation.
- Causes: amyloidosis, radiation fibrosis, sarcoidosis, metastatic tumour, storage disorders.
- Amyloid: pale pink on H&E; Congo red orange-red with apple-green birefringence. AL = light chains (plasma cell neoplasm); AA = SAA from liver (chronic inflammation, TB); ATTR = familial/old age; AANF = senile atrial.
- Endomyocardial fibrosis: restrictive; children, tropics. ARVC: RV fibrofatty replacement, ventricular arrhythmias.

Myocarditis

- Myocardial inflammation, usually with fever; often systolic dysfunction; may have pericarditis.
- Causes: Coxsackie A/B (most common significant cause), other viruses, bacteria, protozoa; immune (rheumatic fever, SLE, drugs, transplant rejection); giant cell myocarditis.
- Viral: dense T-lymphocyte infiltrate with necrosis. Outcome: no symptoms to heart failure or sudden death.

Source: VM 036 Histopathology of cardiomyopathy and myocarditis (2026-2027)

PART 1: LECTURE

Pathology of hypertension [VM 090a]

Topic: Pathology

Definition and causes

- Hypertension = persistent resting blood pressure above 140/90.
- Primary (essential) = 95%, no obvious cause.
- Secondary: renal (renal artery stenosis, glomerulonephritis, polycystic kidney, chronic pyelonephritis); endocrine (primary aldosteronism, Cushing, steroids, phaeochromocytoma, thyroid, acromegaly, pre-eclampsia); coarctation; neurologic.

Benign vs malignant

- Benign: commonest; gradual moderate rise; no symptoms; slow organ damage.
- Malignant (5%): rapid rise, usually > 200/120, symptomatic organ damage; de novo (young Afro-Caribbean men) or on top of benign (accelerated).

Benign hypertension: pathology

- Vessels: accelerated atherosclerosis; wall degeneration leading to aortic dissection and cerebral haemorrhage.
- Hyaline arteriolosclerosis: acellular pink thickening with narrow lumen (plasma leakage + extra matrix).
- Fibroelastic hyperplasia: small arteries; thick intima/media with reduplicated internal elastic lamina.
- Benign nephrosclerosis: small kidneys, finely granular surface, adherent capsule, poor corticomedullary demarcation, more peripelvic fat.
- Microscopy: hyaline arterioles, atrophic hyalinized glomeruli, atrophic tubules plus dilated tubules of working glomeruli. Clinically proteinuria and chronic renal failure.
- Heart: concentric LVH, ischaemic heart disease, later dilatation and LVF.

Malignant hypertension: pathology

- Malignant arteriolosclerosis: fibrinoid necrosis + hyperplastic (onion-skin) arteriolosclerosis from PDGF.
- Malignant nephrosclerosis: normal size (shrunken if accelerated), smooth flea-bitten surface, capsule strips easily.
- Microscopy: glomerular necrosis and haemorrhage.
- Heart: only mild LVH (short course).

Complications

- Vessels: atherosclerosis, aortic dissection, aneurysm rupture. Brain: cerebral (basal ganglia) haemorrhage.
- Heart: LVH, left ventricular failure (pulmonary oedema, heart failure cells, hypoxic encephalopathy), sudden arrhythmic death.
- Left heart failure symptoms: dyspnoea, orthopnoea, paroxysmal nocturnal dyspnoea.

Source: VM 090a Pathology of hypertension (2026-2027)

PART 1: LECTURE

Pathology of COPD, asthma and ILD [004] and lung pathology [033]

Topic: Pathology

Emphysema and chronic bronchitis

- Emphysema: permanent enlargement of air spaces distal to terminal bronchioles, wall destruction without fibrosis.
- Centriacinar: commonest symptomatic type; respiratory bronchioles; upper lobes; smokers.
- Panacinar: whole acinus; lower lobes; alpha-1 antitrypsin deficiency.
- Paraseptal: subpleural distal acinus, upper lobes; spontaneous pneumothorax. Irregular: commonest overall, with scarring, asymptomatic.
- Pathogenesis: smoke attracts neutrophils/macrophages; elastases and oxidants beat anti-elastases (alpha-1 AT).
- Pink puffer (emphysema): barrel chest, dyspnoea, good oxygenation. Blue bloater (+ bronchitis): infections, CO2 retention, cyanosis. Death: respiratory or right heart failure.
- Chronic bronchitis: productive cough 3 months in 2 consecutive years. Gland hypertrophy = hypersecretion; small airway disease = obstruction.

Asthma and bronchiectasis

- Asthma: reversible obstruction, eosinophilic inflammation, smooth muscle hypertrophy, mucus.
- Atopic (commonest): type I IgE, Th2 (IL-4, IL-5); positive skin test. Non-atopic: infection, pollutants.
- Plugs: Curschmann spirals (shed epithelium), Charcot-Leyden crystals (eosinophils). Remodelling: sub-basement membrane fibrosis, goblet cells, muscle hypertrophy.
- Status asthmaticus: attack lasting days, hypercapnia, acidosis, may kill.
- Bronchiectasis: permanent bronchial dilatation (muscle/elastic loss) from infection or obstruction; bilateral lower lobes; copious purulent sputum; CF commonest inherited cause. Complications: cor pulmonale, brain abscess, amyloidosis.

Pneumoconiosis and ILD

- ILD: interstitial inflammation and fibrosis; restrictive pattern.
- 1-5 micrometre particles most harmful; macrophage is key cell.
- Coal: anthracosis, upper-lobe coal nodules, progressive massive fibrosis. Silicosis: upper-lobe whorled collagen nodules, weakly birefringent; TB risk.
- Asbestosis: lower-lobe subpleural fibrosis, asbestos bodies; pleural plaques (commonest); mesothelioma risk.
- Sarcoidosis (type IV): hilar nodes, raised ACE/Ca; compact non-caseating granulomas along lymphatics; asteroid and Schaumann bodies.
- Hypersensitivity pneumonitis (type III + IV): farmer's/bird fancier's lung; BAL lymphocytosis, low CD4/CD8; poorly formed peribronchiolar granulomas.
- IPF: men 60+; fibrosis, honeycomb lung; dry cough, clubbing, crackles.

Pneumonia and immune injury

- Lobar (pneumococcus): congestion, red hepatization (3-4 d), grey hepatization (1 wk), resolution.
- Bronchopneumonia: patchy bilateral basal, extremes of age. Interstitial (viral, mycoplasma): septal lymphocytes, empty alveoli.
- Goodpasture (type II): anti-basement membrane IgG, linear; haemoptysis + haematuria.

Source: 004 COPD, asthma and ILD; C&B 033 Lung pathology (2026-2027)

PART 1: LECTURE

Pathology of pulmonary tuberculosis [010]

Topic: Pathology

Organism and staining

- Human type → pulmonary TB (droplet, most common route); bovine type (raw milk) → tonsil and intestinal TB in children; contact → skin.
- Lipid-rich wall: mycolic acids (60-90 C), cord factor (cytotoxin), sulfolipids (block phagolysosome fusion). Resists stains, many antibiotics, acids/alkalis; survives in macrophages.
- Ziehl-Neelsen: hot carbol-fuchsin → acid-alcohol → methylene blue. Slender, curved, red acid-fast bacilli.
- Risk: overcrowding, poverty, low resistance (DM), butchers, health care workers.

Granuloma (tubercle)

- Bacillus survives inside macrophage → presented to Th1 cells → IFN-γ (+TNF-α) activates and recruits macrophages; macrophage IL-12 boosts Th1.
- Tubercle: central caseous necrosis (homogeneous pink, cheese-like grossly), epithelioid cells, Langhans giant cells (horseshoe nuclei), peripheral lymphocytes and fibrosis.

Primary TB

- Unsensitised host. Ghon complex = Ghon focus (subpleural, middle third of lung, 1-1.5 cm, caseating) + lymphangitis + caseous hilar lymphadenitis.
- Fate: healing by fibrosis (commonest); direct spread to lung/pleura; blood spread (few bacilli destroyed, moderate → single organ, many → disseminated); later reactivation.

Secondary TB

- Sensitised host: reactivation (usual) or re-infection. Apices of upper lobes (high O2 tension); hypersensitivity → walling off → cavitation. Initial focus < 2 cm.
- By resistance: good → healed fibrotic apex; some → chronic fibro-caseous; very low → acute TB (whole-lobe caseation, no fibrosis); blood spread → miliary.
- Progressive TB (elderly, immunosuppressed): cavitating TB (ragged cavity; vessel erosion → haemoptysis; bronchial drainage = open TB), TB bronchopneumonia (airway spread, poor prognosis), miliary (< 2 mm foci, "galloping consumption"), single organ TB.
- Systemic miliary: liver, spleen, marrow, adrenals, meninges, kidney, genital tract; poor prognosis. Isolated organ (bone, kidney, adrenal, meninges; spine = Pott's): good prognosis with treatment.
- Complications: pleural infection, intestinal TB from swallowed sputum, haemoptysis, extensive fibrosis → heart failure.

Immunosuppressed host

- Diffuse caseation or miliary spread instead of fibrosis.
- MAC (atypical, opportunistic): AIDS with low CD4; lungs and GI tract; fever, drenching sweats, weight loss, diarrhoea, RUQ pain; water/hot tubs/soil. Macrophages full of AFB, granulomas and necrosis rare.

Source: Pulmonary tuberculosis [010] (full and short decks)

PART 1: LECTURE

Pathology of infectious diseases [VM 113]

Topic: Pathology

Pneumonia basics

- Patchy or diffuse alveolar inflammation with consolidation. Types by site: lobar, lobular (bronchopneumonia), interstitial.
- Risks: immune defects (neutrophil/complement/humoral → bacterial; cell-mediated → mycobacteria), suppressed cough (coma, anaesthesia), smoking, obstruction, oedema, chronic disease, malignancy.

Lobar pneumonia vs bronchopneumonia

- Lobar: acute diffuse fibrinous inflammation of one or more lobes; adults; pneumococci by droplets; DM and malignancy predispose.
- Stages: congestion (1-2 d) → red hepatization (3-4 d: red, firm, liver-like; neutrophils, RBCs, fibrin) → grey hepatization (~1 wk: grey-brown, dry; RBCs disintegrate, more fibrin, clear zone by alveolar wall) → resolution (1-2 wk).
- Complications: organization (firm lung), abscess, septicaemia (endocarditis, pericarditis, meningitis, arthritis), toxic myocarditis/heart failure.
- Bronchopneumonia: patchy, bilateral, basal; extremes of age; hospital acquired (Staph aureus, Klebsiella, E. coli, Pseudomonas); nodules around bronchioles exuding pus; lysis and complications common.

Lung abscess

- Suppurative (liquefactive) inflammation; late abscess: necrotic pus centre, pyogenic membrane, granulation tissue.
- Acute (4-6 wk) vs chronic (> 6 wk); primary vs secondary.
- Main mechanism: aspiration with mouth anaerobes (periodontal disease, seizures, alcohol, dysphagia, reduced consciousness). Others: septic emboli (often multiple) from bacteraemia or tricuspid endocarditis, infected bulla/cyst, obstruction by bronchial carcinoma.
- Microbiology rarely confirmed: samples contaminated by upper airway flora.
- 90% cure with antibiotics; poor outlook in elderly, HIV, immunosuppression, malignancy, symptoms > 8 wk; ~75% mortality if immunocompromised or obstructed.

Atypical and opportunistic

- Atypical: septal inflammation, no consolidation; wide septa with lymphocytes, histiocytes, plasma cells; empty alveoli. Mycoplasma, Legionella, Chlamydia, Coxiella, Pneumocystis, viruses. Mild; may heal by fibrosis.
- Opportunistic: CMV, measles, Pneumocystis jirovecii (commonest HIV opportunistic infection; foamy alveolar exudate), Aspergillus, Candida.

Aspergillosis (branching septate hyphae)

- Simple aspergilloma: fungus ball in old cavity (TB, cyst), no invasion, immunocompetent.
- Chronic: existing cavity + lowered defences; hyphae invade cavity wall, inflammation, fibrosis, necrosis.
- Invasive: immunocompromised, days-weeks, angioinvasion; nodule of coagulative necrosis with radial hyphae and haemorrhagic rim, or lobular bronchopneumonia.
- ABPA: hypersensitivity to inhaled spores.

Source: Pathology of infectious diseases [VM 113]

PART 1: LECTURE

Pathology of lung tumours [VM 011, 067]

Topic: Pathology

Lung cancer basics

- Bronchogenic carcinoma = 95% of primary lung tumours; peak 50-70 y; most common fatal cancer.
- Risks: smoking (pack-years; 80% of cancers), uranium, asbestos, arsenic, nickel, radiation, radon, air pollution, genetics, scars.
- Late presentation: cough, dyspnoea, weight loss, chest pain, metastases (adrenal, brain, liver, bone), paraneoplastic syndromes.
- Pancoast (superior sulcus): shoulder/ulnar arm pain, hand wasting, Horner syndrome, oedema, hoarseness; mostly SCC or adenocarcinoma.

NSCLC

- SCC (30%, M>F, smokers): central, main bronchus; metaplasia → dysplasia → CIS; keratin pearls, intercellular bridges; hypercalcaemia (PTH-rp); HMW CK+, TTF-1−.
- Adenocarcinoma (30%, F>M, commonest in female non-smokers): peripheral, scar cancers; glands + mucin; clubbing, hypercoagulability; TTF-1+, HMW CK−. Sequence: atypical adenomatous hyperplasia (K-RAS) → adenocarcinoma in situ (≤ 3 cm, along alveolar walls, no invasion) → invasive.
- Large cell (10%): undifferentiated, high grade, poor prognosis, paraneoplastic rare.
- Treatment: surgery if resectable without metastasis; radiotherapy if not resectable; chemo if metastatic (poor response).

Small cell (SCLC)

- Neuroendocrine; 10-20%; 99% smokers; central; small cells, scant cytoplasm, nuclear moulding, high mitoses, necrosis; chromogranin+, dense-core granules.
- ACTH (Cushing), ADH (SIADH). Usually metastatic at diagnosis. Chemo (initially very sensitive), radiotherapy for SVC obstruction, surgery contraindicated. Staged limited vs extensive.

TNM (NSCLC)

- T1 ≤ 3 cm; T2 > 3-5 cm or main bronchus, visceral pleura, atelectasis; T3 > 5-7 cm or chest wall, phrenic, parietal pericardium; T4 > 7 cm or mediastinum, heart, great vessels, carina, etc.
- N1 ipsilateral hilar; N2 ipsilateral mediastinal/subcarinal; N3 contralateral, scalene, supraclavicular. M1a pleural/contralateral lung; M1b single, M1c multiple extrathoracic.

Carcinoid, metastases, pleura, diagnosis

- Carcinoid: < 40 y, M=F, no smoking link; central polypoid endobronchial mass; nests, salt-and-pepper chromatin; atypical = more mitoses + necrosis. Serotonin → flushing, diarrhoea, wheeze, tricuspid/pulmonary valve scarring.
- Metastases: multiple well-defined nodules, usually adenocarcinoma; TTF-1+ = lung (or thyroid), TTF-1− = secondary.
- Mesothelioma: asbestos (latency up to 40 y; smoking no effect); plaques → pleural encasement; epithelioid (best), sarcomatous, biphasic; calretinin+, TTF-1−; 50% die within 1 y.
- Sampling: central → bronchoscopy; peripheral → CT-guided FNA/biopsy; pleural → fluid + biopsy.

Source: Lung tumours lecture; Lung tumours revision

PART 1: LECTURE

Basic chest X-ray interpretation [VM 080]

Topic: Radiology

Routine and technique

- Systematic routine: airways, lungs, heart, hila, bones and soft tissues, hidden areas.
- Check technique first: AP film → heart and mediastinum look enlarged; poor inspiration → large heart, hazy lungs; underpenetrated → too white; overpenetrated → too dark; rotated → one side denser.
- Silhouette sign: an opacity touching the heart, aorta or diaphragm hides that border. Right heart border = right middle lobe; left heart border = lingula; left lower lobe pneumonia leaves the left heart border visible.

Too white

- Consolidation (pus, blood, water, cells, protein): classic sign = air bronchogram; fissure-bounded lobar opacity.
- Interstitial: reticular, coarse thickened interstitium (fibrosis, asbestosis, sarcoid, radiotherapy, drugs).
- Collapse: opacity + volume loss + shift of trachea, mediastinum, hilum or fissure towards it (ET tube, tumour, mucus plug, foreign body). Right upper lobe: horizontal fissure rises. Left lower lobe: sail-shaped retrocardiac opacity.
- Focal lesion/coin: cancer (spiculated mass), metastasis, hamartoma, pneumonia, abscess, infarct.
- Cavity: dark centre ± air-fluid level (TB, Staph, tumour, infarct, collagen-vascular disease).
- Bronchiectasis: tram tracks (cylindrical), air-fluid levels (cystic); CXR not adequate to diagnose.
- Multiple nodules: > 5 mm mostly metastases; < 5 mm metastases, sarcoid (with reticulonodular shadowing and hilar nodes), miliary TB, pneumoconiosis, chickenpox.
- Pleura: effusion blunts costophrenic angle with meniscus rising to axilla; plaques/calcification (asbestos, TB); mass (mesothelioma, metastasis, fibroma).

Too black

- COPD: hyperexpansion, flat diaphragms, wide rib spaces, bullae, but lung markings still present.
- Pneumothorax: lung edge with no markings beyond it.
- Tension pneumothorax: emergency; large pneumothorax, collapsed lung, trachea/mediastinum shifted away, flat diaphragm.

Heart, hila, hidden areas

- Cardiomegaly: heart diameter > 50% of the transverse (thoracic) diameter; heart failure, pericardial effusion, valve disease.
- Heart failure ABCDE: alveolar oedema (bat wing), Kerley B lines, cardiomegaly, dilated upper lobe vessels, effusion.
- Hila: enlarged (bilateral lymphadenopathy, pulmonary hypertension; unilateral → primary carcinoma), too dense, or displaced (collapse).
- Hidden areas: apices, behind the heart (collapse, hiatus hernia with air-fluid level), under the diaphragm (free air, subphrenic collection).

Source: Basic chest X-ray interpretation [VM 080]

PART 1: LECTURE

Radiology of TB [VM 085]

Topic: Radiology

Course of pulmonary TB

- Primary complex → parenchymal and nodal spread → miliary TB (blood spread, widespread granulomas) → healing with fibrosis and calcification → reactivation.
- Past surgical treatment: artificial collapse of the lung.

Primary TB

- Parenchymal consolidation, lymphadenopathy, pleural effusion, Ghon complex.
- Ghon focus may appear as a dense nodular opacity (e.g. right mid-zone) with hilar nodes.
- A small focus may be invisible on CXR; CT can show it (e.g. posterior segment, right upper lobe).
- Pleural effusion: homogeneous opacity rising to the axilla.

Post-primary TB

- Patchy upper-zone consolidation, cavitation, then fibrosis; pleural disease.
- CT: upper lobe cavity, atelectasis/fibrotic bands, traction bronchiectasis.
- Healed: bilateral apical opacities with fibrosis, volume loss, upward retraction, calcific foci.
- Suspect in chronic cough not helped by antibiotics, night sweats, haemoptysis.

Miliary and extrapulmonary TB

- Miliary: innumerable 1-3 mm nodules throughout both lungs (haematogenous).
- Pericardial effusion: enlarged flask-shaped heart.
- Brain tuberculomata.
- Pott's disease: vertebral TB.

At-risk patients and old treatments (lecture cases)

- Migrant children from high-prevalence countries, homeless people, former IV drug users and those on treatment for other illness, elderly patients with past TB.
- Plombage: inert material (Lucite balls, rubber, paraffin wax, gauze) placed under the upper ribs to collapse the upper lobe, as a collapsed lung was thought to heal faster; seen as balls in the upper pleural space.

Source: Radiology TB [VM 085]

PART 1: LECTURE

Arterial imaging [VM 015]

Topic: Radiology

Modalities

- Doppler ultrasound: carotid, upper limb and lower limb arterial studies; non-invasive, no radiation.
- Angiography options: conventional catheter angiography (invasive), CT angiography (CTA) and MR angiography (MRA).
- CTA is non-invasive and fast compared with conventional catheter angiography, but uses X-rays and iodinated contrast.
- MRA gives a vessel map without ionising radiation.

Reading an arterial occlusion

- Look for: site of block, atheromatous calcification (aorta, iliac, leg arteries), collaterals, distal run-off.
- Good collaterals refilling the popliteal artery with good distal run-off = chronic occlusion.
- Poor or absent collaterals with weak distal filling = suggests acute occlusion.
- MRA of leg: tibial artery occlusion with multiple collateral branches = chronic occlusion.
- Bilateral SFA occlusion may coexist with aorto-iliac calcification and a small femoral artery aneurysm.

Anatomy to know for the cases

- SFA occlusion is bypassed mainly by profunda femoris branches linking with popliteal genicular arteries.
- Popliteal artery ends as anterior tibial and posterior tibial (tibioperoneal trunk) arteries; gives genicular branches.

Aneurysms

- Fusiform = circumferential spindle-shaped dilatation (e.g. abdominal aorta, descending thoracic aorta).
- Saccular = one-sided sac-like outpouching (case: infrarenal abdominal aorta).

Intervention

- SFA occlusion: interventional treatment = percutaneous transluminal angioplasty (PTA) and stenting.

Source: Arterial vascular imaging [VM 015] lecture and computer lab

PART 1: LECTURE

Venous imaging [VM 024]

Topic: Radiology

Modalities for DVT

- Duplex/colour Doppler with compression, contrast venography, CT venography, MR venography, impedance plethysmography.
- Duplex ultrasound = investigation of choice for diagnosis and follow-up of acute DVT: non-invasive, painless, available.
- Highly sensitive for occlusive proximal thrombi, less so for calf thrombi. Most lower limb DVTs begin in the soleal veins of the calf.

Duplex findings in DVT

- Non-compressible venous segment (key sign); absent colour flow if fully occlusive.
- Loss of phasic flow on Valsalva; no flow augmentation with calf squeeze; increased flow in superficial veins.
- Acute thrombus: increased vein diameter, anechoic (dark) thrombus.
- Chronic thrombus: decreased vein diameter, echogenic (bright) thrombus.

Doppler technique

- Spontaneity: flow without augmentation. Phasicity: flow changes with respiration.
- Compression: in transverse plane. Augmentation: squeeze distal to probe = patency below the probe.
- Valsalva: deep breath and strain = patency of abdominal and pelvic veins.

Other modalities

- Contrast venography: historical gold standard, now replaced by duplex; invasive, expensive, painful; thrombus = filling defect; most useful in non-post-phlebitic legs.
- CT venography: for veins above the common femoral vein (IVC, iliac) where ultrasound is less accurate; lecture = modality of choice for post-phlebitic limb and deep vein patency.
- MR venography: useful but time, cost, resources; signs include filling defects, vein wall thickening, perivascular oedema.
- Impedance plethysmography: good sensitivity for proximal DVT, poor specificity.

Source: Venous imaging [VM 024] lecture and computer lab

PART 1: LECTURE

Imaging of thromboembolic disease and PE [VM 091]

Topic: Radiology

CT pulmonary angiography (CTPA)

- First-choice test for PE; negative CTPA = no further investigation or treatment for PE.
- Best with underlying cardiorespiratory disease; can confirm an alternative diagnosis.
- PE = filling defect in a contrast-filled pulmonary artery (main, lobar or segmental).
- Pregnancy test first in women of child-bearing age. Not appropriate if D-dimer negative and Wells says PE unlikely.
- May rarely miss a small peripheral PE.

Alternatives

- V/Q scan: good first-line alternative; much lower radiation. Preferred if age < 40, contrast allergy or severe renal impairment.
- PE on V/Q = perfusion defect with normal ventilation. Normal V/Q virtually excludes PE.
- V/Q hard to interpret if CXR abnormal or cardiopulmonary disease (e.g. COPD): use CTPA instead.
- Conventional pulmonary angiography: most accurate but invasive.
- MR pulmonary angiography: contrast-enhanced = no radiation but contrast hazards; non-contrast cardiac-gated = no radiation or contrast, alternative in renal impairment and pregnancy, less accurate than CT.

Supporting tests

- Chest X-ray: often normal; wedge-shaped peripheral opacity (Hampton's hump), small effusions, oligaemia in massive PE; excludes pneumonia, pneumothorax, heart failure.
- Echocardiography: reliably diagnoses massive PE (acute right heart strain or failure).
- Leg Doppler: most PE patients also have DVT; use when probability high (Wells > 4 or positive D-dimer) but CTPA negative.

Treatment link

- IVC filter is life saving in massive DVT to prevent PE; filters can be placed and later removed.

Source: Imaging of thromboembolic disease (PE) [VM 091]

PART 1: LECTURE

ICC-1 spotting: anatomy and radiology

Topic: Radiology

Exam format

- 40 marks: Anatomy.TV video (20 marks, 5-point answer, e.g. artery = beginning, end, branches, surface anatomy, pulsation site) + 10 stations (2 min, 2 marks each).

Thorax anatomy

- Right lung 10 segments: upper 3 (apical, posterior, anterior), middle 2 (lateral, medial), lower 5. Left lung 8: upper 4 (apicoposterior, anterior, superior and inferior lingular), lower 4.
- Bronchial arteries: right from 3rd right posterior intercostal; left from descending thoracic aorta. Bronchial veins: right to azygos, left to superior hemiazygos.
- SVC = union of right and left brachiocephalic veins. Right atrium receives SVC, IVC, coronary sinus.
- Arch of aorta branches: brachiocephalic, left common carotid, left subclavian. Concavity: left bronchus, ligamentum arteriosum, superficial cardiac plexus.
- Ascending aorta gives right and left coronaries. RCA: marginal, posterior interventricular. Apex: left 5th ICS, 9 cm from midline.

Abdomen and limbs

- Coeliac trunk: left gastric, common hepatic, splenic. SMA: inferior pancreaticoduodenal, middle colic, right colic, ileocolic. IMA: left colic, sigmoid, continues as superior rectal.
- Femoral artery: superficial circumflex iliac, superficial external pudendal, superficial epigastric, profunda femoris, descending genicular.

Radiology spots

- TB: Ghon focus; cavity with hilar node. PE: filling defect in pulmonary artery; saddle embolus; poor contrast filling of a zone.
- Arterial: SFA occlusion with collaterals = chronic; poor collaterals = acute; fusiform and saccular aortic aneurysms.
- Venous: compression on Doppler excludes DVT; no colour fill + thrombus = thrombophlebitis; filling defect on venogram = thrombus.

Chest X-ray (ABCDE)

- Adequate PA film: > 6 anterior or > 10 posterior ribs above diaphragm; spinous processes midway between clavicles.
- Effusion: blunted or obliterated costophrenic angle. COPD: flat diaphragms, wide rib spaces, lucency. Pneumothorax: pleural edge, no markings.
- CTR normal ≤ 0.5; mediastinum < 8 cm (wider = think dissection). Right hemidiaphragm higher by up to 1 ICS.

Source: Year 3 ICC-1 Spotting Final

PART 1: LECTURE

Management of disturbed behaviour [VM 031] (lecture and LGW)

Topic: Mental health

Causes and risk factors

- Model: Antecedents → Behaviour → Consequences; aim = prevention → management → harm reduction.
- Organic: unmet needs, pain, delirium/physical illness, head injury, epilepsy, dementia, intoxication (alcohol, cocaine, amphetamines, MDMA, LSD; fluctuates hour by hour; urine drug screen helps) and withdrawal.
- Delirium tremens (alcohol withdrawal): clouded consciousness, disorientation, visual/auditory/tactile hallucinations; medical emergency, mortality 5-10%.
- Psychiatric high-risk symptoms: command hallucinations, delusions/hallucinations focused on a person, violent delusions of control, violent fantasy. Mania is a psychiatric emergency. Antisocial and emotionally unstable personality disorders.

Prediction and first steps

- Warning signs: tense angry face, prolonged restlessness, louder speech, prolonged eye contact, violent psychotic content, blocking escape routes, verbal threats.
- Principles: ensure safety, rule out organic cause, act swiftly, confidentiality does not override serious risk, work with seniors, keep records, "do no harm".
- If delirium suspected → treat underlying cause (e.g. post-op fever, confusion).

Hierarchy of interventions (proportionate to risk)

- Verbal de-escalation / talking down, low-stimulus room → time out → rapid tranquilisation → physical restraint (trained staff only) → seclusion.
- Time out: max 15 min, never a locked room. Seclusion: last resort, never a planned treatment, observe constantly, review every 2 h.
- Leaving A&E: cannot use Section 5(2); hold under Mental Capacity Act if lacks capacity.

Rapid tranquilisation (last resort)

- Before: legal status, history/MSE/exam, flumazenil available. Offer oral before IM.
- Lorazepam 1-2 mg (max 4 mg/24 h) ± haloperidol 5-10 mg (max 30 mg/24 h PO, 18 mg/24 h IM). Non-psychotic: lorazepam; psychotic: lorazepam + antipsychotic; lower doses in elderly.
- After: T, P, BP, RR every 5-10 min for 1 h; ECG formally recommended for all given haloperidol.

Adverse effects

- Akathisia: inner restlessness, pacing → reduce/change drug, propranolol, clonazepam, mirtazapine.
- Acute dystonia (oculogyric crisis, torticollis; young neuroleptic-naive males, haloperidol) → procyclidine.
- NMS: fever, rigidity, altered mental state, autonomic instability, ↑↑CK → stop drug, supportive care, bromocriptine + dantrolene; restart after ≥5 days with an unrelated drug.
- Respiratory depression (RR <10 or SpO2 <90%) → oxygen, flumazenil if benzodiazepine.
- QTc >500 ms → arrhythmia risk; high risk: IV antipsychotics, haloperidol, methadone; worsened by low K, Mg, Ca.

Source: VM-MH017.1 Management of disturbed behaviour lecture; VM-MH017.2 LGW tutor/student notes

PART 1: LECTURE

Depression, anxiety and obsession [VM 042]

Topic: Mental health

Session focus

- Three "neurotic" disorders: depression, anxiety, obsessive-compulsive disorder (OCD).
- For each know: main symptoms, diagnostic criteria, and biological, psychological and social treatment.
- Key references: NICE guidance on depression, anxiety and OCD.

Depression (case: 67-year-old woman 4 weeks after CABG)

- Clues: tearful, withdrawn, not eating, awake at night, not engaging in rehab; normal observations.
- Core symptoms: low mood, loss of interest/enjoyment, low energy; plus poor sleep, appetite, concentration.
- Common after cardiac surgery and slows recovery; seek liaison psychiatry advice.
- Drug choice after a cardiac event: sertraline (SSRI); avoid tricyclics. On antiplatelets/ anticoagulant → watch bleeding (PPI helps).
- Look at social factors (husband berating her).

Anxiety (case: 30-year-old woman collapsing in crowded shops)

- Panic attacks: sudden fear of dying, racing heart, breathlessness, sweating, dry mouth; repeated → panic disorder; in crowded places → agoraphobic features.
- Daily worry from waking for >2 months → generalised anxiety.
- Exclude physical causes first: ECG, bloods, CXR normal.
- Treatment: CBT; if medication needed → SSRI; avoid long-term benzodiazepines.

OCD (case: 20-year-old student washing hands in bleach)

- Contamination obsessions + washing compulsions; time-consuming, impair study and social life.
- Red cracked skin = damage from washing, not infection → antibiotics not indicated.
- Treatment: CBT with exposure and response prevention ± SSRI.
- Address coping with alcohol and cannabis.

Measurement scales

- BDI: self-rated depression inventory.
- HADS: self-rated anxiety and depression in hospital/medically ill patients.
- Hamilton Depression Scale: clinician-rated.

Source: VM-MH018.1 Depression Anxiety and Obsession student notes (case vignettes; answers are standard NICE-based teaching, the notes give questions only)

PART 1: LECTURE

Drug misuse presenting to A&E [VM 075]

Topic: Mental health

Opiates (heroin)

- Overdose: pinned pupils, shallow breathing, cyanosis, low SpO2, loud snoring, unrousable; often mixed with benzodiazepines/alcohol.
- Treat with naloxone; short half-life → may become unconscious again if he leaves. Flumazenil reverses benzodiazepines.
- Withdrawal: sweating, goose-bumps, dilated pupils, runny nose, watery eyes, yawning, cramps, diarrhoea, tachycardia.
- In A&E: no methadone usually; symptomatic (diazepam, loperamide, buscopan). On AMU: methadone/buprenorphine per protocol; confirm prescription with agency and pharmacy; urine drug screen; refer to drug service early.

Other drugs

- Cocaine/crack: stimulant, binge use; hypertension, fits, MI, stroke, gut infarction, rhabdomyolysis; snorting → perforated septum.
- GHB/GBL: GABA-B, sedation, respiratory depression (worse with alcohol); narrow overdose margin. GBL withdrawal → delirium, hallucinations; treat with benzodiazepines + baclofen, may need ITU.
- MDMA: dilated pupils, jaw clenching, ↑temperature; deaths from heatstroke, dilutional hyponatraemia, heart failure.
- Benzodiazepine withdrawal: anxiety, depersonalisation, light/sound sensitivity, fits → diazepam detox.
- Injecting: abscess, cellulitis, septicaemia, acute endocarditis, DVT/PE (groin), arterial injection ischaemia; HIV, hepatitis B/C.

Diagnosis (ICD-10)

- Harmful use: use causing physical or mental damage (e.g. low mood after crack binge).
- Dependence: craving, poor control, use despite harm, priority over other activities, tolerance, ± withdrawal.

Alcohol

- Withdrawal: sweating, tremor, tachycardia, anxiety, vomiting; fits; delirium tremens (disorientation, visual hallucinations) → consider haloperidol.
- Wernicke: confusion, ataxia, nystagmus, ophthalmoplegia → IV thiamine (correct Mg, glucose). Korsakoff: loss of new learning/short-term memory.
- AMU: SADQ (<16 mild, 16-30 moderate, >30 severe); chlordiazepoxide reducing regimen; thiamine/Pabrinex; seizures → rectal diazepam 10 mg.
- A&E intoxication: ABCDE, risk of aspirating vomit, check glucose. GP screening: AUDIT/AUDIT-C, CAGE; brief interventions.

Safe prescribing and drug seeking

- Red flags: requests for specific opioids, vague pain, doctor shopping, "lost" methadone. But may be genuinely ill.
- Avoid unreviewed repeats; use instalment, small, time-limited prescriptions.

Source: VM-MH019.1 Presentations of drug misuse to A&E lecture; VM-MH019.1 Overview of illicit drugs

PART 1: LECTURE

Egyptian Mental Health Act and involuntary management [VM 084]

Topic: Mental health

Capacity

- Ability to understand, retain, weigh information and communicate a decision.
- Presumed unless proved otherwise; task- and time-specific; support decision-making; unwise decision ≠ lack of capacity; best interests and least restrictive option.
- Help capacity: no jargon, pictures, translators, treat pain, time, quiet setting, relative present.
- Loss may be temporary (unconscious), permanent after having it (dementia) or lifelong (severe learning disability).

The Act (2009, presidential decree)

- Tight criteria for detention; court review of detention; notify public prosecutor within 24 h of involuntary admission.
- Voluntary patients consent to treatment; documented, reviewed treatment plans; patients' rights committee in each facility; sanctions for violations.

Admission articles

- Art 10: anyone 18+ can request voluntary admission and discharge (unless involuntary criteria apply).
- Art 11: voluntary patient may be held ≤72 h if serious imminent danger or cannot self-care; only emergency treatment without consent; extendable ≤1 week.
- Art 13: involuntary admission needs a psychiatrist's approval, severe illness, refusal, plus either imminent severe deterioration or serious imminent threat to self/others; notify within 24 h.
- Art 14: non-psychiatrist may admit for assessment ≤48 h on written request (relative to 2nd degree, police, social worker, health inspector, consul, unrelated outside psychiatrist); prosecutor within 24 h.
- Art 15: extend to max 7 days if independent assessment not possible.
- Art 16: >1 week needs two registered psychiatrists' assessments sent to Regional Council within 7 days, or detention ends.
- Art 21: escaped patient → police/prosecution return him.

Treatment articles

- Art 28: refusing involuntary patient can be compelled; review ≤ every 4 weeks; independent assessment if >3 months.
- Art 29: emergency treatment without consent ≤72 h.
- Art 30: ECT only under general anaesthesia + muscle relaxant with written informed consent; refusing involuntary patient needs independent evaluation.

Seclusion, restraint, observation

- Last resort, shortest time, not therapeutic. Restraint: physical, mechanical or chemical.
- Constant (within reach): extremely suicidal. Level 1: every 15 min, stays on unit. Level 2: every 30 min. Level 3: hourly whereabouts.

Source: VM-MH020.1 Egyptian Mental Health Act Involuntary management

PART 1: LECTURE

Obsessive compulsive disorder [VM 088]

Topic: Mental health

Definition (ICD-11 / DSM-5)

- Obsessions: repetitive, persistent intrusive and unwanted thoughts, images or urges (contamination, violent scenes, urge to stab), usually with anxiety; person tries to ignore, suppress or neutralise them.
- Compulsions: repetitive behaviours (washing, checking, ordering) or mental acts (counting, praying, repeating words) done in response to an obsession to reduce anxiety.
- Time-consuming (e.g. >1 h/day) or significant distress or impairment.

Key features

- Obsessions are distressing (often violent or obscene) and recognised as the person's own thoughts (unlike psychosis).
- Compulsions are not enjoyable, do not complete useful tasks, and are meant to prevent an unlikely event.
- Cycle: obsession → anxiety → compulsion → brief relief → repeats.
- Distinguish from anankastic (obsessional) personality traits.

Epidemiology

- Adult prevalence 0.6-2.5%; higher in children.
- Mean onset age 26; M:F 1:1, men present earlier.

Aetiology

- Genetic: first-degree relative → 5-7× risk; up to 80% concordance in identical twins.
- Hyperactive orbitofrontal cortex, anterior cingulate cortex, caudate nucleus; impaired inhibitory response.
- Serotonergic pathways; possible immunological cause (PANDAS: after streptococcal infection in children).

Treatment

- Psycho-education: explain symptoms, simple strategies, signpost support groups.
- Psychotherapy: CBT (main), metacognitive, mentalisation-based, psychodynamic.
- Medication: SSRI (e.g. sertraline, often high dose); clomipramine (tricyclic) if SSRI fails.
- Course can relapse; resistant cases may need long-term therapy or specialist inpatient programmes.

Source: VM-MH021.1 Obsessive Compulsive Disorder (OCD)

PART 1: LECTURE

Reactions to stress including PTSD [VM 089]

Topic: Mental health

Acute stress reaction (ICD-11)

- Follows an extremely threatening or horrific event; ICD-11 does not count it as a mental disorder.
- Emotional, somatic (daze), cognitive (amnesia, depersonalisation, stupor) and behavioural symptoms; autonomic signs (tachycardia, sweating, flushing) common.
- Starts within hours to days; settles within a few days or once removed from the threat.
- Manage: psychoeducation, normalise symptoms, practical support. TCAs, benzodiazepines, anxiolytics: little evidence.

Acute stress disorder (DSM-5) and adjustment disorder

- ASD: actual or threatened death, serious injury or sexual violence + intrusion, negative mood, dissociation, avoidance, arousal; lasts 3 days to 1 month.
- Adjustment disorder: maladaptive reaction to a psychosocial stressor; preoccupation, worry, rumination; impaired functioning.
- Onset usually within 1 month; resolves within 6 months once stressor and consequences end.

PTSD: stressor and features

- DSM-5 exposure: directly experiencing, witnessing in person, learning it happened to a close relative or friend (violent or accidental), repeated exposure to details (first responders).
- Causes: combat, road accidents, terrorism, disasters, hostage-taking, assault.
- 4 core groups: intrusion (flashbacks, nightmares), avoidance, negative cognition and mood (guilt, detachment), arousal (startle, poor sleep and concentration). Duration > 1 month.
- Lifetime prevalence 7.8%; F > M.
- Comorbidity: affective and anxiety disorders, substance misuse, somatisation; family problems, violence; poor physical health.

Aetiology and risk

- Psychological: fear conditioning, appraisal, traumatic memories, maintaining behaviours.
- Biological: raised adrenaline and noradrenaline; low cortisol with HPA hypersuppression; genetic.
- Person: female, Black or Hispanic, family history, previous trauma, low intelligence or self-esteem.
- Stressor: prolonged or repeated, grotesque, intentional, harm to children, loss of function (not injury severity).

Treatment of PTSD

- Psychological: CBT, EMDR, psychodynamic therapy, hypnotherapy.
- Drug: SSRIs. Medication does have a role.

Source: VM 089 Reactions to stress including PTSD

PART 1: LECTURE

Care of the critically ill patient [VM 003]

Topic: Mental health

Recognition and principles

- Most arrests are predictable: deterioration comes before 50-80% of cardiac arrests.
- Hypoxia and hypotension are common warning signs; delays in referral to higher care add to the problem.
- Early warning scores with an escalation protocol detect ward deterioration.
- Complete the initial assessment, treat life-threatening problems as found, reassess, check the effect of each treatment, call for help early (assess, treat, reassess).
- Initial: personal safety, responsiveness, first impression (movement, pulse, breathing effort, temperature).

A: Airway

- Causes: CNS depression, blood, vomit, foreign body, trauma, infection, inflammation, laryngospasm, bronchospasm.
- Signs: noisy breathing (stridor, wheeze, gurgling), choking, see-saw breathing, accessory muscles. A talking patient has an open airway.
- Treat: head tilt, chin lift, jaw thrust; simple adjuncts; advanced (LMA, tracheal tube); oxygen; early capnography.

B: Breathing

- Causes: reduced drive (CNS depression); reduced effort (weakness, nerve damage, chest defect, rib pain); lung disease (pneumothorax, haemothorax, infection, COPD, asthma, PE, ARDS).
- Look (distress, cyanosis, rate), listen (breath sounds), feel (expansion, percussion, tracheal position).
- Treat: oxygen, treat cause (drain pneumothorax), bag-mask if inadequate, monitor SpO2 and rate.

C: Circulation

- Primary (cardiac): ACS, arrhythmia, hypertensive heart disease, valve and hereditary disease. Secondary: asphyxia, hypoxaemia, blood loss, hypothermia, septic shock.
- Assess pulse, capillary refill, BP, organ perfusion (chest pain, mental state, urine output), bleeding; record ECG.
- Treat: IV/IO access and bloods, ECG monitoring, fluid challenge, treat cause, inotropes or vasopressors; aspirin, nitrates, oxygen (if needed), morphine for ACS.

D and E

- D: AVPU or GCS, pupils, lateralising signs; glucose < 4 mmol/L give glucose; lateral position; check drug chart.
- E: undress to examine (injuries, bleeding, rashes), avoid heat loss, keep dignity.

Source: VM 003 Care of critically ill patient

PART 1: LECTURE

People with learning disabilities [VM 014]

Topic: Mental health

Definition and causes

- Criteria: impaired intellectual function + impaired adaptive function + onset in the developmental period.
- DSM-5: IQ < 70 indicates LD; level (mild to profound) set by adaptive functioning.
- Prevalence: mild 1-2%, severe and profound about 0.35%.
- Causes: idiopathic (often), genetic, substance misuse (foetal alcohol syndrome, preventable), perinatal, environmental.

Key syndromes

- Down's: trisomy 21 (also translocation, mosaic); maternal-age related; AVSD, hypothyroidism, leukaemia, coeliac, cataracts; dementia at a younger age, depression.
- Fragile X: commonest inherited LD in boys and single-gene cause of autism; CGG repeats on distal Xq; hyperactivity, social and speech deficits; tremor/ataxia in older carrier males; premature ovarian failure in carrier females; worsens through generations.

Physical health

- More illness, poor access to care, late presentation; 37% of deaths avoidable in one inquiry.
- Epilepsy 22% vs 1%; dementia 21.6% vs 5.7%; schizophrenia 3% vs 1%; hearing problems 40%; thyroid problems.

Mental illness

- Dual diagnosis common; change in behaviour is highly significant.
- Diagnostic overshadowing: blaming new symptoms on the LD. Exclude physical causes (look for pain) first.
- Depression: often missed; atypical (hypersomnia, increased appetite), self-injury or aggression.
- Schizophrenia: more in mild LD; velo-cardio-facial syndrome; simple delusions.
- Challenging behaviour is not a diagnosis; multifactorial.

Assessment and management

- History from several sources; observe in natural settings; serial examinations.
- Bio-psycho-social, multidisciplinary, use mainstream services, assess capacity.
- Drugs: treat illness; 2nd line for behaviour; start low, go slow; never only to sedate; physical check, ECG where indicated, bloods; review at 2 weeks.
- GMC: extra time, simple words or pictures, demonstrate procedures first, reasonable adjustments.
- Safeguarding: watch for abuse and financial exploitation.

Source: VM 014 People with learning disabilities

PART 1: LECTURE

Self-harm and suicide [VM 065]

Topic: Mental health

Self-harm: definition and epidemiology

- NICE: self-poisoning or self-injury, irrespective of apparent purpose. 'Deliberate self-harm' is outdated and stigmatising.
- Lifetime prevalence about 5% (13% in adolescents); much is hidden.
- Risk factors: adolescence (peak women 15-24, men 25-34), female, separated or divorced, low income, abuse, isolation, alcohol (half), prisoners, early psychosis.
- 90% have a psychiatric disorder: depression (2/3), substance misuse, anxiety; personality disorder in half.
- Lithium (bipolar) and clozapine (resistant schizophrenia) reduce self-harm and suicide; SSRI effect controversial.

Methods and paracetamol

- Self-poisoning ~90% (paracetamol, antidepressants, benzodiazepines, often with alcohol); self-injury ~10%.
- Paracetamol: commonest cause of acute liver injury; 10-15 g severe, 5 g if high risk. NAPQI depletes glutathione.
- Activated charcoal if serious overdose within 2 h; N-acetylcysteine is a glutathione precursor.

Assessment

- Everyone needs assessment of needs and risk, in parallel with medical care; refer to liaison psychiatry. Be non-judgemental.
- Intent: precautions against discovery, note or affairs in order, planning, disclosure, regret. Physical severity does not show intent.
- Get collateral history; risk scales not recommended.
- Self-harm raises suicide risk 50-100 times; highest within 6 months.
- Suicide risk after self-harm: older, male, isolated, mental or physical illness, hopelessness, substance misuse, violent method, strong intent.

Risk level and plan

- Low (no intent, impulsive, supportive, no history): back to GP.
- Medium (frequent thoughts, previous attempts, low support): follow-up, contingency plan, crisis team?
- High (clear plan, major depression, violent method, regret it failed): crisis team, admission, Mental Health Act assessment.

Suicide

- About 1 million deaths a year; M:W 4:1 in the West; most were depressed.
- Factors: family history, mental illness, pain or terminal illness, alcohol or drugs (2/3), access to means (doctors), prisoners.
- Asking about suicide does not cause it. Clusters occur: do not publicise methods.

Source: VM 065 Self harm and suicide

PART 1: LECTURE

Retinal vascular diseases [VM 043]

Topic: Ophthalmology

Diabetic retinopathy

- Commonest cause of blindness in working-age adults. Microangiopathy: loss of pericytes, microaneurysms, capillary closure, ischaemia, leakage.
- Risk: duration, poor control, hypertension, renal disease.
- NPDR: microaneurysms, dot and blot haemorrhages, hard exudates.
- Macular oedema: commonest cause of visual loss in NPDR; any stage.
- Severe NPDR: cotton wool spots (nerve fibre layer infarcts), venous beading, IRMA.
- PDR: ischaemic retina releases VEGF; NVD/NVE. Advanced: vitreous haemorrhage, tractional detachment, neovascular glaucoma.
- Treat: glucose control; anti-VEGF for oedema; PRP for severe NPDR (high risk) and PDR (side effects: peripheral and night vision loss); vitrectomy for advanced disease.
- Silent until late: annual retinal check for all diabetics.

Hypertensive retinopathy

- Grade 1 barely detectable arterial narrowing; 2 obvious narrowing with focal irregularity; 3 = 2 + haemorrhages +/- exudates; 4 = 3 + disc swelling.

Retinal vascular occlusion

- Both: sudden painless loss of vision; check BP, lipids, glucose, FBC, ESR/CRP, carotid Doppler, ECG, echo.
- Artery: amaurosis fugax first; CRAO = poor acuity + RAPD; BRAO = field defect. Pale retina, cherry red spot (choroid supplies fovea), pale disc, thin vessels.
- Artery emergency: lower IOP (massage, mannitol, paracentesis); nitrates or 5% CO2.
- Vein: CRVO = poor acuity + RAPD; flame haemorrhages, cotton wool spots, disc swelling, macular oedema, tortuous veins. BRVO: field loss or none.
- Vein treatment: anti-VEGF for macular oedema; PRP for new vessels.

Age-related macular degeneration

- Drusen = lipid in Bruch's membrane. Risk: age, smoking, UV, family history.
- Dry: gradual central loss; drusen then geographic atrophy. Amsler grid, OCT.
- Wet: choroidal new vessels bleed or leak under the retina and scar; distortion, central scotoma.
- All: Amsler grid, stop smoking, sunglasses, vitamins C and E, zinc, beta-carotene. Wet: anti-VEGF, PDT.

Source: VM 043 Retinal vascular diseases

PART 1: LECTURE

Fundus examination skill lab [VM 043]

Topic: Ophthalmology

Preparation and technique

- Start with WIPER: wash hands, greet, introduce, permission (close approach, dark room, bright light), pain check, expose face and neck, patient seated.
- Inspect for glasses/contact lenses, cataract, pupil abnormality, red eye, discharge, lid asymmetry. Check visual acuity and colour vision before the fundus.
- Dim room; large aperture; focusing wheel at zero with index finger on it. Patient removes glasses (not contact lenses) and looks at a distant object.
- Red reflex from arm's length, one pupil at a time. Absent red reflex: cataract, retinoblastoma.
- Right eye = ophthalmoscope in right hand (and vice versa); other hand on forehead, thumb on eyebrow; move in closer.
- Order: disc (colour, contour, cup:disc) then vessels (tortuosity, dilatation) then periphery (look up, down, right, left) then fovea last (look at the light). Thank, explain, document.

Instruments

- Direct: portable, upright image, no dilatation needed. But small field, no stereopsis, spoilt by media opacity.
- Indirect: wide field, better stereopsis. But inverted image, needs dilatation, lower magnification.
- Slit-lamp biomicroscopy: wide field, better stereopsis; inverted image, needs dilatation and skill.

Optic disc

- Normal: vertically oval, pink, cup 30-50% of disc, well-defined contour.
- Papilloedema: passive non-inflammatory swelling from raised ICP, always bilateral. Pink, cup obliterated, blurred margins.
- Optic neuritis: primary inflammation (MS demyelination, TB, vitamin B deficiency). Same disc look: pink, obliterated cup, blurred margins.
- Optic atrophy (e.g. glaucoma): pale disc, increased cup, clear margins.

Vessels and macula

- Tortuous vessels = retinal vein occlusion; haemorrhages = diabetic retinopathy; aneurysms.
- Macula: yellow (xanthophyll), about 6 mm, 2 disc diameters from the disc. Fovea: depressed dark-red centre, 1 disc diameter (about 1.5 mm), cones only.
- Cherry-red spot = retinal artery occlusion; macular oedema = diabetic retinopathy; drusen = age-related macular degeneration.

Source: Fundus exam student notes; Fundus examination skill lab presentation [VM 043]

PART 1: LECTURE

Leprosy [VM 116]

Topic: Dermatology

Organism and spread

- Mycobacterium leprae: chronic granulomatous infection; obligate intracellular, acid-fast rod, not Gram-stainable. Targets macrophages and Schwann cells.
- Not cultured in vitro (mouse foot pad); reservoir is the nine-banded armadillo. Doubling time 12-14 days; incubation 1-20 years (average 5-10).
- Grows best in cool areas: superficial nerves, skin, upper airway mucosa, anterior chamber of the eye, testes.
- Droplet spread needing prolonged close contact with an untreated patient; about 95% naturally immune. Not spread by casual contact, intact skin, pregnancy or sex.

Clinical types (depend on cell-mediated immunity)

- WHO: paucibacillary = 1-5 lesions; multibacillary = more than 5.
- Tuberculoid: 1-2 sharply defined hypopigmented anaesthetic macules, under 10 cm, dry and hairless; face and extensor limbs. Thick, palpable, sometimes tender nerves; early nerve damage. High CMI, smear negative, not infectious.
- Lepromatous: many bilateral symmetrical, poorly defined lesions (macules to nodules), normal sensation. Leonine facies, madarosis, thick ear lobes, nasal collapse. Glove-and-stocking neuropathy (late). Eyes, testes, liver, kidneys, bones affected. Low CMI, many bacilli, infectious.
- Borderline tuberculoid: 5-20 larger lesions, satellites, asymmetrical, less anaesthesia. Mid-borderline: varied lesions, "Swiss cheese". Borderline lepromatous: widespread symmetrical, normal sensation and hair.

Diagnosis

- Skin smear from ear lobes, elbows, knees and lesions. Biopsy: full thickness from the edge of an active lesion (granulomas, giant cells, nerve infiltration). Lepromatous biopsy: many AFB, fat-laden macrophages, few lymphocytes.
- PCR is the most accurate and specific test.
- Lepromin test is not diagnostic; it shows the type: tuberculoid positive (more than 5 mm), lepromatous no reaction.

Treatment and prognosis

- WHO multidrug therapy prevents resistance; patient no longer infectious within 1-2 weeks.
- Paucibacillary: daily dapsone 100 mg + monthly rifampicin 600 mg, 6 months.
- Multibacillary: add clofazimine (50 mg daily, 300 mg monthly); 12-24 months.
- Side effects: haemolysis in G6PD deficiency, allergy, hepatitis, red urine, red skin, ichthyosis.
- Contacts: single-dose rifampicin cuts paucibacillary disease by 50%.
- Skin lesions resolve; nerve damage is permanent. Address stigma, isolation, anxiety and depression.

Source: Leprosy Updated [VM 116]

PART 1: LECTURE

Filariasis [VM 121]

Topic: Dermatology

Organisms and groups

- Nematodes; 4 serious human species: Wuchereria bancrofti, Brugia malayi, Onchocerca volvulus, Loa loa.
- Lymphatic: W. bancrofti, B. malayi, B. timori. Cutaneous: Loa loa, O. volvulus, M. streptocerca. Body cavity: M. perstans, M. ozzardi.
- Lymphatic filariasis (elephantiasis) is spread by mosquitoes (Aedes, Anopheles, Culex, Mansonia). WHO elimination programme uses mass drug administration.

Lymphatic filariasis

- Adults in lymph nodes and vessels (femoral, epitrochlear). Th2 response, raised IgE and IgG4; Wolbachia bacteria add to inflammation.
- Course: asymptomatic microfilaraemia, then acute adenolymphangitis, then chronic irreversible lymphoedema.
- Acute ADL: sudden fever with painful nodes (inguinal, testis, cord), skin peels within a week. Retrograde lymphangitis distinguishes it from bacterial.
- Filarial fever: fever without adenitis. TPE: dry paroxysmal nocturnal cough, wheeze, weight loss, eosinophilia, diffuse CXR infiltrates.
- Lymphoedema: I pitting, reversible; II not reversible, no skin change; III non-pitting, skin thickening; IV warty nodules (elephantiasis).

Diagnosis

- Night blood film 10 pm to 2 am (Loa loa: 10 am to 2 pm).
- Circulating antigen test (W. bancrofti only); antibodies; PCR mainly research.
- Ultrasound "filarial dance" sign; CXR infiltrates in TPE.

Treatment

- DEC is the drug of choice: 6 mg/kg for 12 days; doxycycline 200 mg/day for 6 weeks may be added.
- Lymphoedema: elevation, compression, bed rest, steroids; skin hygiene to prevent infection. Chyluria: low-fat, high-protein diet with MCT.
- Surgery for hydrocele and scrotal elephantiasis.

Onchocerciasis and loiasis

- Onchocerciasis (river blindness): black fly; 99% sub-Saharan Africa. Pruritus, nodules, leopard skin, corneal fibrosis and blindness.
- Diagnosis: skin snips (Africa: gluteal and thigh); slit lamp; Mazzotti test with DEC. Treat with ivermectin 150 mcg/kg every 3 months; moxidectin (age 12+).
- Loiasis: Calabar swellings; worm under the conjunctiva. Daytime blood film.
- Before DEC, count microfilariae (encephalopathy risk). Below 2500/mL: DEC 8-10 mg/kg/day for 21 days. If higher, treat only if symptomatic, after albendazole or apheresis.

Source: Filariasis [VM 121]

PART 1: LECTURE

Poisoning lecture [VM 030]

Topic: Poisoning and toxicology

First steps

- ABC: airway with C-spine control and gag check; O2/ventilation; IV access, bloods, cardiac monitor. If hypotensive with clear lungs: saline or Ringer bolus.
- Empiric: dextrose, oxygen, naloxone (titrated for opioid respiratory depression), thiamine (Wernicke).
- Opioid triad: coma, respiratory depression, pinpoint pupils; look for needle tracks.
- Anticholinergic: delirium, fever, tachycardia, big pupils, dry flushed skin, retention. Sympathomimetic: same but sweating. Cholinergic: secretions, fasciculations.

Antidotes

- Paracetamol: NAC. OP: atropine and pralidoxime. Iron: deferoxamine. Digoxin: Fab. Methanol: ethanol or fomepizole. Benzodiazepine: flumazenil. Opioid: naloxone.
- Methaemoglobin: methylene blue. CO: hyperbaric O2. Cyanide: cyanide kit. TCA, cocaine, salicylate: sodium bicarbonate. Warfarin: PCC, vitamin K. CCB: calcium; beta-blocker and CCB: glucagon.

Decontamination and elimination

- No induced emesis (never salt water). Lavage not routine; may help late for concretions or slow gastric emptying (salicylate, TCA).
- Activated charcoal 50 g, best within 1 h. Not for metals, alcohols, hydrocarbons (aspiration), corrosives, bowel obstruction.
- Whole bowel irrigation (PEG): large iron, slow-release or enteric-coated tablets, packets. Not with ileus, coma or fits unless airway protected.
- Multiple-dose charcoal: theophylline, salicylate, carbamazepine, dapsone, digoxin, TCA.
- Urinary alkalinisation (urine pH 7-8): salicylates, long-acting barbiturates, haemolysis or rhabdomyolysis. Needs normal kidneys; watch K, lung oedema.
- Dialysable drugs: small Vd, low protein binding, small, water-soluble (alcohols, phenobarbital, theophylline, lithium, salicylate).

Key overdoses

- Paracetamol: toxic at 150-250 mg/kg. Often no symptoms at first; liver damage peaks at 72-96 h. Check level on the nomogram and AST/ALT. NAC (glutathione) within 8 h is nearly 100% protective. Risks: alcohol, malnutrition, enzyme inducers.
- TCA: anticholinergic signs, seizures, hypotension, QRS over 100 ms. Give bicarbonate (first line); MDAC.
- Salicylate: tinnitus, vomiting, hyperventilation, fluid loss. Respiratory alkalosis first, then metabolic acidosis. Alkalinise urine (level over 35 mg/dL). Dialyse if level over 100 mg/dL or lung injury, CNS signs, renal failure, refractory acidosis.
- Digoxin: GI symptoms first; bradycardia, AV block, arrhythmias. High K predicts need for Fab. Fab if K over 5, level over 10 ng/mL at 6 h, 10 mg or more taken, or dangerous arrhythmia. Watch for low K after Fab.
- Body packers: if no leak, give lactulose and observe; leak or obstruction may need surgery.

Source: Basics of drug poisoning lecture [VM 030]

PART 1: LECTURE

Poisoning LGW and cases [VM 030, VM 050]

Topic: Poisoning and toxicology

General approach

- High anion gap: MUDPILES (methanol, uraemia, DKA, paraldehyde, iron, lactic acidosis, ethylene glycol, salicylates).
- Coma cocktail: dextrose, naloxone, thiamine. Then GI decontamination, antidotes, elimination, symptom control.

Scorpion and snake

- Scorpion venom causes an autonomic storm (catecholamines plus acetylcholine). Look for high BP, sweating, tremor, arrhythmia, pulmonary oedema, high K, high glucose, acidosis.
- Observe 6 h; admit if systemic signs. Antivenom best within 4 h (up to 24 h): 3-5 ampoules IV/IM, repeat every 30 min. Same dose for children and adults; skin test first.
- Viper (desert): severe swelling, blisters, necrosis. Causes coagulopathy, DIC, haemolysis, myolysis, renal failure and high K. Low platelets and fibrinogen.
- Cobra (Nile Valley): minimal local signs. Fasciculations, then ptosis, diplopia, dysarthria and respiratory paralysis; conscious, sensation intact.
- First aid: rest, immobilise, remove rings. No incision, suction, ice or shocks. Polyvalent antivenom 3-5 vials for any systemic envenomation, never too late. Tetanus cover.

Organophosphates

- OP inhibits AChE, irreversibly after 24-36 h. Carbamates are reversible and rarely fatal.
- Muscarinic DUMBELS; nicotinic MATCH (fasciculations, tachycardia, hypertension); central confusion, convulsions, coma.
- Ix: red-cell (true) and plasma (pseudo) cholinesterase, ABG, ECG, CXR.
- Atropine (muscarinic only): titrate to dry chest and relieved bronchospasm, not heart rate or pupils. Keep atropinised 1-2 days.
- Atropine toxicity: dry hot skin, fever, big fixed pupils, delirium, tachycardia; cool and give fluids and benzodiazepines.
- Oximes reactivate AChE before aging (first 24-48 h); atropine first.
- Intermediate syndrome (2-3 days): neck flexor, proximal and respiratory weakness from muscle necrosis. Delayed neuropathy (2-3 weeks): permanent.

CO and cyanide

- CO binds Hb 200-250x more than O2 and shifts the curve left. Also binds myoglobin and cytochrome oxidase.
- Throbbing headache, coma, MI, pulmonary oedema; cherry-red skin rare. Delayed neuropsychiatric effects after 2-40 days.
- 100% O2: half-life 5 h to 1 h. Hyperbaric O2 (20 min) if confused, COHb over 25%, or fetal distress. Consider cyanide in fire smoke.
- Cyanide (bitter almonds) blocks cytochrome oxidase: red asphyxia, no cyanosis, venous O2 equals arterial, lactic acidosis.
- Cyanide kit: nitrite makes MetHb to bind CN; thiosulfate forms thiocyanate; hydroxocobalamin forms cyanocobalamin.

Source: LGW Poisoning; Poisoning LGW cases; Case 1-3 student notes and handouts [VM 030]

PART 1: LECTURE

Clinical therapeutics: cardiology [VM 110]

Topic: Use of medicine

Antihypertensives

- Dihydropyridines (amlodipine, nifedipine) act on vessels: hypertension, angina incl. vasospastic, Raynaud; AEs oedema, flushing, gingival hyperplasia. Nimodipine = subarachnoid haemorrhage (prevents vasospasm). Verapamil/diltiazem act on heart: AV block, constipation.
- ACE inhibitors: lower angiotensin II, raise bradykinin, lose efferent constriction (lower GFR). Use: hypertension, HF (lower mortality), diabetic proteinuria. AEs: cough, angioedema, hyperkalaemia, teratogen, rise in creatinine; caution in bilateral renal artery stenosis.
- ARBs (losartan): same benefits, no bradykinin rise; used if ACEi cough. Aliskiren = direct renin inhibitor; avoid in pregnancy.
- Hydralazine: arteriolar dilator, safe in pregnancy; reflex tachycardia, drug-induced lupus. Nitroprusside: cyanide toxicity.

Blood thinners

- Heparin: activates antithrombin (IIa, Xa); monitor PTT; reverse with protamine. HIT type 2 at 5-10 days: stop heparin, give argatroban (fondaparinux safe).
- Warfarin: blocks vitamin K epoxide reductase (II, VII, IX, X, C, S); monitor PT/INR; crosses placenta. Early protein C fall = skin necrosis, so bridge with heparin. Reverse: vitamin K; rapid = FFP/PCC.
- Dabigatran (thrombin) reversed by idarucizumab; apixaban/rivaroxaban (Xa) by andexanet alfa.
- Antiplatelets: aspirin (COX, TXA2), clopidogrel/ticagrelor (P2Y12), eptifibatide (GpIIb/IIIa), cilostazol (PDE, claudication).

Diuretics

- Loop (furosemide): ototoxicity, low K and Mg, alkalosis, gout; lose calcium. Thiazides: lower urinary calcium; hyperglycaemia, lipids, uric acid, calcium. Spironolactone: hyperkalaemia, gynaecomastia. Mannitol: raised ICP/IOP; avoid in HF, anuria.

Angina, lipids, heart failure

- Nitrates: veins > arteries, lower preload; headache, reflex tachycardia; contraindicated with PDE-5 inhibitors, RV infarct, HCM. Ranolazine: late Na current, refractory angina.
- Statins: myopathy (worse with fibrates), raised LFTs. Resins: GI upset. PCSK9 inhibitors raise LDL removal.
- Sacubitril/valsartan for HFrEF; never with ACEi (angioedema). SGLT2i: UTI, candida, euglycaemic DKA.
- Digoxin: Na/K ATPase block; toxicity = GI upset, yellow vision, arrhythmias; worsened by low K, high Ca, low Mg, renal failure, verapamil/amiodarone. Antidote: Fab fragments.

Antiarrhythmics

- IA: QT, lupus (procainamide). IC (flecainide): contraindicated post-MI/structural disease. II: beta-blockers. III: amiodarone (check PFT, LFT, TFT). IV: verapamil/diltiazem. Adenosine = SVT drug of choice. Magnesium = torsades, digoxin toxicity. Ivabradine = funny channel; stable angina, HFrEF.

Source: Clinical Therapeutics Cardiology [VM 110] (OCR text)

PART 1: LECTURE

Antiplatelet and anticoagulant therapy [VM 034]

Topic: Use of medicine

Categories and heparins

- Primary prophylaxis = prevent first clot (low-dose heparin after surgery). Therapeutic = treat established DVT to stop propagation/embolism. Secondary prophylaxis = prevent recurrence (warfarin after VTE).
- Heparin accelerates antithrombin against Xa and IIa. Monitor UFH with APTT; LMWH with anti-Xa assay (usually not needed).
- LMWH: longer half-life, once-daily SC, less bleeding, simple dosing, outpatient DVT treatment by weight, HIT very rare.
- UFH preferred: high bleeding risk (renal impairment), need for rapid reversal, massive PE after thrombolysis, LMWH failure. Overdose: stop infusion (half-life 1 h); urgent reversal = protamine.
- Complications: bleeding, HIT, osteoporosis with long use.

Warfarin

- Oral, long half-life; full effect after 3-5 days, so continue heparin until then. Teratogenic. Many drug interactions (antibiotics, fluconazole, aspirin).
- Monitor PT/INR. First DVT/PE: target 2.5 (2-3). Recurrence while therapeutic: target 3.5 (3-4).
- Bleeding risk rises sharply when INR > 5; intracranial bleed most serious.
- INR 6-8, no/minor bleed: stop 24-48 h, recheck, restart lower. INR > 8: stop 48 h, consider vitamin K (oral 2.5 mg or IV 1-2 mg; IV faster, takes at least 6 h).
- Life/sight/limb-threatening bleed: stop warfarin, give PCC (Beriplex) + IV vitamin K. FFP not recommended (incomplete, slow, large volume).

Contraindications and other options

- Absolute: bleeding diathesis, severe hypertension, previous cerebral haemorrhage, CNS trauma, GI bleeding, heparin allergy/HIT. Relative: uraemia, liver disease, alcoholism, SBE. Breastfeeding is not a contraindication.
- IVC filter: anticoagulation contraindicated, recurrence despite adequate anticoagulation, serious complications, free-floating iliofemoral/caval thrombus, very poor lung reserve.
- Thrombolysis: massive PE, phlegmasia cerulea dolens (purple, swollen, painful leg; catheter-directed thrombolysis or thrombectomy).

NOACs and antiplatelets

- Dabigatran (direct thrombin inhibitor); rivaroxaban, apixaban (direct Xa). Advantages: wide therapeutic index, fast onset, predictable, no monitoring. Disadvantages (per lecture): no specific antidote, GI bleeding.
- Fondaparinux: SC pentasaccharide, antithrombin-mediated Xa inhibition only, no thrombin effect; not a NOAC.
- Aspirin 75-81 mg: irreversible COX inhibitor; GI bleeding. Others: ADP receptor and GP IIb/IIIa inhibitors. PDE inhibitors (cilostazol, dipyridamole): raise cAMP, antiplatelet + vasodilator, preferred in PAD.

Source: Antiplatelets and Anticoagulant Therapy [VM 034]; Anticoagulant QA case review

PART 1: LECTURE

Clinical therapeutics: chest [VM 111]

Topic: Use of medicine

Beta-lactams

- Never combine two beta-lactams for the same infection.
- Natural penicillin G/V: Gram-positives (S. pneumoniae, enterococci); penicillin G for neurosyphilis. Anti-staph (nafcillin, oxacillin): MSSA. Amoxicillin/ampicillin: not antipseudomonal. Piperacillin(/tazobactam): Pseudomonas.
- MSSA cover: anti-staph penicillins, BL/BLI combinations, 1st-gen cephalosporin (cefazolin).
- Cephalosporins: 2nd (cefuroxime) Gram-negative, no Pseudomonas; 3rd (ceftriaxone) cross BBB, only ceftazidime covers Pseudomonas; 4th (cefepime) Pseudomonas + MSSA; 5th (ceftaroline) MRSA, not Pseudomonas.
- Carbapenems (imipenem, meropenem, ertapenem): ESBL Gram-negatives, anaerobes, Pseudomonas; poor MRSA. Aztreonam: Gram-negatives incl. Pseudomonas; inhaled in cystic fibrosis.

Other antibiotics and toxicity

- Aminoglycosides (gentamicin, amikacin): Gram-negatives incl. Pseudomonas; streptomycin/amikacin/kanamycin in MDR-TB. Toxic to kidney + ear, neuromuscular block; check audiometry.
- Macrolides (50S): atypicals (Mycoplasma, Chlamydia, Legionella), NTM, H. pylori; QT prolongation.
- Fluoroquinolones (cipro, levo, moxi): QT + tendon. Levofloxacin 750 mg/day antipseudomonal.
- Vancomycin: MRSA, C. difficile, enterococci; no Gram-negatives; kidney + ear, red man syndrome. MRSA: vancomycin, ceftaroline.

Anti-TB (RIPE + S)

- Rifampicin: RNA polymerase; hepatotoxic, enzyme inducer, orange body fluids. Isoniazid: mycolic acid; neuropathy, add B6. Pyrazinamide: hepatotoxic, high uric acid. Ethambutol: optic neuritis, red-green loss. Streptomycin: 30S; ototoxic.

Bronchodilators and anti-inflammatories

- Beta-2 agonists raise cAMP: SABA salbutamol (rescue), LABA formoterol (fast onset), indacaterol/vilanterol 24 h. AEs: tachycardia, tremor, hypokalaemia, tolerance.
- Antimuscarinics (inhaled only): ipratropium (short), tiotropium/umeclidinium (24 h); dry mouth, bitter taste.
- Methylxanthines (oral/IV): PDE block; aminophylline rescue, theophylline maintenance; seizures, arrhythmia, gastritis.
- IV magnesium sulphate 2 g: adjunct in acute severe asthma.
- ICS: oral candidiasis, hoarseness, pneumonia at high dose. Systemic steroids: exacerbations only; infection, osteoporosis, hyperglycaemia, adrenal suppression.
- Roflumilast (oral PDE-4): COPD FEV1 < 50% with exacerbations despite ICS/LABA/LAMA. Azithromycin 250 mg/day long-term: anti-inflammatory. Montelukast: asthma incl. aspirin/exercise-induced.

Inhalers

- MDI (Ventolin) needs coordination; DPI (budesonide) needs inspiratory flow; SMI (Spiriva). 60% use inhalers wrongly. Nebuliser: no coordination, emergencies.

Source: Chest therapeutics [111]; Introduction to Chest Therapeutics [111]

PART 1: LECTURE

Drugs for anaemia [VM 112]

Topic: Use of medicine

Oral iron

- Causes: blood loss (menorrhagia, hookworm, colon cancer), demand (pregnancy, infancy), poor intake or absorption (gastrectomy, coeliac).
- Dose ~200 mg elemental iron daily in 2-3 divided doses for 4-6 months. Absorbed as ferrous form in duodenum; ~25% of ferrous salt absorbed.
- Take 1 h before meals with vitamin C/citrus. Reduced by tea (tannins), phytates, coffee, antacids, milk, calcium, zinc.
- AEs: GI upset, constipation or diarrhoea, metallic taste, teeth staining, dark stools.
- Ferrous sulphate = most common, cheap; ferric salts less bioavailable.

Parenteral iron and toxicity

- Oral is as effective if absorption is normal. Give parenteral if: oral intolerance, haemodialysis on EPO, post-gastrectomy, small bowel resection, proximal IBD, malabsorption, chronic anaemia not maintained orally.
- Iron dextran (IV/deep IM), ferric gluconate, iron sucrose. Anaphylaxis (mainly high-MW dextran): give test dose. Monitor stores; overload treated with deferoxamine.
- Acute poisoning in young children (10 tablets can kill): bloody vomiting/diarrhoea, shock, then metabolic acidosis. Treat: whole bowel irrigation + IV deferoxamine; charcoal does not bind iron.

Folate and vitamin B12

- Folate deficiency: poor diet (alcohol), malabsorption, phenytoin. Also for methotrexate toxicity, haemolytic anaemias.
- Dose 1-5 mg oral daily; pregnancy prophylaxis 400 micrograms daily (neural tube defects). Well absorbed orally; non-toxic.
- B12: hydroxocobalamin preferred (rapid, longer levels). Most deficiency = malabsorption, so IM/deep SC: 100-1000 mcg every 1-2 days for 1-2 weeks, then every 2-3 months for life (every 1-2 weeks for 6 months if neurological signs). Non-toxic.
- Never give folate alone in megaloblastic anaemia: it corrects blood but masks B12 deficiency, letting nerve damage progress.

ESAs and new drugs

- EPO from kidney; low in renal disease, so these patients respond best. Uses: CKD anaemia, AIDS/zidovudine, chemotherapy, before elective surgery or autologous donation, premature infants.
- IV in dialysis, SC otherwise. Target Hb up to but not above 10-12 g/dL; rise no more than 1 g/dL per 2 weeks. Reticulocytes rise ~10 days, Hb 2-6 weeks: no use in acute anaemia. Most need iron.
- Epoetin three times weekly; darbepoetin weekly.
- AEs: hypertension, arthralgia; thrombosis and death if Hb target > 11-12 g/dL.
- Daprodustat: oral HIF-PH inhibitor, raises EPO, lowers hepcidin; dialysis for 4+ months; Hb target not > 11 g/dL.

Source: Drugs for Anaemia with notes [VM 112]

PART 1: LECTURE

Biochemical assay of cholesterol [VM 029]

Topic: Biochemistry

Lipid profile

- Group of tests to assess risk of coronary heart disease, heart attack, hypertension and stroke from atherosclerosis.
- Standard profile: total cholesterol, HDL-C ('good'), LDL-C ('bad'), triglycerides; extended profile adds VLDL-C.
- Sample: serum or plasma after a 12-hour fast (water only).
- After a fatty meal serum is turbid (lipaemic) from chylomicrons; lipoprotein lipase ('clearing factor') clears it within hours.

Lipids and reference values

- Triglycerides: glycerol + 3 fatty acids; main dietary lipid; carried from gut in chylomicrons; excess stored in adipose tissue. Fasting TG < 150 mg/dL.
- Cholesterol: from diet and de novo synthesis; cell membranes, steroid hormones, vitamin D, bile acids. Desirable total < 200 mg/dL.
- Lipoproteins (chylomicrons, VLDL, LDL, HDL) carry hydrophobic lipids in blood.
- LDL carries cholesterol from liver to organs; oxidised LDL deposits in arteries; main drug target. Optimal < 100 mg/dL.
- HDL carries cholesterol from tissues to liver; protective. Low = < 40 mg/dL (men), < 50 mg/dL (women).

Assay principle (enzymatic colorimetry)

- Reagent: cholesterol esterase, cholesterol oxidase, peroxidase, phenol, 4-aminoantipyrine.
- Esterase: cholesterol esters to free cholesterol + fatty acids. Oxidase: cholesterol to H2O2. Peroxidase: H2O2 + phenol + 4-aminoantipyrine to red quinoneimine dye.
- Colour depth is proportional to concentration (more molecules absorb more light). Visible light 380-750 nm; UV < 380; IR > 750.

Practical steps and calculation

- Spectrophotometer = spectrometer (selects wavelength) + photometer (measures intensity). Warm up 15 min.
- Tubes: blank (10 microlitres water), standard (10 microlitres standard), test (10 microlitres serum), each + 1000 microlitres reagent; mix, 5 min at room temperature; read at 546 nm.
- Blank: identical except no analyte; sets zero and removes interference. Wipe cuvette, zero with blank, then read test and standard.
- Concentration (mg/dL) = (A test / A standard) x concentration of standard.
- Checklist: hand gel, gloves, check name/age on sample, check kit expiry. Micropipette: set volume, fit tip, press to first stop, immerse, release; check for bubbles; expel by pressing to second stop.

Source: Biochemical assay of cholesterol [VM 029]

PART 2: TOPIC SUMMARY

Pathology

Blood flow disorders

- Virchow's triad: endothelial injury, stasis or turbulence, hypercoagulability. Stasis = commonest cause of venous thrombosis. Factor V Leiden: young patient with recurrent venous clots.
- Lines of Zahn = thrombus formed in flowing blood during life. Venous thrombi occlude and embolise; arterial thrombi form on ruptured plaques.
- Infarct = coagulative necrosis (brain: liquefactive). Pale wedge: heart, spleen, kidney. Red: lung, dual supply, venous occlusion.
- Nutmeg liver = right heart failure. Dry gangrene: artery only, clear line; moist: artery and vein, no line.

Atherosclerosis, aneurysm, MI

- Fatty streak (earliest, reversible) then fibrous plaque; foam cells = macrophages full of oxidised LDL. Commonest site: lower abdominal aorta.
- AAA: below renal arteries; operate if over 5 cm or enlarging. Berry: circle of Willis, SAH. Charcot-Bouchard: hypertension. Dissection: hypertension, Marfan, not atherosclerosis.

MI age	Finding
1-3 days	Coagulative necrosis, neutrophils
3-7 days	Yellow centre, red rim; rupture risk 2-4 days
10-14 days	Granulation tissue
Over 2 months	Dense scar; late ventricular aneurysm

Heart muscle and hypertension

- Dilated cardiomyopathy (commonest): systolic failure; genetic, alcohol, anthracyclines, past myocarditis.
- Hypertrophic: asymmetric septum, myocyte disarray; sudden death in young people on exercise. Restrictive: amyloid (Congo red, apple-green birefringence). Myocarditis: Coxsackie, T-lymphocytes.
- Benign hypertension: hyaline arteriolosclerosis, small granular kidneys. Malignant: fibrinoid necrosis, onion-skin arterioles, flea-bitten kidneys.

Lung and TB

- Centriacinar emphysema: smokers, upper lobes. Panacinar: alpha-1 antitrypsin, lower lobes. Asthma: Curschmann spirals, Charcot-Leyden crystals. Silica and coal: upper lobes; asbestosis: lower lobes, asbestos bodies. Sarcoid: non-caseating granulomas.
- Lobar pneumonia: congestion, red then grey hepatization, resolution. Abscess: aspirated mouth anaerobes. Aspergilloma: fungus ball in an old cavity.
- TB granuloma: caseation, epithelioid cells, Langhans giant cells. Primary = Ghon complex (subpleural focus + hilar nodes); secondary = apical cavity; miliary foci under 2 mm.

Lung tumours

Type	Clue
Squamous	Central, smoker, keratin pearls, high calcium
Adenocarcinoma	Peripheral, women and non-smokers, TTF-1 positive
Small cell	Central, smoker, SIADH, ACTH; chemotherapy, no surgery
Mesothelioma	Asbestos, calretinin positive, TTF-1 negative

PART 2: TOPIC SUMMARY

Radiology

Chest X-ray basics

- Check technique: AP film and poor inspiration make the heart look big. Adequate film: over 6 anterior or 10 posterior ribs above the diaphragm.
- Silhouette sign: right heart border lost = right middle lobe; left heart border lost = lingula.
- Cardiomegaly: CTR over 0.5. Mediastinum over 8 cm: think dissection.
- Heart failure ABCDE: alveolar (bat wing) oedema, Kerley B lines, cardiomegaly, dilated upper lobe vessels, effusion.
- Hidden areas: apices, behind the heart, under the diaphragm (free air).

Finding	Key sign
Consolidation	Air bronchogram
Collapse	Opacity, volume loss, shift towards; left lower lobe = sail behind heart
Effusion	Blunt costophrenic angle, meniscus
Pneumothorax	Lung edge, no markings beyond; tension = shift away
COPD	Flat diaphragms, wide rib spaces, markings still present

TB imaging

- Primary: consolidation, hilar nodes, effusion, Ghon complex.
- Post-primary: upper zone consolidation, cavity, fibrosis with volume loss; healed = apical fibrosis and calcification.
- Miliary: countless 1-3 mm nodules. Flask-shaped heart = pericardial effusion. Plombage balls = old collapse treatment.

Arterial imaging

- Doppler ultrasound: no radiation. CTA: fast, but X-rays and iodinated contrast. MRA: no radiation. Catheter angiography: invasive.
- Good collaterals with good run-off = chronic occlusion; poor collaterals = acute.
- SFA occlusion: profunda femoris links to genicular branches; treat by angioplasty and stent.
- Fusiform = spindle all round; saccular = one-sided sac.

Venous imaging (DVT)

- Duplex ultrasound = test of choice. Key sign: vein does not compress.
- Acute clot: wide vein, dark thrombus. Chronic: narrow vein, bright thrombus.
- Venography: old gold standard (filling defect). CT venography for iliac veins and IVC; the slide also calls it the choice for the post-phlebitic limb.

Pulmonary embolism

- CTPA first: filling defect in a pulmonary artery; negative CTPA ends the work-up. Pregnancy test first; not needed if Wells unlikely and D-dimer negative.
- V/Q: under 40, contrast allergy, renal failure; normal scan excludes PE; hard to read if CXR abnormal.
- CXR often normal (Hampton's hump). Echo: right heart strain in massive PE. Leg Doppler if high probability but CTPA negative.
- IVC filter in massive DVT to prevent PE.

PART 2: TOPIC SUMMARY

Mental health

Disturbed behaviour

- Rule out organic causes first: delirium, pain, intoxication, withdrawal. Delirium tremens: confusion, hallucinations, 5-10% mortality.
- Warning signs: angry face, restlessness, loud speech, threats, blocking exits.
- Steps: de-escalation, time out (max 15 min), rapid tranquilisation, restraint, seclusion (last resort, review every 2 h).
- Rapid tranquilisation: offer oral first. Lorazepam 1-2 mg, plus haloperidol 5-10 mg if psychotic. Observations every 5-10 min for 1 h; ECG with haloperidol.

Side effect	Action
Acute dystonia	Procyclidine
Akathisia	Reduce drug, propranolol
NMS (fever, rigidity, high CK)	Stop drug, dantrolene, bromocriptine
Benzodiazepine respiratory depression	Oxygen, flumazenil

Depression, anxiety, OCD, stress

- Depression after a cardiac event: sertraline, avoid tricyclics.
- Panic and generalised anxiety: exclude physical causes; CBT, then SSRI; avoid long-term benzodiazepines.
- OCD: intrusive thoughts known to be one's own (unlike psychosis); CBT with exposure and response prevention, SSRI, clomipramine if SSRI fails.
- PTSD: intrusion, avoidance, negative mood, arousal for over 1 month; CBT, EMDR, SSRIs. Acute stress reaction settles within days.

Drugs and alcohol

- Opioid overdose: pinpoint pupils, slow breathing; naloxone is short-acting, so may re-sedate.
- MDMA: hyperthermia, low sodium. Cocaine: MI, stroke, fits.
- Alcohol withdrawal: chlordiazepoxide reducing regimen, thiamine. Wernicke: confusion, ataxia, eye signs; IV thiamine. Korsakoff: loss of new memory.

Self-harm and suicide

- Over 90% have a psychiatric disorder. Self-harm raises suicide risk 50-100 times, highest within 6 months.
- Intent: planning, note, precautions against discovery. Physical severity does not show intent. Risk scales not recommended.
- Asking about suicide does not cause it. Lithium and clozapine reduce suicide.

Egyptian Mental Health Act (2009)

- Capacity: understand, retain, weigh, communicate; presumed; an unwise choice is not incapacity.
- Involuntary admission: psychiatrist approval, prosecutor told within 24 h; non-psychiatrist may hold 48 h; over 1 week needs two psychiatrists. Emergency treatment without consent up to 72 h. ECT only under anaesthesia with written consent.

Other

- Critically ill: ABCDE, treat as found, reassess. Learning disability: IQ under 70 plus adaptive deficit from childhood. Beware diagnostic overshadowing; look for pain. Down's: early dementia. Fragile X: commonest inherited cause in boys.

PART 2: TOPIC SUMMARY

Ophthalmology

Diabetic retinopathy

- Commonest cause of blindness in working-age adults; silent until late, so annual retinal check.

Stage	Signs	Treatment
Non-proliferative	Microaneurysms, dot and blot haemorrhages, hard exudates	Glucose and BP control
Severe NPDR	Cotton wool spots, venous beading, IRMA	PRP if high risk
Proliferative	New vessels (VEGF); vitreous haemorrhage, detachment	PRP, vitrectomy
Macular oedema	Any stage; commonest cause of vision loss	Anti-VEGF

- PRP costs peripheral and night vision.

Hypertensive retinopathy

- Grade 1 slight arterial narrowing; 2 obvious narrowing; 3 plus haemorrhages or exudates; 4 plus disc swelling.

Retinal vessel occlusion (sudden painless loss of vision)

- Artery: pale retina, cherry red spot, thin vessels; often amaurosis fugax first. Emergency: lower eye pressure (massage, mannitol, paracentesis).
- Vein: tortuous veins, flame haemorrhages, cotton wool spots, disc swelling. Anti-VEGF for oedema, PRP for new vessels.
- Work-up: BP, lipids, glucose, ESR/CRP, carotid Doppler, ECG, echo.

Macular degeneration

- Drusen. Dry: slow central loss. Wet: new vessels bleed, distortion; anti-VEGF. All: Amsler grid, stop smoking.

Fundus examination

- Right eye: scope in right hand, your right eye. Red reflex first; absent = cataract or retinoblastoma.
- Order: disc, vessels, periphery, fovea last.
- Direct scope: upright image, no dilatation, small field. Indirect: wide field, inverted image, needs dilatation.

Disc	Look
Normal	Pink, oval, cup 30-50%
Papilloedema	Pink, blurred, no cup; bilateral, raised ICP
Optic neuritis	Same look; inflammatory (MS)
Optic atrophy	Pale, large cup, clear edge

PART 2: TOPIC SUMMARY

Dermatology

Leprosy

- Mycobacterium leprae: acid-fast, cannot be cultured; infects skin and nerves in cool areas. Droplet spread after long close contact; about 95% are immune.
- Type depends on cell-mediated immunity.

	Tuberculoid	Lepromatous
Lesions	1-2, sharp edge, hypopigmented	Many, symmetrical, poorly defined
Sensation	Lost	Normal (late glove-and-stocking)
Nerves	Thick, damaged early	Late
Bacilli	Few, not infectious	Many, infectious
Lepromin test	Positive	Negative

- Lepromatous extras: leonine face, lost eyebrows, thick ear lobes.
- WHO: paucibacillary 1-5 lesions, multibacillary over 5.
- Diagnosis: skin smear, biopsy from active edge; PCR most accurate. Lepromin shows type, not diagnosis.
- Treatment: paucibacillary = dapsone daily + rifampicin monthly for 6 months. Multibacillary = add clofazimine, 12-24 months. Not infectious after 1-2 weeks. Nerve damage is permanent.
- Dapsone: haemolysis in G6PD deficiency. Clofazimine: red skin.

Filariasis

- Lymphatic (W. bancrofti, Brugia): mosquito; fever with painful nodes and retrograde lymphangitis, then chronic lymphoedema up to elephantiasis.
- Tropical pulmonary eosinophilia: night cough, wheeze, eosinophilia.
- Diagnosis: night blood film (10 pm to 2 am); antigen test; ultrasound "filarial dance".
- Treatment: DEC 6 mg/kg for 12 days, with or without doxycycline. Lymphoedema: elevation, compression, skin hygiene. Surgery for hydrocele.
- Onchocerciasis: black fly, itchy nodules, leopard skin, river blindness; skin snips; ivermectin.
- Loa loa: Calabar swellings, eye worm, daytime film. Count microfilariae before DEC (encephalopathy risk).

PART 2: TOPIC SUMMARY

Poisoning and toxicology

First steps

- ABC, then coma cocktail: dextrose, naloxone, thiamine. No induced vomiting.
- Charcoal 50 g, best within 1 h; not for metals, alcohols, hydrocarbons, corrosives.
- Whole bowel irrigation: iron, slow-release tablets, packets. Urine alkalinisation: salicylate. High anion gap: MUDPILES.

Toxidrome	Clues
Opioid	Coma, slow breathing, pinpoint pupils
Anticholinergic	Delirium, fever, big pupils, dry flushed skin, retention
Sympathomimetic	Same, but sweating
Cholinergic	Secretions (DUMBELS), fasciculations

Antidotes

- Paracetamol: NAC. Opioid: naloxone. Benzodiazepine: flumazenil. Organophosphate: atropine + pralidoxime. Iron: deferoxamine. Digoxin: Fab. Methanol: fomepizole or ethanol. TCA and salicylate: bicarbonate. Methaemoglobin: methylene blue. Beta-blocker or CCB: glucagon. Warfarin: PCC, vitamin K.

Key overdoses

- Paracetamol: toxic at 150-250 mg/kg; few early symptoms, liver damage peaks at 72-96 h. Use the nomogram; NAC within 8 h is nearly 100% protective.
- TCA: QRS over 100 ms, fits, low BP: bicarbonate first.
- Salicylate: tinnitus, hyperventilation; respiratory alkalosis then metabolic acidosis. Dialyse if over 100 mg/dL.
- Digoxin: vomiting, bradycardia, heart block; high K predicts need for Fab.

Organophosphates

- Block cholinesterase: muscarinic DUMBELS, nicotinic fasciculations and weakness, confusion, fits.
- Atropine until the chest is dry (not by heart rate or pupils); oxime early, before aging.
- Intermediate syndrome at 2-3 days (neck and breathing weakness); delayed neuropathy at 2-3 weeks.

Bites and stings

- Scorpion: autonomic storm (high BP, sweating, lung oedema); antivenom best within 4 h, same dose for children.
- Viper: swelling, blisters, bleeding (DIC). Cobra: little local sign, ptosis then breathing paralysis.
- No incision, suction or ice. Antivenom is never too late.

CO and cyanide

- CO: headache, coma; 100% O2 cuts half-life from 5 h to 1 h; hyperbaric if confused, COHb over 25% or fetal distress. Late neuropsychiatric effects.
- Cyanide: bitter almonds, no cyanosis, lactic acidosis; hydroxocobalamin, nitrite, thiosulfate.

PART 2: TOPIC SUMMARY

Use of medicine

Cardiovascular drugs

- ACE inhibitors: cough, angioedema, high K, teratogenic; ARB if cough. Never ACEi with sacubitril/valsartan.
- Dihydropyridines: ankle oedema, flushing. Verapamil/diltiazem: heart block, constipation. Hydralazine: safe in pregnancy.
- Nitrates: never with PDE-5 inhibitors. Loop: low K, ototoxic. Thiazide: high glucose, urate, calcium. Spironolactone: high K, gynaecomastia.
- Digoxin toxicity: GI upset, yellow vision, arrhythmia; worse with low K; antidote Fab.
- Amiodarone: check lung, liver, thyroid. Adenosine for SVT; magnesium for torsades. Flecainide: not after MI.

Anticoagulants

Drug	Target	Monitor	Reversal
Unfractionated heparin	Antithrombin (IIa, Xa)	APTT	Protamine
Warfarin	Vitamin K factors	INR 2-3	Vitamin K; PCC if serious bleed
Dabigatran	Thrombin	None	Idarucizumab
Apixaban, rivaroxaban	Xa	None	Andexanet alfa

- One slide says NOACs have no specific antidote; the cardiology slides list the two above.
- LMWH: once daily, less bleeding, HIT rare. UFH preferred in renal failure or when fast reversal is needed.
- Warfarin takes 3-5 days: overlap with heparin. Target INR 2.5; 3.5 if recurrence on treatment. INR over 8: stop, consider vitamin K. FFP not recommended.
- HIT (5-10 days): stop heparin, give argatroban. Antiplatelets: aspirin (COX), clopidogrel (P2Y12), cilostazol (claudication).

Chest drugs

- Never combine two beta-lactams. Pseudomonas: piperacillin/tazobactam, ceftazidime, cefepime, carbapenems, aminoglycosides. MRSA: vancomycin, ceftaroline.
- Aminoglycosides, vancomycin: kidney and ear. Macrolides, quinolones: long QT; quinolones: tendons.
- TB: rifampicin orange fluids, enzyme inducer; isoniazid neuropathy (give B6); pyrazinamide high urate; ethambutol red-green vision.
- SABA rescue; LABA, LAMA maintenance; IV magnesium 2 g in severe asthma; ICS cause oral thrush. 60% use inhalers wrongly.

Anaemia

- Oral iron: ferrous sulphate, about 200 mg elemental daily for 4-6 months, before food with vitamin C; tea and antacids reduce it.
- Child iron overdose: deferoxamine, bowel irrigation; charcoal useless.
- B12: IM hydroxocobalamin for life. Never folate alone in megaloblastic anaemia. Pregnancy folate 400 micrograms.
- IV iron if oral fails or malabsorption; test dose (dextran anaphylaxis).
- EPO in CKD: Hb target not above 10-12 g/dL; needs iron; no use in acute anaemia.

PART 2: TOPIC SUMMARY

Biochemistry

Lipid profile

- Assesses risk of coronary disease and stroke from atherosclerosis: total cholesterol, HDL, LDL, triglycerides (extended profile adds VLDL).
- Sample: serum or plasma after a 12-hour fast (water only). After a fatty meal serum is milky from chylomicrons; lipoprotein lipase clears it.

Lipid	Target
Total cholesterol	Under 200 mg/dL
LDL ("bad", main drug target)	Under 100 mg/dL
HDL ("good", protective)	Low if under 40 (men), under 50 (women)
Fasting triglycerides	Under 150 mg/dL

- LDL carries cholesterol from liver to tissues; oxidised LDL deposits in arteries. HDL carries it back to the liver.

Assay principle (enzymatic colour test)

- Cholesterol esterase frees cholesterol from esters; cholesterol oxidase makes H2O2; peroxidase joins H2O2 with phenol and 4-aminoantipyrine to form a red dye.
- Colour depth is proportional to concentration. Read at 546 nm (visible light 380-750 nm).

Practical steps

- Spectrophotometer: warm up 15 min. Three tubes, each 1000 microlitres reagent plus 10 microlitres of water (blank), standard or serum. Mix, 5 min at room temperature.
- Blank sets zero and removes interference. Wipe cuvette, zero with blank, read standard and test.
- Concentration = (A test / A standard) x standard concentration.
- Micropipette: press to first stop to draw, second stop to expel; check for bubbles. Check patient name and kit expiry.

ICC-1 Vascular surgery: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. Acute arterial ischaemia and vascular trauma [089] (lectures and LGW)
2. Chronic arterial ischaemia [088] (lecture and LGWs)
3. Venous vascular diseases [087] (lecture and LGWs)
4. Aneurysms [085] (lecture and LGW)
5. Vascular history taking and examination [104]
6. Extracranial, renal and mesenteric ischaemia [080]
7. Chronic limb swelling [084] (lecture and LGWs)
8. Principles of limb ablation and supportive therapy [082]
9. Diabetic foot [083] (lecture and LGW)
10. Anatomy recap: vascular lower limb

Part 2: topic summaries

1. Principles of arterial and venous diseases
2. Aneurysms
3. Vascular presentations

PART 1: LECTURE

Acute arterial ischaemia and vascular trauma [089] (lectures and LGW)

Topic: Principles of arterial and venous diseases

Definition and causes

- Acute limb ischaemia (ALI) = sudden fall in limb perfusion threatening viability; acute if symptoms < 2 weeks, chronic if > 2 weeks. Time-sensitive.
- Causes: embolism, thrombosis, trauma. ~80% of emboli are cardiac; atrial fibrillation is the commonest cause (also recent MI mural thrombus, valve disease, endocarditis, myxoma). Stopping warfarin in AF raises embolic risk.
- Embolus: sudden, no prior claudication, embolic source. Thrombosis: on an atherosclerotic plaque, graft or aneurysm; prior claudication (acute-on-chronic).

Clinical picture

- 6 Ps: pain, pallor, pulselessness, paraesthesia, paralysis, poikilothermia. Pain = usual presenting symptom; pulselessness = defining feature; paralysis = most ominous (may be irreversible).
- Colour: pale, then cyanosed, then mottled; fixed mottling = late. Empty veins, cold limb with a level, slow capillary refill.
- Sensory loss order: light touch, vibration, proprioception, then deep pain/pressure (late). Intrinsic foot muscles weaken first.
- Pale, cold, pulseless foot = ischaemia even with weakness. Saddle embolus mimics paraplegia: no sensory level and absent pulses.
- Category III (irreversible): anaesthesia, paralysis/rigid muscles, no arterial or venous Doppler signal.

Investigations

- Little time for tests. Duplex: quick, operator dependent, misses aorta/iliacs. CTA: fast, shows site and cause. DSA: invasive, before thrombolysis. MRA: no radiation, slow.

Management

- Start IV heparin + analgesia + optimise underlying condition, then revascularise.
- Embolus, viable limb: minimal work-up (ECG, CXR, bloods), Fogarty balloon embolectomy via common femoral arteriotomy.
- Thrombosis: thrombolysis if focal lesion; bypass if extensive.

Complications

- Reperfusion injury: hyperkalaemia (arrhythmia), myoglobin causing acute tubular necrosis, ARDS, myocardial depression.
- Compartment syndrome after revascularisation: tense, pain on passive stretch, pulses may be present; urgent fasciotomy (calf, anterior tibial compartment).

Vascular trauma

- Blunt, penetrating, iatrogenic. Hard signs (absent pulses, active bleeding, expanding/pulsatile haematoma, bruit/thrill, distal ischaemia): operate now, no imaging.
- Soft signs (small stable haematoma, nerve injury, unexplained hypotension, past bleeding, proximity): measure ABI; ABI < 0.9 = arteriography.
- Knee dislocation with pale pulseless foot = popliteal artery injury.
- ABCDE, control bleeding; repair directly, by graft, endovascular, or ligation (damage control).

Source: Acute Arterial Ischaemia and Vascular Trauma lecture [089]; LGW Acute Limb Ischaemia

PART 1: LECTURE

Chronic arterial ischaemia [088] (lecture and LGWs)

Topic: Principles of arterial and venous diseases

Causes and sites

- Over 45: atherosclerosis. Under 45: presenile atherosclerosis in diabetics; in non-diabetics Buerger's disease, Raynaud's, arteritis.
- Aortoiliac: buttock, thigh and calf claudication; Leriche syndrome = impotence + claudication + buttock wasting (absent femoral pulses).
- SFA (adductor canal, commonest site): calf claudication only; profunda collaterals supply thigh. Claudication is one level below the block.
- Infrapopliteal (diabetes, Buerger's): poor collaterals, so limb-threatening; worst with aortoiliac disease (multilevel).

Clinical features

- Claudication: pain on exercise relieved by rest. Neurogenic pseudoclaudication: bilateral, paraesthesia, back pain, provoked by standing still, more downhill, normal pulses.
- Rest pain: burning forefoot pain at night, worse with leg horizontal, relieved by hanging leg down; opioid-resistant; leads to gangrene without reconstruction.
- Arterial ulcer: punched out, very painful, no granulation, lateral ankle/toe tips, absent pulses, shiny hairless skin.

Investigations

- ABI = higher ankle pressure / higher arm pressure. > 1.2 calcified (diabetes, unreliable); 1.0-1.2 normal; 0.5-0.9 claudication; 0.1-0.4 critical ischaemia (CLI: ABI ≤ 0.5).
- Doppler waveform: triphasic, then biphasic, then monophasic.
- Healing needs ankle > 70 mmHg / toe > 50 mmHg. Imaging (CTA, or MRA if kidney function poor) only when intervention planned.

Management

- Claudication is usually benign (5% amputation in 5 years); 50% improve with stopping smoking + exercise. Commonest death = heart attack/stroke.
- Best medical therapy: stop smoking, statin, antiplatelet, BP and diabetes control, exercise, foot (podology) care. Screen heart and carotids (duplex, ECG/echo).
- Intervene for lifestyle-limiting claudication, rest pain, tissue loss. Endovascular poor for long, heavily calcified, orifice or uncrossable lesions.
- Grafts: synthetic (Dacron, ePTFE) or biological (autologous vein).
- Compartment syndrome after arterial repair: fasciotomy.

Buerger's disease

- Young male (< 40) heavy smokers; small/medium distal vessels (tibial, plantar, radial); upper and lower limbs; Raynaud's, thrombophlebitis migrans.
- Angiography of all four limbs: distal segmental skip lesions, corkscrew collaterals.
- Stopping smoking arrests (does not reverse) it; sympathectomy for rest pain/ulcers; bypass rarely feasible; amputation for gangrene.

Source: Chronic Arterial Ischaemia lecture (Dr Hesham Amer) [088]; LGW Chronic Arterial Ischaemia; Chronic Ischaemia LGW

PART 1: LECTURE

Venous vascular diseases [087] (lecture and LGWs)

Topic: Principles of arterial and venous diseases

Venous return

- Helped by calf muscle pump, one-way valves, negative intrathoracic pressure. Veins: superficial, deep, perforators.
- Venous insufficiency = failure of valves, patency or calf pump.

Deep vein thrombosis

- Virchow's triad: endothelial damage, stasis, hypercoagulable blood. Risks: age, surgery (pelvis, hip, leg), cancer, immobility, pregnancy, pill, factor V Leiden, protein C/S or antithrombin III deficiency, antiphospholipid.
- Often silent; symptoms ~3rd-4th day after surgery: swelling, pain, tenderness. Whole-leg swelling = iliofemoral. Homans' sign obsolete.
- Wells < 2 (low): D-dimer; if positive, compression ultrasound. Wells > 2: ultrasound directly. Duplex is the investigation of choice.
- Treatment: LMWH + warfarin/NOAC together; stop heparin when INR 2-3.
- PE with anticoagulation contraindicated: IVC filter.
- Prophylaxis: early mobilisation, hydration, intra-operative calf compression, stockings, SC LMWH from before surgery while at risk.
- Complications: PE, post-thrombotic leg, venous ulcer, phlegmasia alba/cerulea dolens.

Varicose veins

- Dilated, elongated, tortuous veins; mostly primary; more in women and with prolonged standing. Secondary: pelvic mass, previous DVT.
- Medial thigh/calf = great saphenous; posterolateral calf = small saphenous. Examine standing; cough thrill at SFJ.
- Trendelenburg test: high thigh tourniquet keeps veins empty = SFJ incompetence; otherwise move lower.
- Duplex = investigation of choice (junctions, perforators, deep veins).
- Complications: pain, swelling, bleeding, venous eczema (itchy red patch), thrombophlebitis, lipodermatosclerosis, ulcer.
- Treatment: stockings (minor, elderly, pregnant, unfit); sclerotherapy (small/medium below knee); endovenous laser; surgery (> 4 mm, complications, failed compression) = SFJ ligation + stripping.

Venous ulcers

- ~85% of chronic leg ulcers; 15-30% also arterial disease (mixed ulcer).
- Venous: gaiter area above medial malleolus, sloping edges, pigmentation, lipodermatosclerosis, eczema. Arterial: punched out, toes/lateral malleolus, absent pulses. Neuropathic: under metatarsal heads, thick keratinised edge.
- Ix: duplex, ABPI, CBC, ESR/CRP, lipids, glucose.
- Treatment: mobilise, elevate, compression (key; safe only if ABPI > 0.8). After healing: class I-II stockings; vein surgery if no deep incompetence.

Source: Venous Vascular Diseases lecture [087]; LGW Venous Vascular Disease; Varicose Veins LGW; Venous Ulcers LGW

PART 1: LECTURE

Aneurysms [085] (lecture and LGW)

Topic: Aneurysms

Definition and types

- Aneurysm = permanent localised dilatation of an artery to 1.5 x its normal diameter; can occur anywhere. Commonest = abdominal aortic (AAA), ~95% infrarenal.
- Classified by aetiology (atherosclerotic, traumatic, congenital, inflammatory, dissecting), structure (true/false), shape (fusiform/saccular) and site.
- True = wall has all 3 arterial layers. False (pseudo) = haematoma communicating with the lumen through a partial tear; wall = adventitia + compressed tissue.
- Pseudoaneurysm: mostly iatrogenic after puncture (femoral commonest), post-surgical (anastomotic), or infected in IV drug users. Duplex to diagnose; small asymptomatic = conservative; > 2.5 cm = US-guided thrombin injection; or open repair.

AAA: presentation and investigation

- Most found incidentally on U/S; pulsatile mass, back pain, compression (vein, nerve, ureter, vertebrae). Smoking = key risk.
- Rupture: abdominal pain + back pain + hypotension; mortality > 80%. Can mimic pancreatitis, peritonitis, duodenal perforation, MI. Contained retroperitoneal rupture gives flank/groin pain, mistaken for ureteric colic.
- U/S = screening and surveillance; CTA/MRA = planning (CTA also shows thoracic aorta).
- SVS screening: all men ≥ 65; men ≥ 55 with family history; women ≥ 65 only with family history or smoking.

AAA: management

- Repair asymptomatic AAA at 5.5 cm (rupture risk > 5% surgical mortality). Open or EVAR depending on anatomy, age, comorbidity, surgeon, equipment, patient preference.
- EVAR: lower early mortality (1.2% vs 4.6%) but needs lifelong surveillance for migration, iliac limb occlusion, endoleak (flow into the sac).
- Complications of both: renal dysfunction, paraplegia, ischaemic colitis, graft infection; aorto-enteric fistula (rare, more after open) = GI bleeding.
- Ruptured AAA: ABCDE, permissive hypotension, urgent CTA, immediate surgery.
- Control smoking, lipids and BP to slow growth.

Visceral and peripheral aneurysms

- Splenic: commonest visceral; mostly women, linked to pregnancy. Treat if symptoms, growth, > 2.5 cm, or pregnancy (planned or current). Vaccinate as for splenectomy.
- Renal artery: 2nd commonest visceral; hypertension; repair if hard-to-control BP or > 2 cm.
- ~90% of peripheral aneurysms are femoral/popliteal; popliteal ~70%. Limb (not life) threatening: thrombosis, distal embolism (blue toe), vein/nerve compression.
- Prominent wide popliteal pulse = suspect aneurysm; 50% bilateral, check the aorta. Repair femoral > 2.5 cm, popliteal > 2 cm; exclusion + bypass (in situ saphenous vein preferred) or covered stent.

Source: Aneurysms lecture [SU.VAS.10.1]; LGW Aneurysms [SU.VAS.11.2]

PART 1: LECTURE

Vascular history taking and examination [104]

Topic: Vascular presentations

History

- Analyse symptoms: onset, provoking/relieving factors, quality, radiation, severity, timing. Claudication: distance, site (buttock, thigh, calf, foot), relief with rest. Rest pain: disturbs sleep, relieved by position.
- Buttock claudication + new erectile dysfunction = aorto-iliac occlusion (Leriche's syndrome).
- Pain after meals + weight loss = coeliac/SMA/IMA ischaemia. TIA, fleeting blindness, speech problems = carotid disease.
- Pack-years = packs (20 cigarettes) per day x years. Ask about past and passive smoking. Family history: AAA in 1st-degree relative, hypercoagulable states, early MI.
- Drug history: antiplatelets, anticoagulants, statins, OCP/HRT, antihypertensives.

Arterial examination

- General: radial pulse rate/rhythm, BP both arms, abdomen for visible/expansile aortic pulsation and bruit above the umbilicus.
- Colour: waxy white = acute severe; blue fixed mottling = acute irreversible; dark purple shiny = chronic ischaemia. Trophic changes: shiny skin, hair loss, brittle nails.
- Ischaemic ulcer: very painful, deep, punched out, toes or lateral ankle, grey granulation.
- CRT normal < 2 s; > 5 s = significant ischaemia.
- Pulses proximal to distal: femoral at mid-inguinal point (halfway ASIS to pubic symphysis); popliteal deep, patient prone, knee flexed; posterior tibial behind medial malleolus; dorsalis pedis lateral to EHL (absent in 10%). Doppler if not felt.
- Buerger's angle < 20 degrees = severe ischaemia; purple on dependency = positive Buerger's test.

Venous examination (standing)

- General: pulse, BP, neck veins, abdominal dilated veins. Veins crossing the groin = secondary varicose veins.
- Venous ulcer: large, shallow, sloping edges, medial ankle (gaiter), mild pain. Also eczema, haemosiderin, atrophie blanche, lipodermatosclerosis (inverted champagne bottle).
- Saphena varix: soft bluish swelling below and lateral to pubic tubercle, cough thrill, vanishes on lying.
- Hard, tender, cord-like vein = thrombophlebitis. Bruit over a varicosity that does not collapse on lying = AV malformation.

Special tests and duplex

- Trendelenburg: tourniquet at SFJ, stand. Fills from above on release = SFJ incompetence; fills from below with tourniquet on = incompetent perforators.
- Modified Perthes': tourniquet + walking; veins distend with severe pain = occluded deep system.
- Duplex before any VV treatment: source of reflux, suitability for ablation, deep vein function.

Source: Vascular history and examination LGW notes, checklists and VV notes [SU.VAS.01.2]

PART 1: LECTURE

Extracranial, renal and mesenteric ischaemia [080]

Topic: Vascular presentations

Carotid disease

- Stroke = focal (occasionally global) loss of cerebral function > 24 h with vascular cause; TIA < 24 h. 80% ischaemic, 20% haemorrhagic.
- Main cause of carotid territory stroke: thromboembolism from plaque at the ICA origin. Risks: HTN, DM, smoking, IHD, TIA, hyperlipidaemia.
- Symptoms: hemisensory/motor signs, dysphasia, amaurosis fugax (monocular blindness); FAST.
- Duplex = first line; CTA/MRA before surgery for arch and intracranial vessels; also bloods, echo, 24 h ECG tape.
- Management: best medical therapy + CEA or stenting; symptomatic = intervene ideally within 2 weeks (7-day stroke risk up to 8-10%).
- CEA complications: stroke 2-3%, bleeding/haematoma, cranial nerve injury (XII, X, IX).

Renal disease and dialysis access

- Atherosclerotic stenosis = commonest renal artery disease (1-5% of hypertensives); low perfusion activates RAAS. Treat medically +/- endovascular or surgical revascularisation.
- RRT: haemofiltration, peritoneal dialysis, haemodialysis, transplant.
- Vein mapping: artery > 2 mm, vein > 2.5 mm. Native AVF 6 months before need; graft 6-8 weeks before.
- Rule of 6s: flow ≥ 600 mL/min, < 6 mm deep, diameter ≥ 6 mm, straight length for 2 needles.
- Sites: radiocephalic, ulnobasilic, brachiocephalic, brachiobasilic transposition.
- Complications: stenosis, thrombosis, infection, failure to mature, steal (4 stages), high-output heart failure, aneurysm.

Mesenteric ischaemia

- Acute: embolism (usually SMA just beyond middle colic; limited bowel, better survival), thrombosis (at vessel origin; extensive bowel), NOMI, venous thrombosis.
- NOMI causes: low output (CHF, MI, sepsis, major surgery), vasopressors, ergotamines, cocaine, digitalis.
- Acute features: sudden severe pain out of proportion to signs, urge to defecate, fever, vomiting; raised WCC and lactate. Complications: sepsis, bowel death, death.
- CTA (or contrast CT) = investigation of choice. Treat: revascularisation (embolectomy, bypass, thrombolysis, angioplasty/stent) + resect dead bowel.
- Chronic: pain ~30 min after eating, lasting 1-3 h; food fear, weight loss; can become acute-on-chronic. CTA.
- Venous thrombosis: > 80% have a cause; hypercoagulable states (polycythaemia vera most common), tumour, infection, cirrhosis, trauma, pancreatitis.

Source: Extracranial, renal and mesenteric ischaemia [SU.VAS.15.1]

PART 1: LECTURE

Chronic limb swelling [084] (lecture and LGWs)

Topic: Vascular presentations

Causes and mechanisms

- Oedema mechanisms: increased permeability (cellulitis, angioedema, sepsis); increased capillary pressure (venous insufficiency, heart failure); low oncotic pressure (liver disease, nephrotic syndrome, protein-losing enteropathy, malabsorption); impaired lymph drainage (lymphoedema).
- Unilateral = local (DVT, cellulitis, filariasis); bilateral = systemic (heart, kidney, liver failure). Painful = DVT, cellulitis; painless = systemic, venous insufficiency, lymphoedema.
- Drugs: CCBs (~50% get oedema), NSAIDs, steroids.
- Myxoedema: mucopolysaccharides bind water = non-pitting boggy oedema, dry coarse skin; check TSH.
- Lipoedema: always bilateral, symmetrical, feet spared.
- Klippel-Trenaunay: lateral venous complex, capillary naevus, hypoplastic deep veins, limb lengthening.

Lymphoedema

- High-protein fluid from defective lymph drainage. Pitting at first; later hard and non-pitting. Stemmer's sign: cannot pinch skin at base of 2nd toe.
- Primary: congenita (infancy, males, bilateral; Milroy's = congenital + hereditary); praecox (commonest, females, after menarche, unilateral, to the knee); tarda (adult onset).
- Secondary: malignancy, surgery (axillary clearance), radiotherapy; filariasis (Wuchereria bancrofti) in developing countries. Exclude cancer recurrence in the axilla.
- Stages: 1 reversible, soft pitting, settles with elevation; 2 irreversible, hard; 3 elephantiasis with papillomas.
- Lymphoscintigraphy: tracer in first web space, gamma camera.
- Treatment: skin care, massage (manual lymph drainage), bandages, garments, pneumatic compression, exercise, complete decongestive therapy; surgery in severe cases. Antibiotics for infection.
- Arm care after axillary surgery: protect skin, no injections/blood tests/BP on that arm, avoid heat and tight items, lose weight.
- Lymphangiosarcoma: rare, years later, multiple subcutaneous nodules; distinguish from recurrent breast cancer.

Post-thrombotic syndrome

- After DVT: aching, heaviness, itching, swelling, secondary varicose veins, brown ankle pigmentation, ulcer. Iliofemoral obstruction: whole-leg swelling, abdominal wall collaterals, venous claudication.
- Secondary VV clues: previous DVT, irregular pattern, veins crossing the groin, ulceration.
- Duplex: deep veins, missed DVT, perforators; bloods for clotting problems.
- Treatment: mobilise, elevate, compression to heal ulcer, antibiotics if infected, then lifelong compression stockings.

Investigations

- CBC, urine and renal function, LFT, albumin, lipids, TSH, D-dimer (DVT); CXR, ECG, echo; abdominal U/S; venous duplex.

Source: Chronic limb swelling lecture [SU.VAS.12.1]; LGWs Chronic limb swelling, Lymphoedema, PTS [SU.VAS.13.2]

PART 1: LECTURE

Principles of limb ablation and supportive therapy [082]

Topic: Vascular presentations

Why and who

- PVD is the leading cause of lower limb amputation in the Western world; diabetics are 8-12 times more likely to need amputation.
- Causes: critical limb ischaemia, sepsis, trauma, neoplasm, congenital.
- Outcome is poor: mortality approaches 50% at 2 years in diabetics; half of diabetics with a major amputation lose the other leg within 5 years.
- CLI: revascularise first (angioplasty or bypass); amputate if this fails, or primarily by choice or comorbidity.

Diabetic foot and level choice

- Foot sepsis with palpable pulses: good prognosis with early surgery. Absent pulses: revascularise as early as possible.
- Digital gangrene: amputate digit + metatarsal head, leave wound open; infection is often more extensive than it looks, so warn of a higher level at consent.
- Level depends on rehab potential, vascular disease and likely healing; proximal levels need more walking energy. Check blood supply before toe amputation.

Operative principles

- Do not undermine flaps; tourniquet cuts blood loss without harming healing.
- Divide nerves cleanly, away from bone ends (prevents neuroma). Muscle that does not bleed or contract to diathermy = devitalised, go higher.
- Severe sepsis: guillotine amputation, complete later. Suction drain for major amputation; avoid stump bandaging (skin breakdown).

Levels

- Ray: metatarsal neck, tennis-racquet incision. Transmetatarsal: plantar flap, excellent walking.
- BKA: long posterior flap (Burgess); tibia cut 14 cm below knee joint (10-12 cm below tuberosity); minimum 7.5 cm. Fibula 2 cm above tibia. Skew flaps if posterior flap skin is not viable.
- AKA: mid-femur, at least 15 cm above tibial plateau (lever arm + room for prosthetic knee); shortest 8 cm below inferior pubic ramus; equal fishmouth flaps.
- Hip disarticulation: high mortality, poor ambulation.

After surgery

- Early: DVT, flap necrosis, infection, pain, haematoma, flexion contracture, psychological. Late: bulbous stump, bone erosion, neuroma, osteomyelitis, ulcer.
- Post-amputation pain about 80%, phantom pain about 70%; no proven treatment. Epidural = best peri-operative analgesia but does not prevent phantom pain. Refer to pain team if first-line oral agents fail.
- Rehab starts immediately: physiotherapy prevents flexion contractures; stump socks shrink the healed stump; PPAM aid (40 mmHg) for early walking; limb fitting at about 6 weeks.

Source: Principles of Limb Ablation and Supportive Therapy [082]

PART 1: LECTURE

Diabetic foot [083] (lecture and LGW)

Topic: Vascular presentations

Burden and causes

- 20% of diabetes admissions are for foot problems; about 15% of diabetics get a foot ulcer each year; 85% of diabetic amputations follow an ulcer. Ulcers are precipitated by trauma.
- Two main causes: neuropathy and ischaemia. Neuropathic 45-60%, ischaemic about 10%, neuro-ischaemic 25-45%; infection is the common final presentation.
- Sensory: injury not felt (tight shoes). Motor: weak short flexors vs long extensors = claw toes, prominent metatarsal heads, ulcer there. Autonomic: less sweating, dry cracked skin.
- Diabetes gives 4-fold PAD risk; diabetes + PAD = highest amputation risk. Microvascular: thick basement membrane, leaky capillaries, sticky platelets.
- Neuropathic ulcer: at pressure points, with a ring of callus.

Wagner grades

- 0 intact skin (impending); 1 superficial; 2 deep to tendon, bone or ligament; 3 osteomyelitis; 4 gangrene of toes/forefoot; 5 gangrene of whole foot.

Acute infected foot

- First control the diabetes (sliding scale, correct ketoacidosis).
- Cultures + swabs, then IV antibiotics covering Staph aureus, strep, pseudomonas, anaerobes. Surface swabs may miss the organism; aspirate tissue fluid for deep culture.
- Tenderness or crepitus = extensive infection. Toe infection spreads to the web space, showing only as mild dorsal redness and tenderness.
- Plain X-ray: osteomyelitis or gas. Drain collections and remove dead tissue within 24 hours.
- Then assess nerves and vessels: absent pulses + low ankle pressure = high-risk foot. Duplex, then CT angiogram; revascularise (angioplasty or surgery) unless healing.

Wound care and prevention

- Do not suture: heal by secondary intention (sutures cut skin edge flow and trap infection). Elevate, avoid oedema.
- Negative pressure therapy: removes fluid, boosts microcirculation, shrinks wound. Total contact cast (with window) for early walking; must be expertly fitted.
- Ulcer care: off-loading, debridement, dressings, antibiotics, revascularisation, amputation; also larvae therapy and BMT. Needs MDT (ortho, vascular, endocrine, podiatry, ID).
- Screen with 10 g monofilament, pulse palpation, ankle pressure.

Charcot foot

- Neuropathy-driven bone fragmentation and joint destruction; medial arch collapse = rocker-bottom foot.

Source: Diabetic Foot lecture [083]; Diabetic Foot LGW [083]

PART 1: LECTURE

Anatomy recap: vascular lower limb

Topic: Vascular presentations

Arteries: hip and thigh

- Aorta, common iliac, then external iliac (lower limb) and internal iliac (pelvis; gives superior and inferior gluteal).
- Femoral artery starts behind the inguinal ligament at the mid-inguinal point, runs through the femoral triangle and adductor canal, then becomes popliteal.
- Branches: superficial circumflex iliac, superficial epigastric, superficial external pudendal; deep external pudendal, profunda femoris, descending genicular.
- Profunda femoris: largest branch, main thigh supply; gives lateral and medial circumflex femoral and 4 perforating arteries.
- Femoral head: mainly retinacular arteries from the circumflex femorals (especially medial); minor supply from artery in ligament of head (obturator).

Anastomoses

- Cruciate (back of thigh, below greater trochanter): transverse limb = transverse branches of medial and lateral circumflex; vertical limb = ascending medial circumflex + descending inferior gluteal above, first perforating below. Links internal iliac and profunda.
- Around ASIS: superior gluteal, ascending lateral circumflex, deep circumflex iliac, superficial circumflex iliac.

Clinical points

- Bleeding control: press femoral artery against the superior pubic ramus, midway between ASIS and pubic symphysis.
- Femoral artery catheter in the femoral triangle for left heart or renal angiography.

Leg and foot

- Popliteal artery: from adductor magnus opening to lower border of popliteus, then splits into anterior (smaller) and posterior (larger) tibial arteries.
- Anterior tibial ends midway between malleoli as dorsalis pedis: superficial, EHL tendon medial, EDL and nerve lateral; enters sole at 1st interosseous space.
- Posterior tibial ends under the flexor retinaculum as medial and lateral plantar. Order front to back: tibialis posterior, FDL, vessels, nerve, FHL ("Tom, Dick, Very Nice Harry").

Veins

- Superficial veins start at the dorsal venous arch; deep veins: femoral and popliteal.
- Great saphenous: in front of medial malleolus, up medial leg and thigh, through cribriform fascia to femoral vein 4 cm below and lateral to pubic tubercle; 15-20 valves.
- Small saphenous: behind lateral malleolus with sural nerve, between gastrocnemius heads; 5-10 valves.
- GSV uses: venous cut-down in front of medial malleolus; reversed graft for coronary bypass (valves). Varicose veins: perforator valve incompetence, deep to superficial flow.

Source: Anatomy Recap ICC1 Vascular Lower Limb

PART 2: TOPIC SUMMARY

Principles of arterial and venous diseases

Acute versus chronic arterial ischaemia

	Acute limb ischaemia	Chronic limb ischaemia
Time course	Under 2 weeks, sudden	Over 2 weeks, gradual
Commonest cause	Embolus (AF, 80% cardiac) or thrombosis on plaque	Atherosclerosis (Buerger's if young smoker)
Key symptom	6 Ps; paralysis = most ominous	Claudication, then rest pain at night, then tissue loss
First steps	IV heparin + analgesia, then revascularise	Stop smoking, exercise, statin, antiplatelet
Procedure	Fogarty embolectomy (embolus); thrombolysis or bypass (thrombosis)	Angioplasty/stent or bypass for disabling claudication, rest pain, ulcer

- Claudication pain is one level below the block: SFA (commonest site) = calf; aortoiliac = buttock and thigh (Leriche: + impotence, absent femoral pulses).
- ABI: 1.0-1.2 normal; 0.5-0.9 claudication; 0.1-0.4 critical ischaemia; over 1.2 = calcified vessels (diabetes), unreliable.
- After revascularisation watch for reperfusion injury (hyperkalaemia, acute tubular necrosis) and compartment syndrome (pain on passive stretch, pulses may persist): urgent fasciotomy.
- Trauma: hard signs (absent pulse, active bleeding, expanding haematoma, bruit) = straight to theatre; soft signs = ABI, arteriography if under 0.9. Knee dislocation + pulseless foot = popliteal injury.

Venous disease

- DVT: Virchow's triad. Wells under 2 = D-dimer first; over 2 = duplex directly. Treat LMWH with warfarin or a NOAC; IVC filter only if anticoagulation is contraindicated.
- Varicose veins: great saphenous = medial; small saphenous = posterolateral calf. Duplex before treatment. Trendelenburg test controls the SFJ.
- Treatment ladder: stockings, sclerotherapy, endovenous laser, SFJ ligation and stripping.

Ulcer comparison

	Venous	Arterial	Neuropathic
Site	Gaiter area, medial malleolus	Toes, lateral malleolus	Under metatarsal heads
Look	Shallow, sloping edges, pigmentation	Punched out, no granulation	Callus ring
Pain	Mild	Severe	Painless
Key treatment	Compression (only if ABPI over 0.8)	Revascularise	Off-load pressure

PART 2: TOPIC SUMMARY

Aneurysms

Definitions

- Aneurysm = permanent localised dilatation to 1.5 times normal diameter. True = all three layers; false (pseudoaneurysm) = haematoma communicating with the lumen, usually after femoral puncture.
- AAA is the commonest; about 95% are infrarenal. Smoking is the key risk factor.

Abdominal aortic aneurysm

- Usually found by chance on ultrasound; may cause a pulsatile mass or back pain.
- Rupture: abdominal + back pain + hypotension; mortality over 80%; can mimic ureteric colic, pancreatitis or MI. Treat with ABCDE, permissive hypotension, CTA if stable, immediate repair.
- Ultrasound = screening and surveillance; CTA = planning repair. Screen all men from 65.
- Repair asymptomatic AAA at 5.5 cm. EVAR has lower early mortality than open repair but needs lifelong surveillance (endoleak, migration).
- Late open-repair danger: aorto-enteric fistula (GI bleed).

Other aneurysms

Site	Key facts	Repair when
Popliteal	Commonest peripheral; 50% bilateral; check the aorta; threatens the limb (thrombosis, emboli)	Over 2 cm
Femoral	Second commonest peripheral	Over 2.5 cm
Splenic	Commonest visceral; young women, pregnancy	Symptoms, growth, over 2.5 cm, pregnancy
Renal	Hypertension	Hard-to-control BP or over 2 cm
Femoral pseudoaneurysm	After puncture	Over 2.5 cm: ultrasound-guided thrombin injection

PART 2: TOPIC SUMMARY

Vascular presentations

History and examination

- Claudication distance and site; rest pain eased by hanging the leg down; post-meal pain with weight loss = mesenteric ischaemia; TIA or amaurosis fugax = carotid disease.
- Pulses from proximal to distal: femoral at the mid-inguinal point, popliteal deep with knee flexed, posterior tibial behind medial malleolus, dorsalis pedis lateral to EHL (absent in 10%).
- Buerger's angle under 20 degrees = severe ischaemia. Capillary refill over 5 s = significant ischaemia.
- Venous examination standing; Trendelenburg test for SFJ incompetence; modified Perthes' test for deep vein occlusion.

Carotid, renal and mesenteric disease

- Carotid: duplex first; symptomatic significant stenosis = endarterectomy within 2 weeks. CEA risks: stroke 2-3%, cranial nerves XII, X, IX.
- Dialysis access: native fistula made 6 months before need; rule of 6s for a mature fistula.
- Acute mesenteric ischaemia: pain out of proportion to signs, raised lactate; CTA; revascularise and resect dead bowel. Chronic: pain 30 min after meals, food fear, weight loss.

Chronic limb swelling

- Unilateral = local (DVT, cellulitis, lymphoedema); bilateral = systemic (heart, kidney, liver failure, drugs such as CCBs).
- Lymphoedema: Stemmer's sign; primary praecox = commonest (girls after menarche); secondary = cancer, surgery, radiotherapy, filariasis. Treat with skin care, massage, compression.
- Post-thrombotic syndrome: heaviness, pigmentation, ulcer after DVT; lifelong compression.

Diabetic foot and amputation

- Neuropathy + ischaemia + infection. Wagner 0 intact, 1 superficial, 2 deep, 3 osteomyelitis, 4 forefoot gangrene, 5 whole foot.
- Infected foot: control glucose, cultures, IV broad antibiotics, drain within 24 h, then assess pulses and revascularise. Never suture: heal by secondary intention.
- Amputation: revascularise first if possible. BKA tibia cut about 14 cm below the knee joint (long posterior flap); AKA at least 15 cm above the tibial plateau. Physiotherapy prevents flexion contracture; phantom pain about 70%.

Anatomy anchors

- Femoral artery at the mid-inguinal point; profunda femoris = main thigh supply and key collateral; cruciate anastomosis links internal iliac and profunda.
- Great saphenous vein joins the femoral vein 4 cm below and lateral to the pubic tubercle.

ICC-1 Hematology: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. Thrombophilia, inherited and acquired [008b, 008c]
2. Haemostasis [008a] (revision notes and LGW)
3. DVT and pulmonary embolism [017]
4. Approach to anaemia [052]
5. Iron deficiency anaemia [069a]
6. Megaloblastic anaemia [069b]
7. Autoimmune haemolytic anaemia [077]
8. Haemoglobin assay and blood indices [060] (prelab and lab)
9. Blood film preparation [066] (prelab and lab)
10. Haematology cases LGW [092]
11. Clinical chemistry tutorial [027]

Part 2: topic summaries

1. Thrombophilia, DVT and pulmonary embolism
2. Anaemias (megaloblastic, iron deficiency, haemolytic)

PART 1: LECTURE

Thrombophilia, inherited and acquired [008b, 008c]

Topic: Thrombophilia, DVT and pulmonary embolism

Basics

- Thrombophilia = inherited or acquired tendency to venous thrombosis. Virchow's triad: stasis, vessel wall damage, blood changes. In VTE, coagulability and stasis matter most.
- Provoked VTE (transient, removable risk factor) = lower recurrence; unprovoked (no factor, or persistent factor) = high recurrence.

Inherited

- Antithrombin deficiency: AD, recurrent VTE < 40, pregnancy VTE; highest risk of the natural anticoagulant defects. Clue = heparin resistance (no APTT prolongation on heparin). Acquired low AT: nephrotic syndrome, DIC, severe liver disease.
- Protein C deficiency (AD): VTE in heterozygous adults; homozygous = neonatal purpura fulminans/DIC. Warfarin skin necrosis: protein C (half-life ~6 h) falls before II, VII, IX, X.
- Protein S: cofactor for APC; free PS (40%) active, C4b-bound (60%) inactive.
- Factor V Leiden = most common: G1691A, Arg506Gln, partial resistance to APC; mainly venous. ~92% of APC resistance.
- Prothrombin G20210A: raised prothrombin level, at least 2-fold VTE risk.
- Hyperhomocysteinaemia: arterial and venous thrombosis; acquired causes B12/folate deficiency, smoking, renal damage, drugs, age.

Acquired

- Antiphospholipid syndrome: thrombosis and/or pregnancy loss. Lupus anticoagulant prolongs APTT in vitro (binds reagent phospholipid), not corrected by normal plasma, yet causes thrombosis.
- Sapporo: ≥ 1 clinical + ≥ 1 lab criterion. Clinical: thrombosis; fetal death ≥ 10 wk; premature birth ≤ 34 wk; ≥ 3 miscarriages < 10 wk. Lab: LA, anticardiolipin, anti-beta2 GPI; positive twice ≥ 12 weeks apart.
- HITT: IgG to PF4-heparin, platelet FcγRIIa activation; platelets fall 4-10 days after unfractionated heparin. Check baseline platelets and APTT before heparin.
- Myeloproliferative: polycythaemia (viscosity); essential thrombocythaemia (platelets > 1000).

When and how to test

- Suspect if: VTE < 40-45 y, arterial thrombosis < 30 y, unusual site (cerebral vein), recurrent VTE, strong family history, ≥ 3 miscarriages.
- Basic tests (CBC, PT, APTT, TT, fibrinogen, D-dimer) and history rule out most acquired causes. D-dimer: high sensitivity, low specificity.
- Test inherited thrombophilia if young (< 40), thrombosis-prone family, or result changes treatment (pregnancy, children).
- PCR panel: FV G1691A, prothrombin G20210A, MTHFR C677T/A1298C.
- Timing: at least 1 month after anticoagulation ends; avoid pregnancy, pill/HRT, illness. DNA tests any time.

Source: ICC-1 [008]b, [008]c slides; [008] thrombophilia student notes

PART 1: LECTURE

Haemostasis [008a] (revision notes and LGW)

Topic: Thrombophilia, DVT and pulmonary embolism

Platelets

- Count 150-410 x10^9/L, lifespan 10 days, anucleate discs; made by megakaryocyte fragmentation; thrombopoietin is the main regulator.
- High shear: GPIb-IX-V binds VWF, then firm adhesion. Low shear: GPIa/IIa binds collagen directly.
- Aggregation: GPIIb/IIIa + fibrinogen bridges (defective in thrombasthenia). Also mediates clot retraction.
- Dense granules: ADP, ATP, 5-HT, calcium. Alpha granules: PF4, PDGF, fibrinogen, VWF. TXA2 = main amplification loop and vasoconstrictor.
- Inhibitors of platelets: endothelial prostacyclin (PGI2) and nitric oxide.

Coagulation

- Extrinsic tenase: TF + VIIa + PL + Ca -> Xa (initiation). Intrinsic tenase: IXa + VIIIa + PL + Ca -> Xa. Prothrombinase: Xa + Va + PL + Ca -> thrombin burst.
- TFPI switches off initiation (complex with VIIa, TF, Xa).
- All enzymes are serine proteases except XIII (transglutaminase); XIIIa cross-links fibrin.
- Contact factors (XII, prekallikrein, HMWK): deficiency prolongs APTT but no bleeding. XI deficiency may bleed at surgery.
- Vitamin K-dependent: II, VII, IX, X, protein C, protein S (gamma-carboxylation). Warfarin = vitamin K antagonist. Deficiency causes: obstructive jaundice, antibiotics, malabsorption.

Natural anticoagulants and fibrinolysis

- Antithrombin: main inhibitor of thrombin and Xa; action enhanced by heparin.
- Thrombin-thrombomodulin activates protein C; APC + protein S inactivate Va and VIIIa.
- t-PA converts fibrin-bound plasminogen to plasmin. PAI inhibits t-PA; alpha2-antiplasmin inhibits circulating plasmin. Therapeutic t-PA: MI, stroke, massive PE.

Sequence and tests

- After injury: vasoconstriction -> platelet contact -> platelet plug -> fibrin clot -> fibrinolysis.
- Bleeding time 3-8 min; prolonged in thrombocytopenia, platelet defects (aspirin), VWD, vascular defects.
- PT (VII, X, V, II, I): monitors warfarin. INR = (patient PT / mean normal PT)^ISI.
- APTT 26-40 s (VIII, IX, XI, XII + common): monitors unfractionated heparin.
- Isolated long PT = factor VII; isolated long APTT = VIII, IX, XI or XII.
- TT: affected by fibrinogen, heparin, FDPs.

Source: ICC-1 [008]a haemostasis student notes; ICC1[008] Haemostasis LGW

PART 1: LECTURE

DVT and pulmonary embolism [017]

Topic: Thrombophilia, DVT and pulmonary embolism

DVT: diagnosis

- VTE = DVT and/or PE. Risks: surgery, immobility, age > 60, obesity, pregnancy, pill/HRT, cancer, prior VTE, thrombophilia, dehydration.
- Signs (pain, unilateral swelling, warmth, calf tenderness) are not reliable. Differentials: cellulitis, ruptured Baker's cyst, calf haematoma, muscle tear, lymphoedema.
- Three steps: clinical probability (Wells), imaging, D-dimer.
- Compression duplex ultrasound = investigation of choice (less sensitive for calf). Venography = old gold standard; CT venography for iliac/IVC.
- D-dimer: high sensitivity, low specificity; value = negative result with low pre-test probability.

DVT: treatment

- Uncomplicated: home treatment. Admit for limb-threatening DVT, high bleeding risk, IV analgesia, CrCl < 30, HIT history.
- DOACs preferred over VKAs; if warfarin, overlap with LMWH/UFH.
- No routine thrombolysis; consider (catheter-directed) for phlegmasia cerulea dolens or young iliofemoral DVT.
- IVC filter only if anticoagulation contraindicated; retrievable, remove later. Routine stockings not recommended.
- Duration: provoked 3-6 months then stop; unprovoked or persistent risk (cancer) 3-6 months then indefinite if bleeding risk low-moderate.
- Complications: PE (30-50% of proximal DVT), post-thrombotic syndrome (valve damage), phlegmasia, paradoxical embolism (ASD).
- Upper limb DVT: catheters, thoracic outlet, Paget-Schroetter (effort); ultrasound.

PE: diagnosis

- RV strain -> low LV filling -> hypotension/shock. Symptoms: dyspnoea, pleuritic pain, haemoptysis, syncope. Commonest sign: tachypnoea.
- Modified Wells: > 4 likely (go to imaging); ≤ 4 unlikely (D-dimer first).
- CTPA = main test; better with cardiorespiratory disease. V/Q if CTPA unsuitable; normal V/Q virtually excludes PE; hard to read if CXR abnormal/COPD.
- ECG: commonest sinus tachycardia; RBBB, RAD, S1Q3T3 less common; normal ECG does not exclude. ABG: hypoxaemia, hypocapnia.
- Troponin, BNP and echo = risk stratification, not diagnosis.

PE: treatment

- Severity categories A-E (2026 AHA/ACC): A-B early discharge; C-D admit; E critical care and reperfusion.
- Reperfusion (systemic or catheter thrombolysis, thrombectomy, embolectomy) for E and selected D only.
- LMWH over UFH when parenteral needed; DOACs over VKAs. Duration 3-6 months, extended if unprovoked or persistent risk.

Source: ICC-1 [017] DVT and pulmonary embolism slides

PART 1: LECTURE

Approach to anaemia [052]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Definition and mechanisms

- Anaemia = Hb below normal for age and sex: men < 13, women < 12, pregnancy < 11, 6-12 y < 11.5, birth < 14 g/dL.
- Mechanisms: reduced marrow production, increased destruction (haemolysis), loss (bleeding).
- Dehydration can mask anaemia; pregnancy and splenomegaly (bigger plasma volume) can mimic it.

Classification

- By red cell size (usual first step):
- Microcytic (MCV < 80, MCH < 27): iron deficiency, thalassaemia, some chronic disease, lead, sideroblastic.
- Normocytic: many haemolytic anaemias, chronic disease, acute blood loss, renal disease, mixed deficiency, marrow failure.
- Macrocytic (MCV > 100): megaloblastic (B12, folate) or non-megaloblastic (alcohol, liver disease, hypothyroidism, myelodysplasia, aplastic anaemia, reticulocytosis).
- Leucoerythroblastic film (normoblasts + myelocytes): marrow infiltration; also severe bleeding, acute haemolysis.
- Mixed iron + folate/B12 deficiency: indices may be normal, film dimorphic.

Clinical

- Adaptation: higher cardiac output, flow to brain and heart, raised 2,3-DPG (right shift).
- Severity of symptoms depends on Hb level, speed of onset and comorbidity.
- Acute bleeding: Hb may be near normal early; manage by blood loss and cardiovascular state.
- Mucous membrane pallor is more reliable than skin pallor; hyperdynamic circulation; heart failure if severe.
- History clues: menorrhagia, bowel change in elderly (colorectal cancer), bone pain (myeloma), arthralgia (chronic disease), diet, drugs, gastrectomy.

Investigations

- FBC indices, other cell lines (pancytopenia = marrow problem or hypersplenism), blood film, reticulocytes, haematinics, U&Es, LFTs.
- Reticulocytes 0.5-2.5% (25-125 x10^9/L). High in chronic haemolysis and after bleeding (rise 2-3 days, peak 6-10 days). Low = marrow failure or lack of erythropoietin (CKD).
- Bone marrow not always needed. Aspirate: cell detail, M:E ratio, iron stain. Trephine: cellularity, fibrosis, infiltrates.
- Further tests by cause: haemolysis screen (DAT), endoscopy/colonoscopy, imaging, electrophoresis for myeloma.

Treatment

- Find the cause first; treat the patient, not the Hb. Mild anaemia often needs no correction; sickle cell or MDS patients tolerate low baseline Hb (treat only a significant fall).
- Transfusion only if Hb very low, symptomatic or acute bleeding; it is not a cure.

Source: ICC-1 [052] lecture and student notes; haematology anaemia handout

PART 1: LECTURE

Iron deficiency anaemia [069a]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Iron basics and stages

- Body iron 40-50 mg/kg; ~75% in haemoglobin. No physiological way to excrete iron: extra loss = bleeding.
- Absorption: haem > Fe2+ > Fe3+. Reduced by high gastric pH (gastrectomy, vagotomy, H2 blockers, calcium antacids), coeliac, Crohn's, phytates, tannins; helped by vitamin C (ascorbate).
- Order of deficiency: stores depleted -> film/marrow changes -> anaemia (last). Treat even if Hb normal.
- On treatment the order reverses: Hb normalises before stores refill, so continue iron for several months.

Causes

- Chronic blood loss: menorrhagia (commonest in UK), GI loss (peptic ulcer, gastric/colon cancer), hookworm (commonest worldwide), intravascular haemolysis (haemosiderinuria), blood donation.
- Poor intake or absorption (coeliac, gastrectomy); increased demand (pregnancy, prematurity, growth spurt).
- Older patient or family history of bowel cancer with unexplained IDA: exclude malignancy.

Clinical features

- Tiredness, exertional dyspnoea; angular stomatitis, koilonychia (spoon nails), glossitis, pharyngeal web dysphagia (rare), irritability, poor cognition in children.

Investigations

- FBC: low MCV, MCH, MCHC; thrombocytosis common. Film: microcytic hypochromic cells, pencil cells, target cells.
- Iron studies: LOW iron, HIGH TIBC, LOW ferritin; transferrin saturation (iron/TIBC x 100) < 15%.
- Ferritin is an acute phase protein: may be normal/high in acute illness despite deficiency.
- Soluble transferrin receptor rises early in IDA and separates IDA from ACD (not routine).
- Perls' Prussian blue marrow stain: reliable but invasive, rarely needed.
- Find the cause: coeliac screen (anti-tTG, duodenal biopsy), gastroscopy/colonoscopy, capsule endoscopy, stool for ova/parasites, gynae assessment.

Treatment and differentials

- Ferrous sulphate 200 mg three times daily for 4-6 months, plus treat the cause. Side effects: bloating, constipation, diarrhoea.
- Poor response: poor compliance (commonest), malabsorption, wrong diagnosis (thalassaemia trait), ongoing bleeding.
- IV iron if oral iron not tolerated or ineffective: whole dose in a day, but Hb rises at the same rate.
- Thalassaemia trait: microcytosis with normal iron studies; Hb usually > 9; HbA2 raised.
- ACD (e.g. RA): LOW iron, LOW TIBC, normal/HIGH ferritin; iron trapped in macrophages.
- Sideroblastic: ring sideroblasts in marrow; MCV low, normal or high; primary (MDS) or alcohol, drugs, toxins.

Source: ICC1 [069] Iron deficiency anaemia slides; ICC-1 [052, 069] lecture notes

PART 1: LECTURE

Megaloblastic anaemia [069b]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Basics

- Megaloblastic = macrocytic anaemia (MCV > 100 fl) from failed DNA synthesis; main causes B12 and folate deficiency.
- Non-megaloblastic macrocytosis: alcohol, liver disease, hypothyroidism, aplastic anaemia, reticulocytosis, cytotoxic drugs (hydroxyurea), myelodysplasia.

Vitamin B12

- Animal foods only (eggs, dairy, fish); stores 3-4 mg in liver last about 3 years, so deficiency is slow.
- Freed in stomach, binds intrinsic factor (gastric parietal cells), absorbed in terminal ileum, carried by transcobalamin II.
- Needed to make methionine from homocysteine (activates folate for DNA) and methylmalonyl CoA -> succinyl CoA.
- Causes: vegan diet; pernicious anaemia (antibodies to IF and/or parietal cells); ileal disease/ resection; gastrectomy; bacterial overgrowth; alcohol; tropical sprue; fish tapeworm; repeated nitrous oxide; transcobalamin II deficiency.
- Neurology even without anaemia: peripheral neuropathy, subacute combined degeneration, optic atrophy, dementia.

Folate

- Green vegetables, liver, beans; stores about 10 mg last only about 4 months; heat labile.
- Absorbed in duodenum and jejunum; needs B12 to become active THF (MTHFR step).
- Causes: poor diet (elderly, alcohol, poverty), coeliac, pregnancy, chronic haemolysis, malignancy, psoriasis; drugs: anticonvulsants, sulphasalazine, methotrexate, trimethoprim, pyrimethamine; dialysis.
- Deficiency causes neural tube defects; adult neurology less common.

Investigations

- FBC: macrocytosis, anaemia may be absent, pancytopenia if severe.
- Film: oval macrocytes and hypersegmented neutrophils (> 5% with > 5 lobes, "shift to right") appear early, even before anaemia; tear drop cells; low reticulocytes.
- Raised unconjugated bilirubin and LDH = ineffective erythropoiesis.
- Serum B12 + folate together; red cell folate best for stores. IF antibodies specific for PA; parietal cell antibodies are not.
- Bone marrow (hypercellular, megaloblasts) usually not needed.

Treatment

- Hydroxocobalamin 1 mg IM 3x weekly for 2 weeks, then 1 mg IM 3-monthly for life if malabsorption; oral cyanocobalamin if dietary.
- Never give folic acid before B12 (can worsen neurology). Treat promptly.
- Severe anaemia: short course of folate and iron may be needed as new red cells surge.
- Folate deficiency: folic acid 5 mg daily for 3-4 months; 400 µg daily before conception prevents NTDs.

Source: ICC-1 [069] Deficiency anaemias part 2, megaloblastic anaemia

PART 1: LECTURE

Autoimmune haemolytic anaemia [077]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Types

- Haemolytic anaemia = shortened red cell lifespan. Inherited: membrane (spherocytosis), enzyme (G6PD), Hb (sickle, thalassaemia). Acquired: immune or non-immune.
- Warm AIHA: IgG +/- complement, active at 37 C; spherocytes; mostly extravascular. Causes: idiopathic, SLE, RA, lymphoma/CLL, HIV, drugs.
- Drugs: methyldopa = true AIHA; penicillin = antibody to drug on red cell; rifampicin = complement via immune complex.
- Cold AIHA: active at room temperature or below, in cool extremities; complement-mediated, mostly intravascular; film agglutination. Causes: Mycoplasma, EBV, lymphoproliferative disease.
- Alloimmune: delayed transfusion reaction (extravascular), ABO-incompatible transfusion (intravascular), haemolytic disease of newborn.
- Non-immune: MAHA with schistocytes (HUS, TTP, DIC, HELLP, vasculitis), mechanical valves/ grafts, malaria, clostridia, burns, PNH, march haemoglobinuria.

Clinical

- Anaemia, jaundice, splenomegaly, pigment gallstones (chronic), plus features of the cause.

Four questions in investigation

- 1. Is it haemolysis? Raised reticulocytes (polychromasia), raised unconjugated bilirubin, raised LDH, low haptoglobin, raised urinary urobilinogen; marrow erythroid hyperplasia (rarely needed).
- 1. Immune or not? DAT (direct Coombs): positive = immune, negative = non-immune. Indirect test = free antibodies, used before transfusion.
- 1. Where? Extravascular = macrophages, mainly spleen. Intravascular = haemoglobinaemia, haemoglobinuria, haemosiderinuria, very low haptoglobin, very high LDH, methaemalbumin (Schumm's test).
- 1. Cause: haemolysis days after transfusion + DAT positive = delayed reaction (new alloantibody); AIHA + nodes + lymphocytosis = lymphoid malignancy; fever + agglutination = cold AIHA (Mycoplasma IgM).
- TTP: fever, neurology, renal impairment, low platelets, fragments; low ADAMTS13 is diagnostic.
- PNH: anaemia, abdominal pain, thrombosis, intermittent haemoglobinuria.

Management

- Folic acid 5 mg daily (high turnover uses folate).
- Transfuse only if severe, life-threatening or troublesome.
- Treat the specific cause.

Source: ICC-1 [077] Haemolytic anaemia

PART 1: LECTURE

Haemoglobin assay and blood indices [060] (prelab and lab)

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Sample and safety

- Gloves (peel off wrist to fingers), buttoned lab coat; sharps to sharps bin, disposables to red biohazard bag; spills: 10% hypochlorite for 10 min.
- CBC sample: EDTA, well mixed, run within 6 h (or store up to 24 h at 4 C).

Cyanmethaemoglobin method (ICSH reference)

- Drabkin's solution: red cells lyse; potassium ferricyanide oxidises Hb to methaemoglobin; potassium cyanide converts it to cyanmethaemoglobin.
- Stand 5-10 min; zero at 540 nm with a blank; absorbance at 540 nm is proportional to Hb.
- Hb = (absorbance test / absorbance standard) x standard concentration; or read from a standard curve made with HiCN standard when many samples.
- Micropipette: fit tip, press to first stop, release to draw up, check bubbles, expel at second stop.
- Check: Hb (g/dl) x 3 is roughly the PCV.

Red cell indices

- MCV = PCV x 10 / RBC (fl); MCH = Hb x 10 / RBC (pg); MCHC = Hb x 100 / PCV (g/dl).
- Uses: classify anaemia (micro-, normo-, macrocytic) and quality control.
- RDW = variation in red cell size (anisocytosis); raised in iron deficiency, megaloblastic anaemia and haemolysis.
- Poikilocytosis = abnormal shapes (on the film).
- Each lab sets its own reference ranges; approximate values: Hb men 13-17, women 12-15 g/dl; MCV 80-100 fl; MCH 27-32 pg.

Physiological variation

- Highest Hb at birth. Pregnancy: plasma expands more than red cells, lower limit Hb 11 g/dl.
- High altitude and smoking (carboxyhaemoglobin) raise Hb and PCV.

Other CBC points

- Reticulocytes 0.5-2.5%: reflect marrow activity; mature in 2-3 days (1 day in blood).
- Shift to left (myelocytes, bands > 6%): severe infection, marrow infiltration.
- Nucleated red cells beyond the neonatal period = marrow stress or damage; leukoerythroblastic = immature red and white cells.
- Low Hb, WBC and platelets together = pancytopenia.

Source: Hb estimation assay; ICC-1 [044] prelab; [060] Hb assay checklists

PART 1: LECTURE

Blood film preparation [066] (prelab and lab)

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Lab safety

- Gloves before handling blood; remove them before touching notebooks or microscopes and before leaving the lab; peel from wrist to fingers.
- Sharps and slides go in the sharps bin; disposables in red biohazard bags.
- Blood spill: pour 10% sodium hypochlorite, cover with absorbent towel, leave at least 10 min, repeat, then wash.
- Skin penetrated: wash with soap and water and encourage bleeding. Eyes: rinse gently with water or saline. Mouth: spit and rinse. Report at once.

Making the film

- Sample: EDTA blood or finger/heel prick (choice depends on the case).
- Drop (about 20 uL) near the end of a clean slide; pusher at 30-45 degrees (checklist: 30-40); pull back until blood spreads across the width, then push forward quickly and smoothly.
- Good film covers 2/3 to 3/4 of the slide, smooth, no irregularities. Wrong angle pushes large cells (monocytes) to the tail and spoils the differential.
- Air dry, stain (Wright-Giemsa most common, or Leishman), wash, air dry.

Examining the film

- x10 first: large abnormal cells (blasts) and RBC distribution. Then x40 and x100 (oil): RBC morphology, WBC differential, platelets.
- Good stain: RBCs pink-salmon; neutrophil granules lilac; eosinophil red-orange; basophil dark blue-black; clean background.
- Normal RBC: biconcave, anucleate, central pallor, 7-8 um, MCV 80-100 fL, MCHC 31-37 g/dL. Platelets 150-450 x10^9/L, 1-4 um.

Abnormal red cells and their links

- Target cells: thalassaemia. Spherocytes (no central pallor): hereditary spherocytosis. Tear drop cells: myelofibrosis, thalassaemia major.
- Acanthocytes: abetalipoproteinaemia. Basophilic stippling: lead. Schistocytes (fragments): microangiopathic haemolysis. Agglutination: cold agglutinins.
- Elliptocytes (pencil cells): hereditary elliptocytosis; also increased in iron deficiency.
- B12 deficiency: macrocytes (MCV > 100) and hypersegmented neutrophils.
- Leuko-erythroblastic film: normoblasts + myelocytes.

Reticulocytes and marrow iron

- Young RBCs with RNA remnants; supravital stain (new methylene blue or brilliant cresyl blue). Normal 0.5-2.5%.
- High in chronic haemolysis and after bleeding (rise 2-3 days, peak 6-10 days). No rise in anaemia = poor marrow or no erythropoietin.
- Prussian blue marrow: iron deficiency = no iron anywhere; chronic disease = iron in macrophages, none in red cell precursors.

Source: [066] prelab session; [066] film benchaid; [066] revision benchaid; blood film checklist

PART 1: LECTURE

Haematology cases LGW [092]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Thrombophilia work-up (young DVT, family history)

- Delay thrombophilia tests until at least 1 month after anticoagulation ends: the acute clot and anticoagulants distort results, and results do not change acute management.
- Screen: PT, APTT, TT, protein C, protein S, antithrombin. If all normal: PCR panel for factor V Leiden (G1691A), prothrombin G20210A, MTHFR.
- Factor V Leiden: autosomal dominant; Arg506Gln makes factor Va resistant to activated protein C (APC resistance).
- Risk of venous thrombosis about 4-fold (heterozygous) and about 20-fold (homozygous). Offer screening to affected relatives.

Antiphospholipid syndrome (recurrent miscarriage)

- Three consecutive early miscarriages with no anatomical or hormonal cause: weakly positive lupus anticoagulant suggests APS but is not diagnostic.
- Next: anticardiolipin and anti-beta-2 glycoprotein 1 antibodies.
- Repeat after 12 weeks; still positive with high titre = diagnostic.
- Confirmed APS: anticoagulant therapy as soon as she becomes pregnant.

Autoimmune haemolytic anaemia

- Normocytic normochromic anaemia + jaundice = suspect haemolysis.
- Confirm haemolysis: bilirubin up, haptoglobin down. Reticulocytes up (13%) = effective marrow response.
- Haemoglobinuria (pink-red urine) with high bilirubin and low haptoglobin = intravascular haemolysis.
- Film: spherocytes and red cell agglutination (cold agglutinins). Direct antiglobulin test (DAT) positive is diagnostic of AIHA.
- Cold type antibodies are usually IgM; about half are secondary (lymphoma, viral infection).

TTP in pregnancy

- Fever + neurological change + renal impairment + low platelets + fragmented red cells (microangiopathic haemolysis).
- Normal clotting excludes DIC; normal liver tests exclude HELLP; normal BP argues against pre-eclampsia.
- Cause: antibodies to ADAMTS13, so very large VWF multimers circulate; rare inherited ADAMTS13 deficiency.
- Treatment: early daily plasma exchange with FFP. Untreated mortality high; relapse 30-60%, mostly in the first month. Newer: rituximab, recombinant ADAMTS13.

Source: ICCA [092] haematology cases LGW Qs and As

PART 1: LECTURE

Clinical chemistry tutorial [027]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Lipid profile

- Reference (fasting): total cholesterol < 5.2 mmol/L (< 200 mg/dL); TG < 1.7 mmol/L (< 150 mg/dL); HDL-C > 1 mmol/L (> 40 mg/dL).
- Convert: cholesterol and HDL-C mg/dL = mmol/L x 38.7; TG mg/dL = mmol/L x 88.57.
- Case: cholesterol 6.7 (259 mg/dL) high, TG 1.4 (124) normal, HDL-C 0.83 (32) low = hypercholesterolaemia with normal TG.

Secondary causes (exclude before treating)

- High cholesterol (LDL): hypothyroidism, chronic liver disease, nephrotic syndrome; progestins, thiazides, glucocorticoids, ciclosporin.
- High TG: obesity, type 2 DM, pregnancy, acute hepatitis; alcohol, oestrogens, antifungals, isotretinoin.
- Low HDL: malnutrition, obesity, metabolic syndrome; anabolic steroids, tobacco.

Assessment and management

- Look for familial hypercholesterolaemia signs (tendon xanthomata, arcus, xanthelasma); absence does not exclude FH.
- History: smoking, alcohol, DM, drugs, past PVD or stroke, family history of premature CVD. Exam: BMI, waist, BP, xanthomata, bruits, pulses.
- Manage: exclude secondary causes, treat hypertension, weight loss with dietician, then estimate 10-year risk to decide on drugs.
- QRISK2: UK algorithm for 10-year CVD risk, preferred by NICE since 2010; calibration varies in other populations. Lifestyle and other risk factors first before statins.
- New lab method: convert to reference-method value by regression (HDL-C 0.83 = 1.0 mmol/L; QRISK 8.07% to 6.7%). Method can change the statin decision.

Cardiac markers

- Typical pain + normal ECG does not exclude MI (ECG not always positive).
- Total CK is not heart-specific. Troponin is heart-specific but rises 4-8 h after onset: negative at 3 h does not exclude MI; repeat later.
- CK-MB also rises at 4-8 h; also in skeletal muscle; CK-MB/total CK > 5% supports MI.
- At 12 h: raised troponin T confirms myocardial damage.
- Re-infarction: troponin stays raised > 1 week, so not useful; CK-MB normalises in 2-3 days, so a new rise (ratio > 5%) = new or extended infarct.

Source: ICC-1 [027] clinical chemistry tutorial Q and answers

PART 2: TOPIC SUMMARY

Thrombophilia, DVT and pulmonary embolism

How clotting is balanced

- Platelets: adhesion (GPIb to VWF), aggregation (GPIIb/IIIa with fibrinogen), amplification by TXA2. Coagulation: tissue factor + VIIa starts it; prothrombinase (Xa + Va) gives the thrombin burst; XIIIa cross-links fibrin.
- Natural brakes: antithrombin (thrombin and Xa, boosted by heparin); protein C + protein S (inactivate Va and VIIIa); TFPI; fibrinolysis (t-PA makes plasmin).
- Tests: PT (VII, X, V, II, I) monitors warfarin; APTT monitors unfractionated heparin; isolated long APTT = VIII, IX, XI or XII (XII causes no bleeding).

Thrombophilia at a glance

Condition	Key fact	Clue
Factor V Leiden	Commonest; Arg506Gln, APC resistance	Young venous clot, family history
Prothrombin G20210A	High prothrombin level	DNA test
Antithrombin deficiency	Highest risk natural anticoagulant defect	Heparin resistance
Protein C deficiency	Short half-life	Warfarin skin necrosis; neonatal purpura fulminans
Antiphospholipid syndrome	Clots and pregnancy loss	Long APTT in vitro but thrombosis; repeat antibodies after 12 weeks
HITT	Antibody to PF4-heparin	Platelets fall 4-10 days after heparin

- Test when: VTE under 40, unusual site, recurrent, strong family history, 3 or more miscarriages. Wait at least 1 month after anticoagulation ends (DNA tests any time).

DVT and PE pathway

- Clinical probability first (Wells): unlikely = D-dimer (a negative result excludes); likely = imaging straight away.
- DVT: compression duplex ultrasound. PE: CTPA (V/Q if CTPA unsuitable). ECG usually sinus tachycardia; S1Q3T3 is uncommon.
- Treat: DOAC preferred (or LMWH overlapping with warfarin). Provoked 3-6 months; unprovoked or cancer = consider indefinite.
- Thrombolysis only for massive PE (shock) or limb-threatening DVT. IVC filter only if anticoagulation is impossible.

PART 2: TOPIC SUMMARY

Anaemias (megaloblastic, iron deficiency, haemolytic)

First step: size of the red cells

Microcytic (MCV under 80)	Normocytic	Macrocytic (MCV over 100)
Iron deficiency, thalassaemia trait, anaemia of chronic disease, sideroblastic, lead	Acute bleeding, haemolysis, chronic disease, renal failure, marrow failure	Megaloblastic (B12, folate); alcohol, liver disease, hypothyroidism, myelodysplasia, reticulocytosis

- Reticulocytes high = bleeding or haemolysis (marrow working); low = marrow failure or no erythropoietin (renal disease).
- Treat the cause, not the number. Transfuse only for severe, symptomatic anaemia or acute bleeding.

Iron deficiency versus look-alikes

	Iron	TIBC	Ferritin	Other
Iron deficiency	Low	High	Low	Pencil cells, raised platelets
Chronic disease	Low	Low	Normal or high	Iron trapped in macrophages
Thalassaemia trait	Normal	Normal	Normal	Raised HbA2, mild anaemia

- Causes: menorrhagia (UK), hookworm (worldwide), GI blood loss. Older patient with unexplained iron deficiency = scope for cancer.
- Ferrous sulphate 200 mg three times daily for 4-6 months, continue after Hb normalises to refill stores. Poor response: compliance first.

Megaloblastic anaemia

- B12: animal foods; stores last about 3 years; needs intrinsic factor, absorbed in terminal ileum. Pernicious anaemia: intrinsic factor antibodies. Neurology (subacute combined degeneration) even without anaemia.
- Folate: stores last about 4 months; poor diet, pregnancy, haemolysis, anticonvulsants, methotrexate.
- Film: oval macrocytes, hypersegmented neutrophils; raised LDH and bilirubin.
- Never give folic acid alone before B12. Hydroxocobalamin IM; folic acid 5 mg daily; 400 µg before conception.

Haemolysis in four questions

- Is it haemolysis? Reticulocytes, unconjugated bilirubin and LDH up; haptoglobin down.
- Immune? DAT (direct Coombs) positive = immune.
- Where? Intravascular: haemoglobinuria, very low haptoglobin. Extravascular: spleen.
- Cause? Warm IgG (SLE, CLL, drugs; spherocytes) versus cold IgM (Mycoplasma, EBV; agglutination). TTP: fever, neurology, renal failure, low platelets, fragments, low ADAMTS13; plasma exchange urgently.

Lab skills

- Hb by cyanmethaemoglobin method, read at 540 nm. MCV = PCV x 10 / RBC; MCH = Hb x 10 / RBC; MCHC = Hb x 100 / PCV.
- Blood film: drop near the end, spreader at about 30-45 degrees, smooth push; read at x10 then oil. Target cells = thalassaemia; spherocytes = hereditary spherocytosis or warm AIHA; schistocytes = MAHA; basophilic stippling = lead.
- Clinical chemistry: lipid units (cholesterol mg/dL = mmol/L x 38.7); troponin rises 4-8 h after MI and stays up over a week, so use CK-MB for re-infarction.